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2018.6.18

航空宇宙設計最適化

Design optimization in aerospace engineering

大山　聖
Akira Oyama

スケジュール　Schedule

4/16	[1] 設計最適化問題とは？	What is design optimization?
4/23	[2] 設計最適化手法（その１）	Design optimization methods (part 1)
5/7	[3] 設計最適化手法（その２）	Design optimization methods (part 2)
5/14	[4] 設計最適化手法（その３）	Design optimization methods (part 3)
5/21	[5] 設計最適化手法（その４）	Design optimization methods (part 4)
5/28	休講	
6/4	[6] 飛行機の設計	aircraft design
	JAXA第三研究ユニット長 嶋英志氏	
6/11	[7] 多目的設計探査	Multiobjective design exploration
6/18	[8]【活用事例】Solar-DとDESTINYの軌道計画	
	Trajectory design of Solar-D and DESTINY missions	
6/25	[9] ロバスト設計・不確定性定量評価　robust design /uq	
	東北大学流体科学研究所准教授　下山幸治氏	
7/2	[10]【活用事例】H-III ロケットのターボポンプ設計＆自動車設計	
	H-III Rocket turbopump design and car structure design	
7/9	[11] 超音速機設計	supersonic aircraft design
	JAXA航空プログラムグループ超音速機チーム牧野好和氏	

Outline

多目的設計探査の工学での利用

Applications of MODE in space engineering

- 太陽極域観測衛星計画の軌道設計

  Trajectory design of sun polar region observation satellite

- DESTINYミッションの軌道設計

  Trajectory design of DESTINY mission

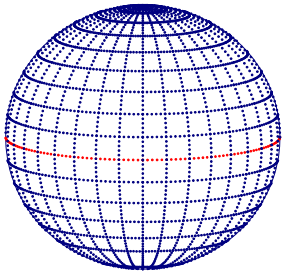
太陽極域観測衛星計画は，次期太陽観測衛星SOLAR-C計画の一案として検討されてきたが，2010年度末に別案が採択されることとなり，SOLAR-Dとして検討が継続されることが決まっている．

本資料では，この検討が行われた2009年度から2010年度にかけて作成された資料により構成されていることに注意

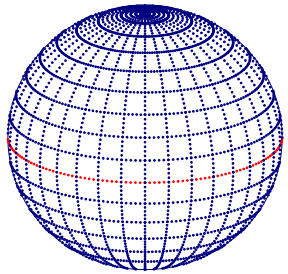
Sun polar region observation mission

目的は太陽極域の観測

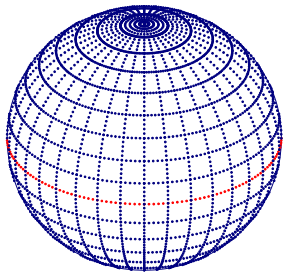
Observation of polar region of the sun



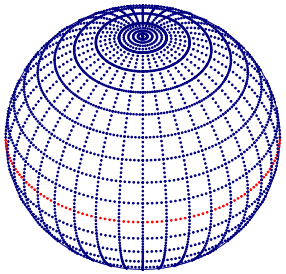
Lat. 10degN



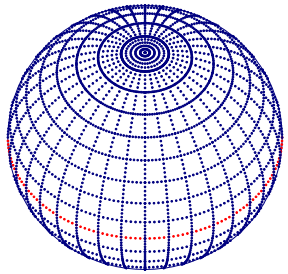
Lat. 20degN



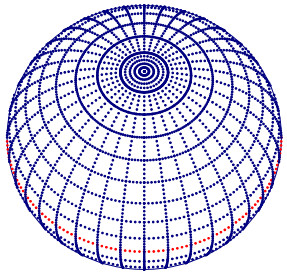
Lat. 30degN



Lat. 40degN



Lat. 50degN



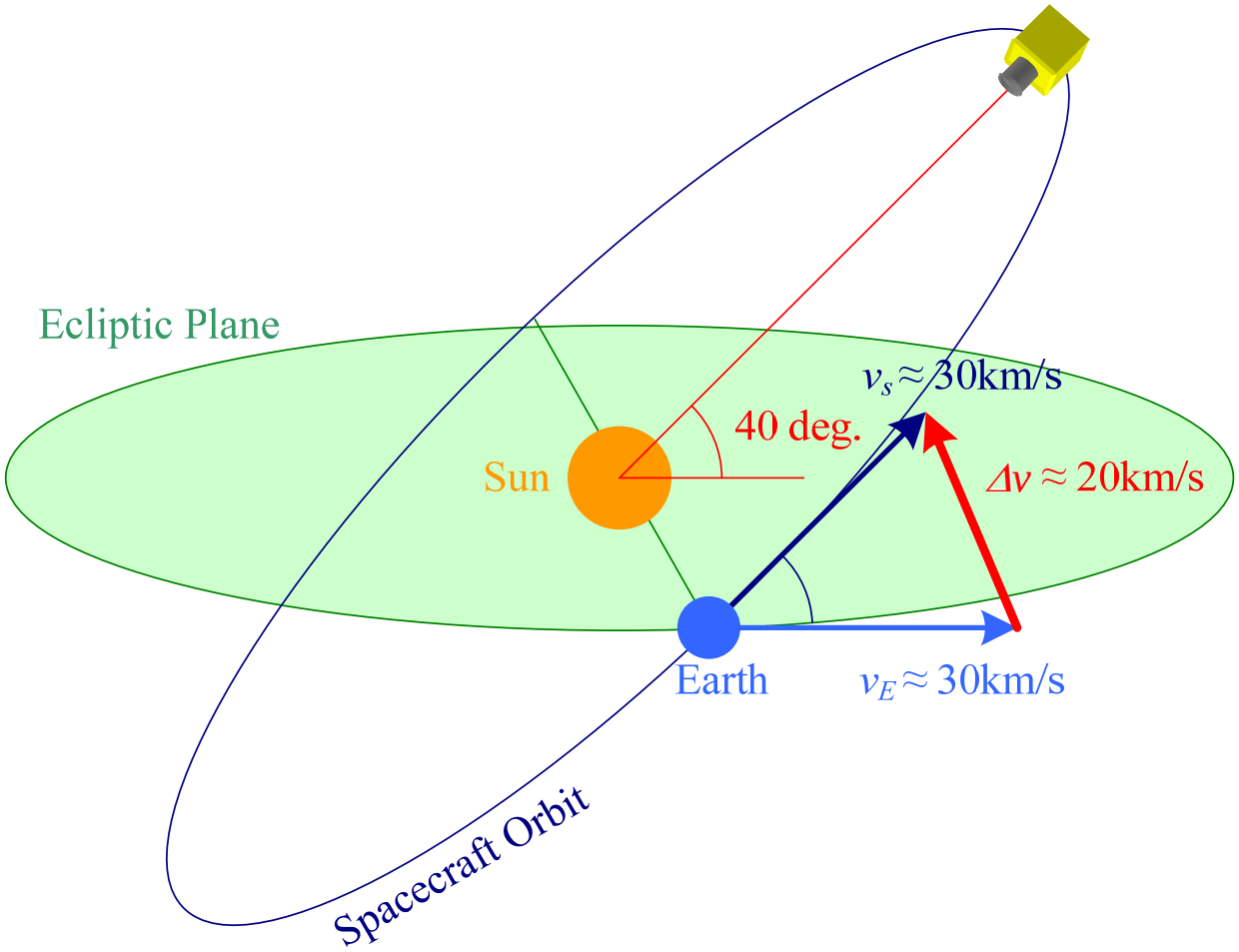
Lat. 60degN

View of the Sun from the High Solar Latitude

Sun polar region observation mission

高傾斜角の軌道からの観測が要求される

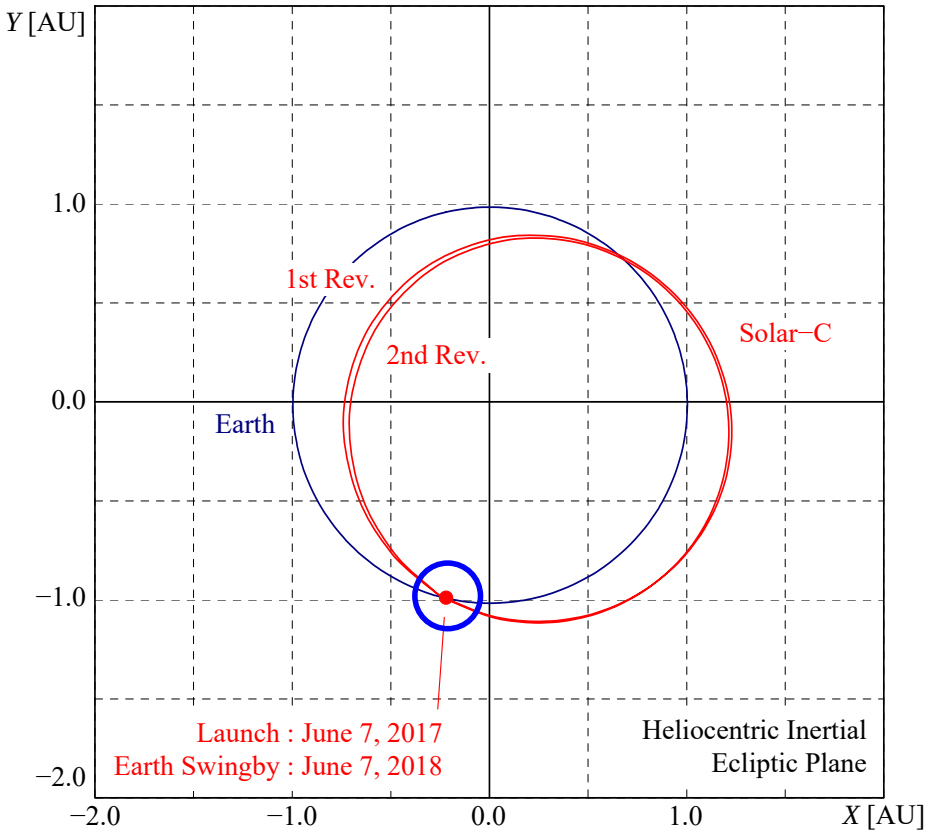
Observation from a trajectory with a high inclination angle is required

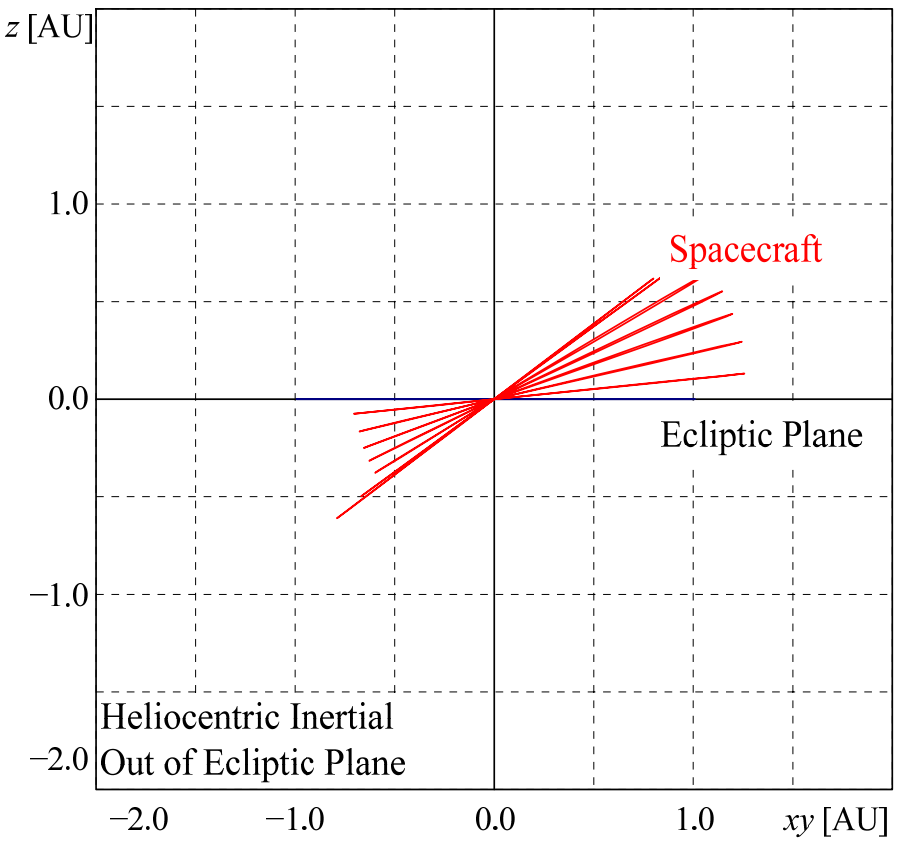


Sun polar region observation mission

イオンエンジンと地球スィングバイを組み合わせた新規軌道の設計が必要とされる

Design of new trajectory based on ion engine and Earth swing-by is required



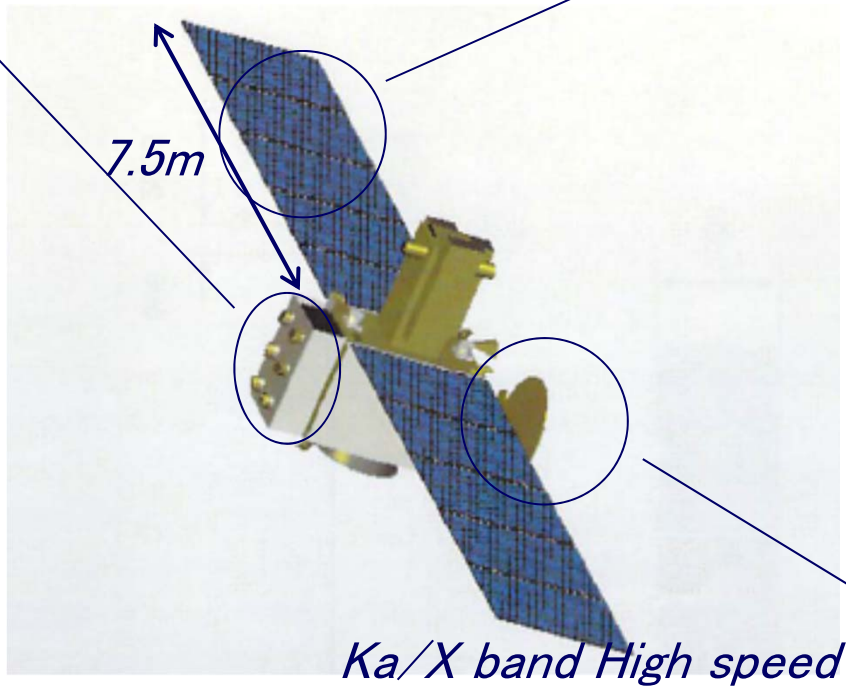


Spacecraft that realizes the mission

Launch mass : 1200kg

Light weight solar paddle

M20 Ion engine system







Ka/X band High speed communication system

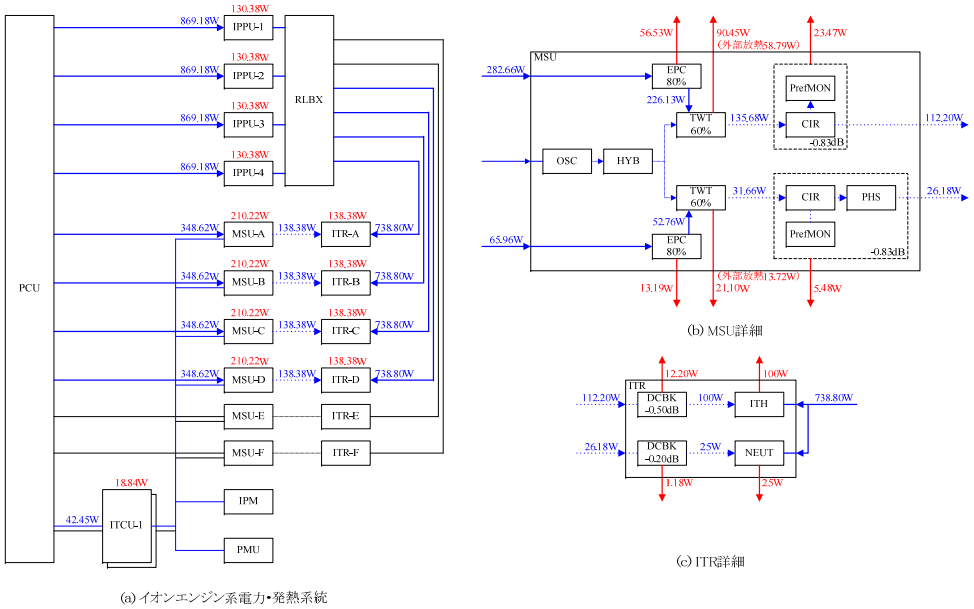
SEP option

Launch by H2A202 + solid upper stage motor

μ20 Ion engine system

- 「はやぶさ」で使用されたμ10の2倍の口径，約3倍の推力をもつμ20（JAXAにて研究開発中）の使用を想定。

  μ20 ion engine has twice the diameter of μ10 ion engine used in Hayabusa mission and about 3 times thrust



(a) イオンエンジン系電力・発熱系統

(b) MSU詳細

(c) ITR詳細

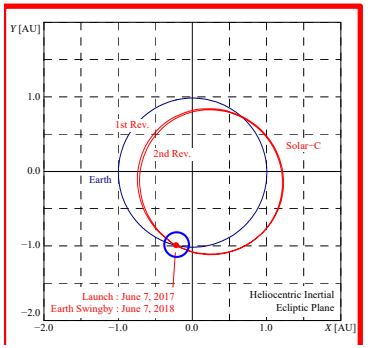
No.	機器構成	略称	搭載台数	質量 [kg]		
				1台	合計	統合
1	スラスタ制御器	ITCU	2	4.47	8.94	8.94
2	電源	IPPU	4	6.00	24.00	24.00
3	リレーボックス	RLBX	1	1.60	1.60	1.60
4	発振器	OSC	6	0.50	3.00	39.00
5	増幅器_ITH（TWT）	TWT_ITH	6	1.30	7.80	
6	増幅器_ITH（電源）	EPC_ITH	6	1.70	10.20	
7	増幅器_NEUT（TWT）	TWT_NEUT	6	1.30	7.80	
8	増幅器_NEUT（電源）	EPC_ITH	6	1.70	10.20	
9	スラスタヘッド	ITH	6	5.00	30.00	90.61
10	DCブロック_ITH	DCBK_i	6	↑	↑	
11	RW_伝送系_ITH	CBL_ion	6	0.82	4.92	
12	中和器	NEUT	6	0.50	3.00	
13	DCブロック_NEUT	DCBK_n	6	↑	↑	
14	RF_伝送系_ITH	CBL_neut	6	0.82	4.92	
15	IESプレート X	IES-PLT X	1	7.00	7.00	
16	IESプレート Y	IES-PLT Y	2	3.50	7.00	
17	IESプレート Z	IES-PLT Z	2	2.40	4.80	
18	ヒートパイプ	ITA-TINT	-	-	8.47	
19	IESポインティング機構	IPM	1	11.50	11.50	
20	ITA機械計装	ITA-MINT	6	1.50	9.00	
21	推進剤供給系	PMU	-	90.00	90.00	90.00
	イオンエンジンシステム		1		254.15	254.15

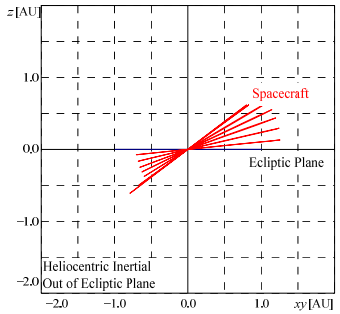
イオンエンジン系の電力・発熱系統
Power and heating system

イオンエンジン系の機器構成と質量バジェット
Equipment configuration and mass budget

軌道計画　Trajectory plan

Date	Event	m	v_∞	i_{SEQ}	Δv_∞	t_{IES}
2017/05/31	Launch	1200.0 kg	7.3 km/s	7.3 deg.	2.7 km/s	7131 / 5214 h
2018/06/07	Swingby #1	1129.1 kg	10.0 km/s	7.2 deg.		
				19.1 deg.	2.4 km/s	7365 / 5568 h
2019/06/07	Swingby #2	1055.4 kg	12.4 km/s	20.3 deg.		
				26.1 deg.	2.1 km/s	6852 / 5316 h
2020/06/06	Swingby #3	988.1 kg	14.5 km/s	27.5 deg.		
				31.0 deg.	1.4 km/s	4186 / 3606 h
2021/06/06	Swingby #4	945.2 kg	15.9 km/s	31.6 deg.		
				36.5 deg.	0.7 km/s	3711 / 2039 h
2022/06/07	Swingby #5	921.9 kg	16.6 km/s	37.8 deg.		
				40.0 deg.	0.0 km/s	0 / 0 h
2023/06/07	Swingby #6	921.9 kg	16.6 km/s	40.0 deg.		
				40.0 deg.	0.0 km/s	0 / 0 h
Final State		**921.9 kg**	**16.6 km/s**		**9.3 km/s**	**29244 / 21743 h**





m ：質量
v_∞ ：無限遠速度
α, δ ：無限遠速度方向（方位角、上下角）
i_{SEQ} ：太陽赤道に対する傾斜角
Δv_∞ ：無限遠速度の増分
t_{IES} ：イオンエンジン運転時間（のべ/1台あたり）

Difficulty

Difference between chemical propulsion and electric propulsion.

	Chemical	Electric
Thrust [N]	$O(10^3)$	$O(10^{-2})$
Operation time [s]	$O(10^3)$	$O(10^8)$
Requirement	None	Sun light
Knowledge	Well known	No enough theory and experience

- Chemical propulsion: Large thrust, short operation time, quick control, well known, easy to design.
- Electric propulsion: Small thrust, long operation time, slow control, not known very much, difficult to design.

Problem in trajectory design

最適な軌道を求める上での知見が不足している

Knowledge on finding the optimum trajectory is not enough

- 太陽極域観測のための軌道は従来の軌道とは大きく異なる

  Trajectory for solar polar region observation is quite different from the conventional trajectories.

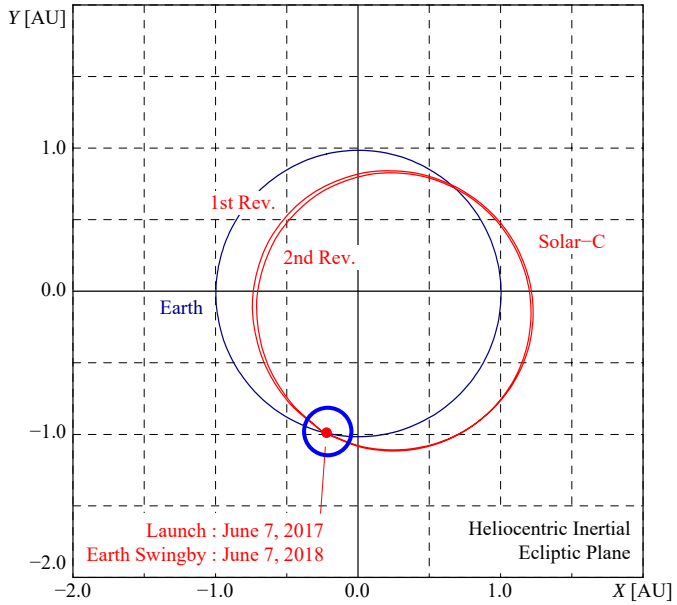
- イオンエンジンを使った軌道設計法については，まだ十分な知見が得られていない

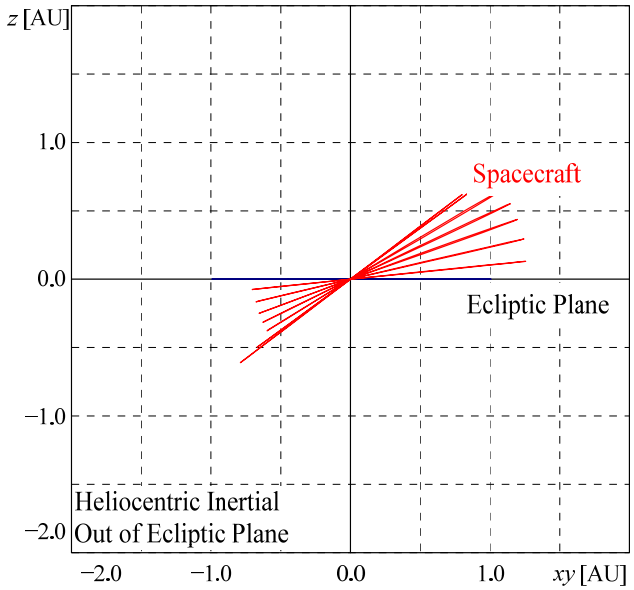
  Knowledge on the trajectory design using ion engine is not sufficient

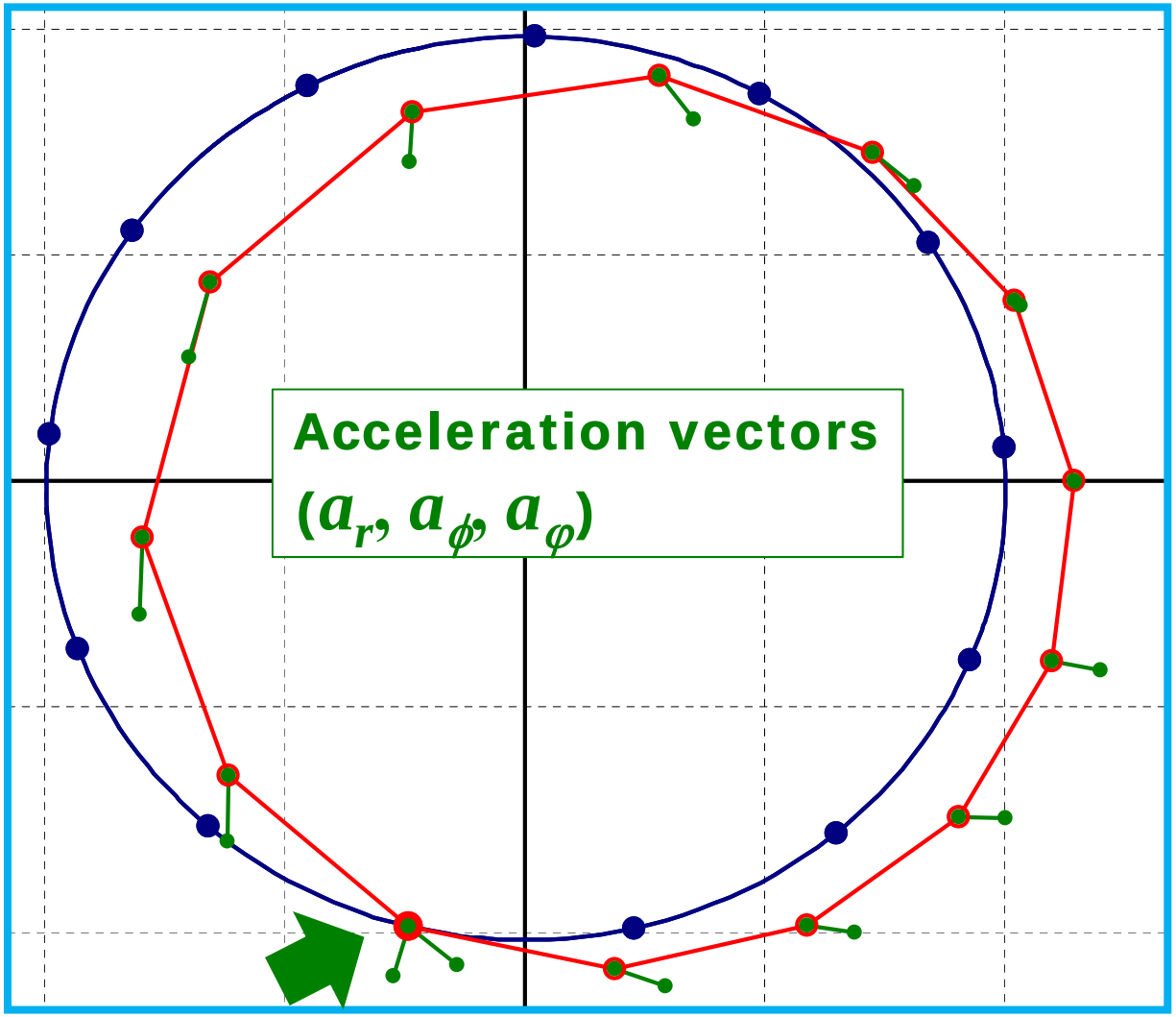
→多目的設計探査の試行（2009年に開始）

Multiobjective design exploration of the trajectory started in 2009

従来の軌道設計方法 Conventional trajectory design method

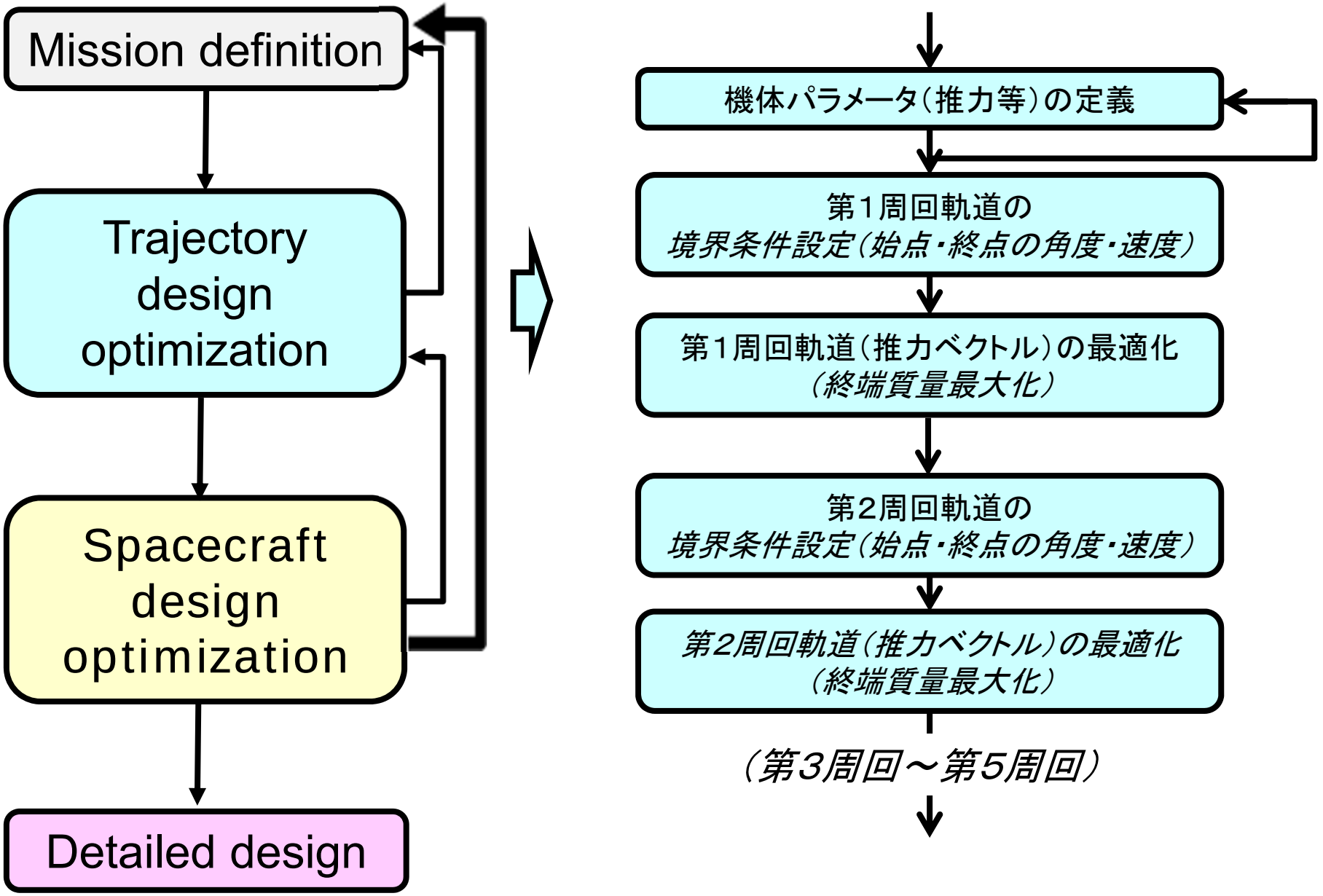






Earth swing-by parameters

従来の軌道設計方法 Conventional trajectory design method



従来の設計方法の課題 Problem in traditional design

- 機体の設計変数およびスイングバイの設計変数（終点速度，終点角度(2)，始点速度，始点角度(2)）は専門家がこれまでの知見をもとに与える

  Design parameters for spacecraft design and swing-by are determined by specialist (not optimized)

- 各周期の軌道（＝イオンエンジンの推力制御プロファイル）は逐次二次計画法で最適化して求める

  Trajectory of each cycle is obtained by sequential quadratic programming

  - 目的：各周期の終端質量の最大化

    Objective: maximization of mass at the end of each cycle

  - 設計変数：14点における加速度ベクトル(a_r, a_ϕ, a_φ)

    Design parameters: acceleration vector at 14 points

従来の設計方法の課題 Problem in traditional design

問題点 Problems

- 終端質量の最大化以外の目的に基づいた最適化されていない

  Single objective optimization (maximization of mass)

- 機体の設計変数とスィングバイの設計変数は最適化されていない（試行錯誤）

  Design parameters for spacecraft design and swing-by are not optimized

研究の目的 Objectives of this research

- 1周期目（打ち上げから1回目の地球スィングバイまで）の軌道の多目的設計探査をこない，軌道設計への多目的設計探査の有効性を示す

  Conduct multiobjective design exploration of the trajectory design of the first cycle

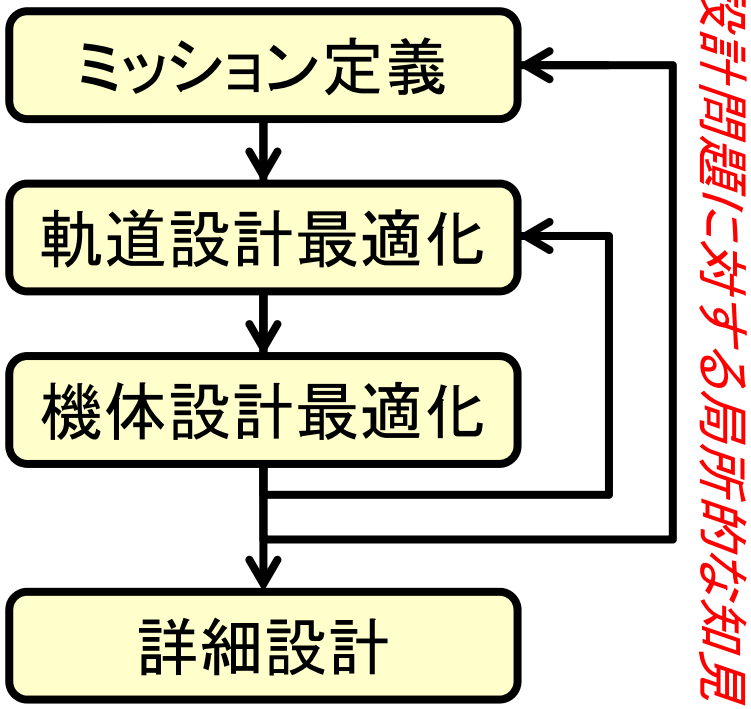
  - より優れた解の発見　to find better trajectory design
  - 設計期間の短縮　to shorten the design period
  - 設計問題に関する新しい知見　to find new knowledge

- 5周期目までの軌道の設計探査を行い，高傾斜角軌道最適化に関する設計知見を獲得する

  Conduct multiobjective design exploration of the trajectory design of the five cycles

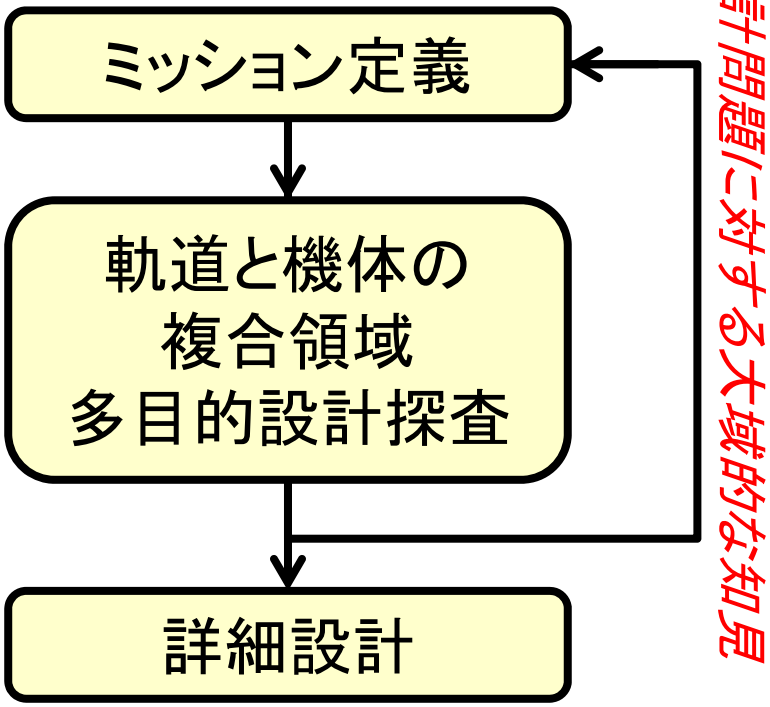
新しい設計方法 New design method

これまでの設計方法
Conventional method



逐次的で，*局所的な知見に基づく*試行錯誤による最適化

新しい設計方法
New method



効率的，かつ，大域的な知見に基づく多目的進化計算による最適化

第１周回の軌道の多目的設計探査

MODE of 1st trajectory

Design problem definition

従来手法で設計されたノミナル軌道を
ベースに設計問題を定式化する
Design problem is defined based on the nominal trajectory

- Launch condition
  - Launcher: H2A with a solid motor upper stage
  - Initial mass: 1200kg
  - Launch date: 2017.6.7
  - The first earth swing-by date: 2018.6.7
  - Initial relative velocity magnitude: 7.7 km/s
  - Azimuth angle at the first swing-by: -94.285 deg.
  - Elevation angle at the first swing-by: 19.793 deg.
- Spacecraft
  - Operation rate is 0.875

設計目的　design objectives

- 終端質量の最大化 （搭載可能ペイロード最大化に相当）

  Maximization of mass of the spacecraft at the end (this corresponds to maximization of science payload mass)

- 終端における対地球相対速度の最大化 （軌道傾斜角最大化に相当）

  Maximization of the relative speed to the Earth at the end (this corresponds to maximization of trajectory inclination angle)

- 太陽との最小距離の最大化 （熱防御システム重量最小化に相当）

  Maximization of minimum distance to the sun (this corresponds to minimization of thermal protection system weight)

制約条件　constraint conditions

- 地球と衝突しないこと

  Do not collide with the earth

- 離心率が0．3を超えないこと

  Eccentricity should not exceed 0.3

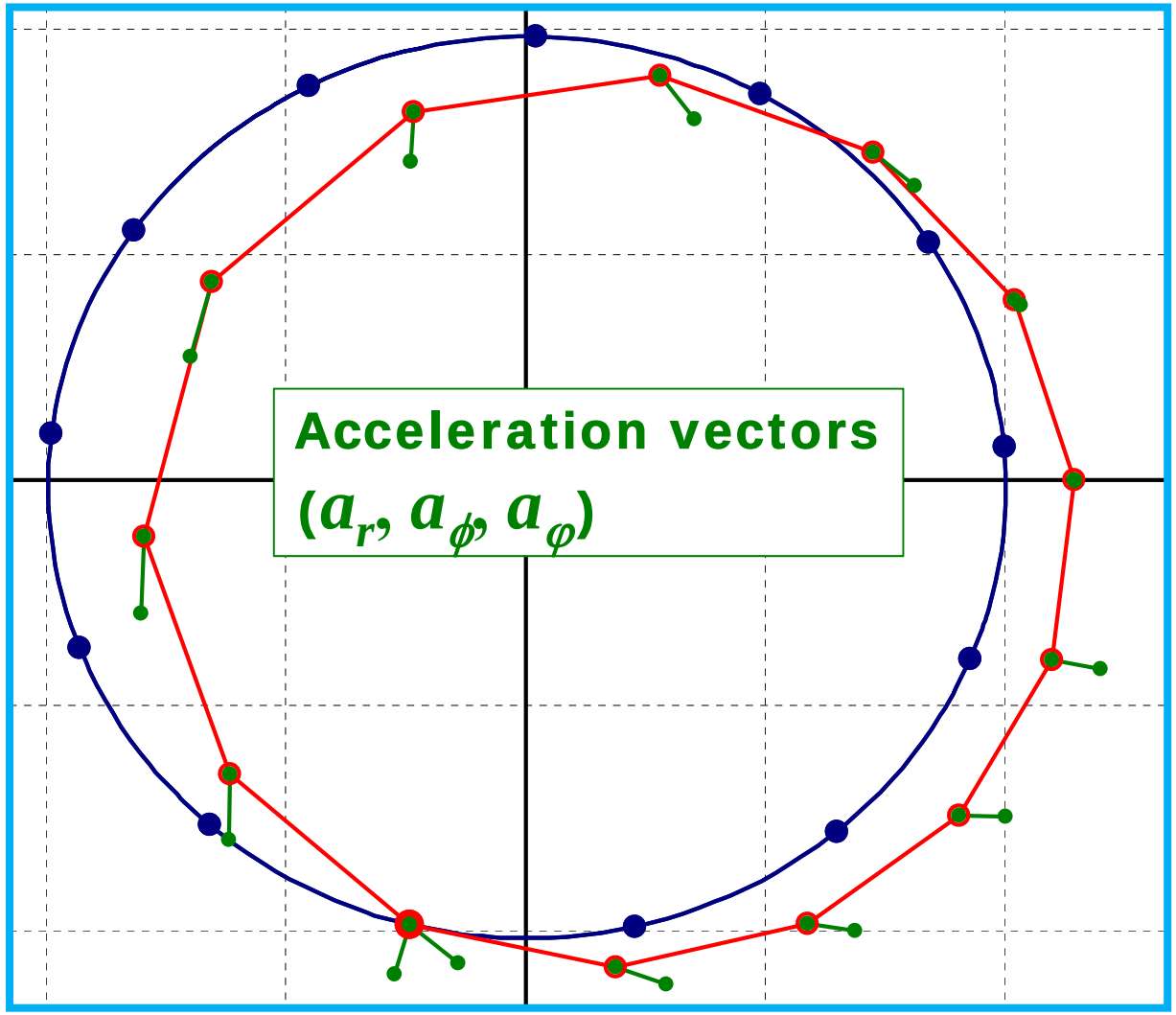
- 最大推力 x 比推力 が一定値（ノミナル値と同値）であること（必要電力一定に相当）

  Maximum thrust x specific impulse = constant

  (this corresponds to required power is constant)

設計変数　Design parameters

- Thrust control profile for the first cycle
  - Acceleration (a_r, a_ϕ, a_φ) at given points (14 points)



設計変数　Design parameters

- Thrust control profile for the first cycle
  - Acceleration (a_r, a_ϕ, a_φ) at given points (14 points)
- Azimuth angle of the velocity at the departure

  -118.12 [deg.] < Az < -78.12 [deg.] (nominal 98.12 [deg.])
- Elevation angle of the velocity at the departure

  1.80 [deg.] < El < 41.80 [deg.] (nominal 21.80 [deg.])
- Relative velocity magnitude in the first Earth swing-by

  9.2 [km/s] < V < 11.2 [km/s] (nominal 10.2[km/s])
- Specific impulse (design parameter of spacecraft!)

  2400 [s] < Isp < 3600 [s] (nominal 3000[s])

Number of design parameters is 3x14+4=46

従来のフローチャート conventional flowchart

試行錯誤 Trial and error

機体とスィングバイに関する設計変数の値
- *始点のAzimuth角(方位角)*
- *始点のElevation角(仰角)*
- *終端の対地球相対速度(速さ)*
- *比推力*

目的関数の値 objective function value
- *終端質量*
- *終端における対地球相対速度(速さ)*
- *最小動径半径*

イオンエンジン推力制御プロファイルの最適化モジュール
- *設計変数: 14点で定義される加速度ベクトル(a_r, a_ϕ, a_φ)*
- *目的関数: 終端質量↑*
- *最適化手法:逐次2次計画法*

新しいフローチャート New flowchart

多目的最適化モジュール MOEA

- 設計変数：始点方位角、始点仰角、終端速さ、比推力
- 目的関数：終端質量↑ 、終端速さ↑、最小動径半径↑
- 最適化手法：多目的進化計算(NSGA-II)

機体とスィングバイに関する設計変数の値

- 始点のAzimuth角（方位角）
- 始点のElevation角（仰角）
- 終端の対地球相対速度（速さ）
- 比推力

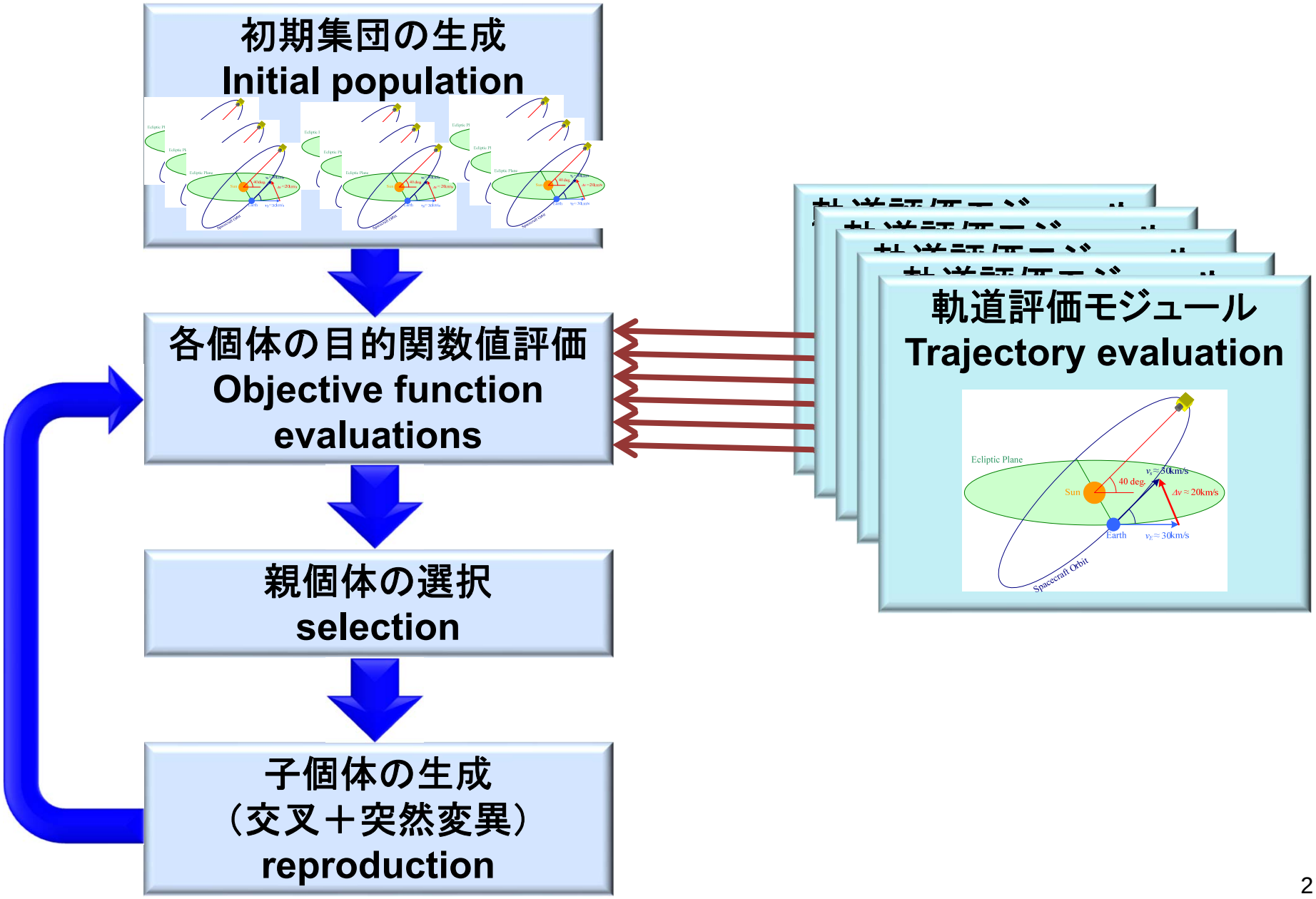
目的関数の値 objective function values

- 終端質量
- 終端における対地球相対速度（速さ）
- 最小動径半径

イオンエンジン推力制御プロファイルの最適化モジュール

- 設計変数： 14点で定義される加速度ベクトル(a_r, a_ϕ, a_φ)
- 目的関数： 終端質量↑
- 最適化手法：逐次2次計画法

NSGA-IIを使って最適化　Optimization using NSGA-II

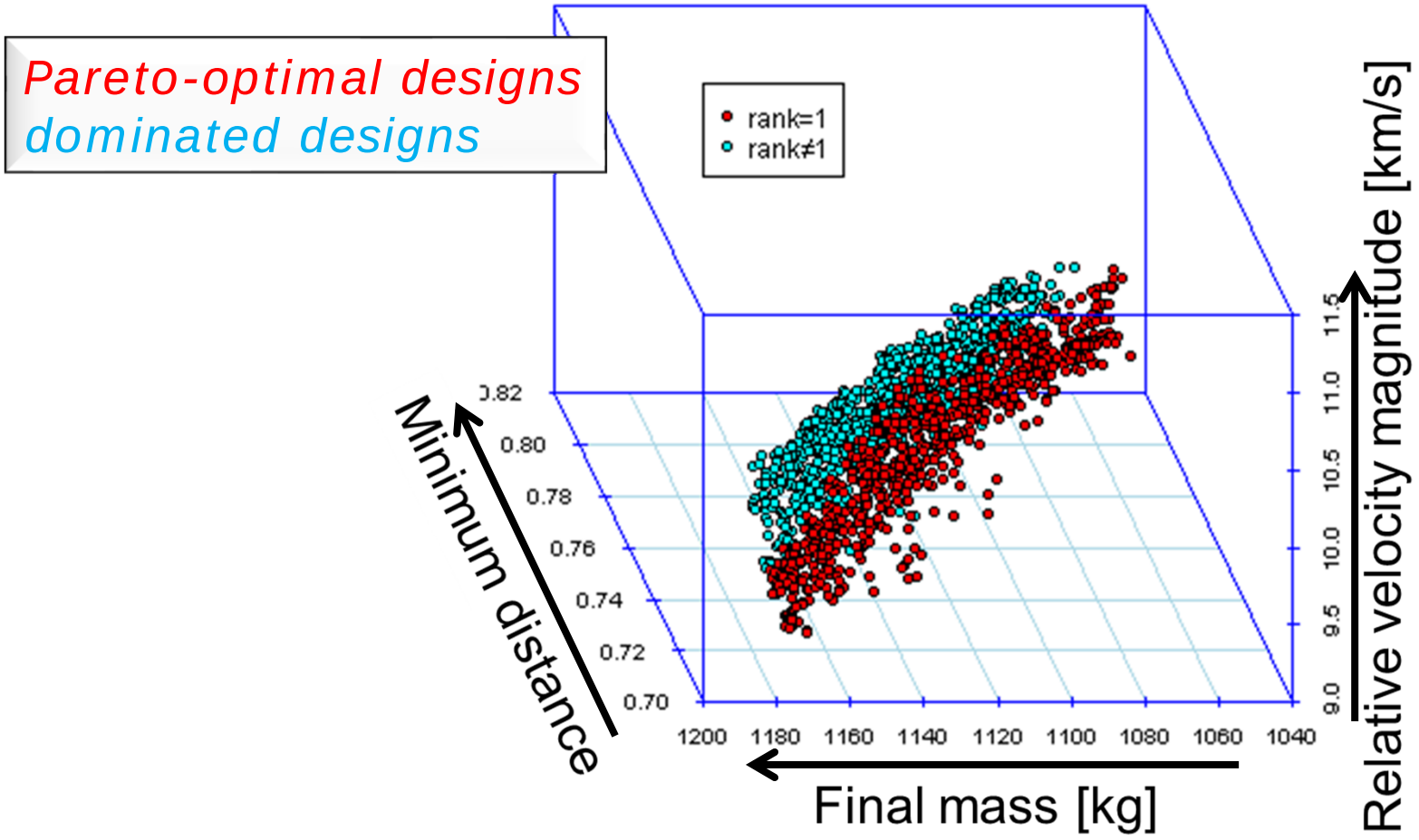


評価回数と計算時間 number of evaluations and computational time

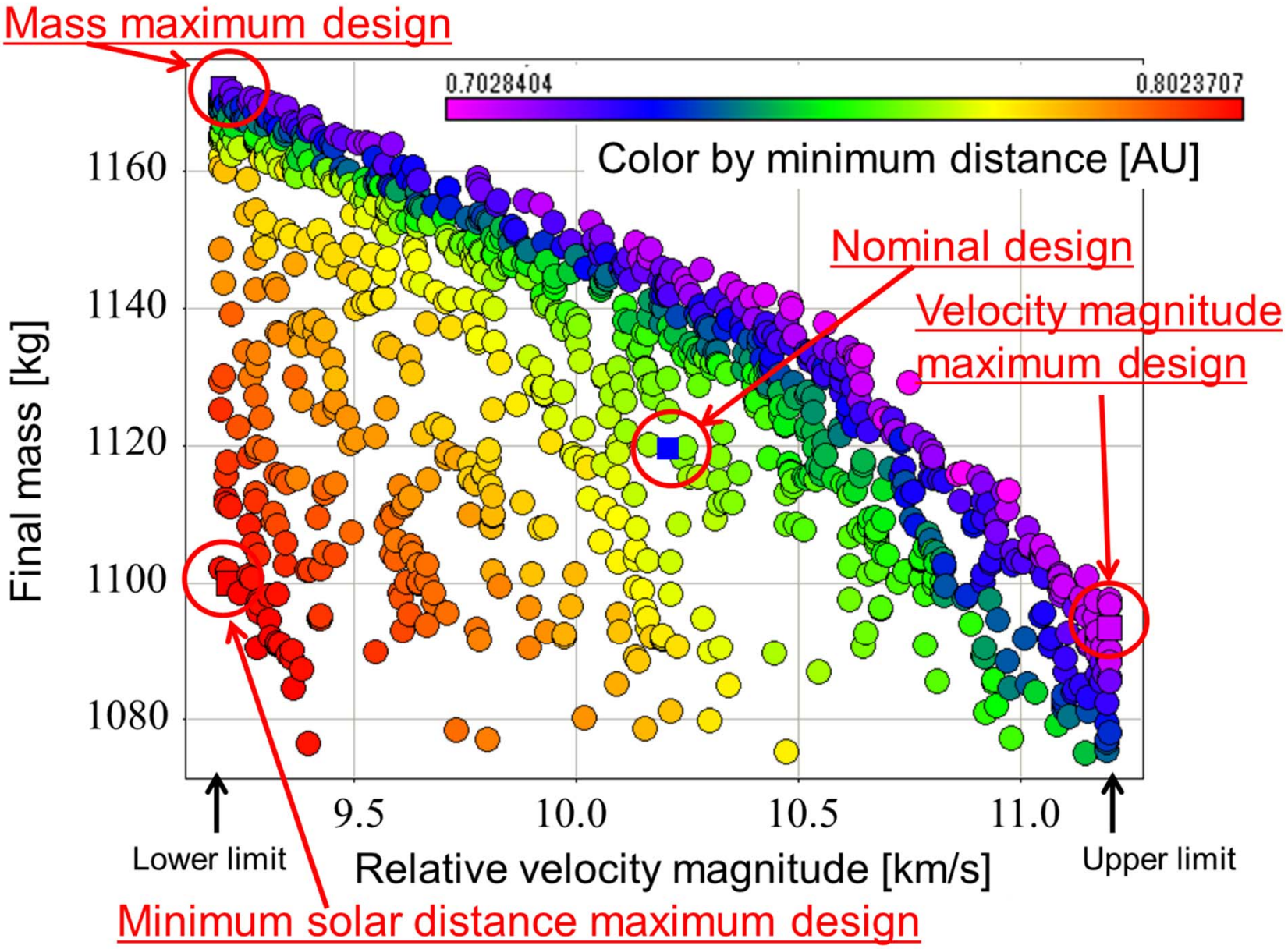
- イオンエンジン推力制御プロファイル最適化モジュールの実行回数は5760回（初期集団サイズ640，集団サイズ128，世代数40）
  Number of design evaluations is 5,760
  (initial population size =640, population size =128, number of generations =40)
  - 得られた軌道の数：5,370
    Number of obtained designs: 5,370
  - 制約条件を満たした軌道の数：3,215
    Number of feasible designs: 3,215
  - パレート最適解の数：639
    Number of Pareto-optimal designs: 639
- 計算時間はCore i7(3.0GHz)マシンで約2週間
  Computational time was two weeks on Core i7 (3GHz) machine

結果　results

- Obtained designs: 5370
- Feasible designs: 3215
- Pareto-optimal designs: 639

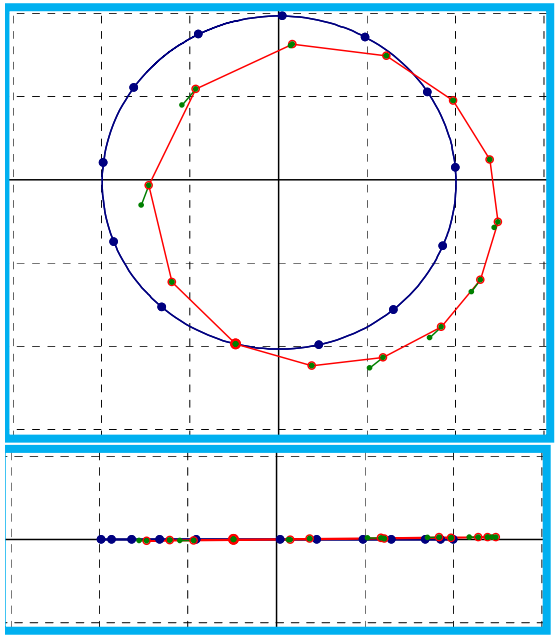


Tradeoff among three design objectives



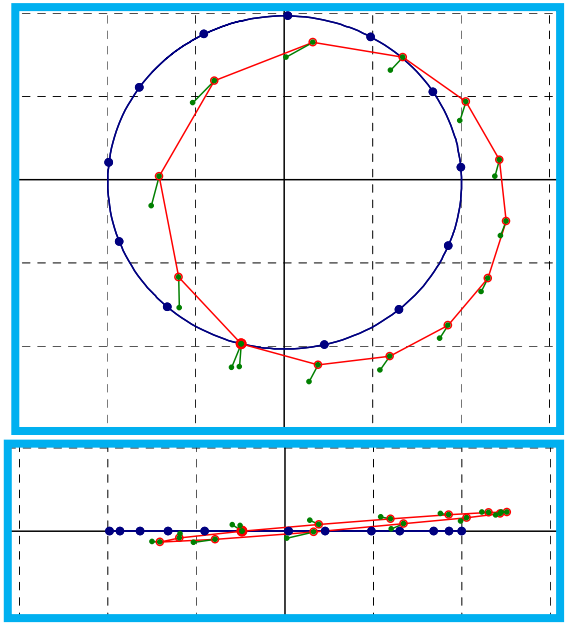
Tradeoff among three design objectives

Mass max.



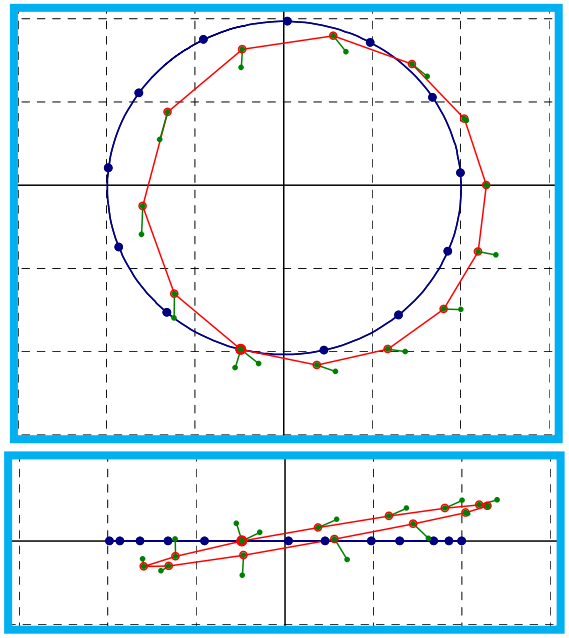
- **0.72AU**×1.29AU
- ***Final mass 1172kg***
- ***V 9.2km/s (lower limit)***
- *ΔV* 0.83km/s
- Az -95.1
- **El 2.4 (lower limit)**
- ***Inclination 7.9***
- **Isp 3598s (upper limit)**
- ***Absolute thrust 100mN***
- ***Thrusting Time 3265h***

Velocity magnitude max



- **0.71AU**×1.30AU
- *Final mass 1094kg*
- ***V 11.2 km/s (upper limit)***
- *ΔV* 2.38 km/s
- **Az -94.1**
- **El 18.8**
- ***Inclination 12.2-12.7***
- Isp 2608 s
- *Absolute thrust 138mN*
- ***Thrusting Time 7500h***

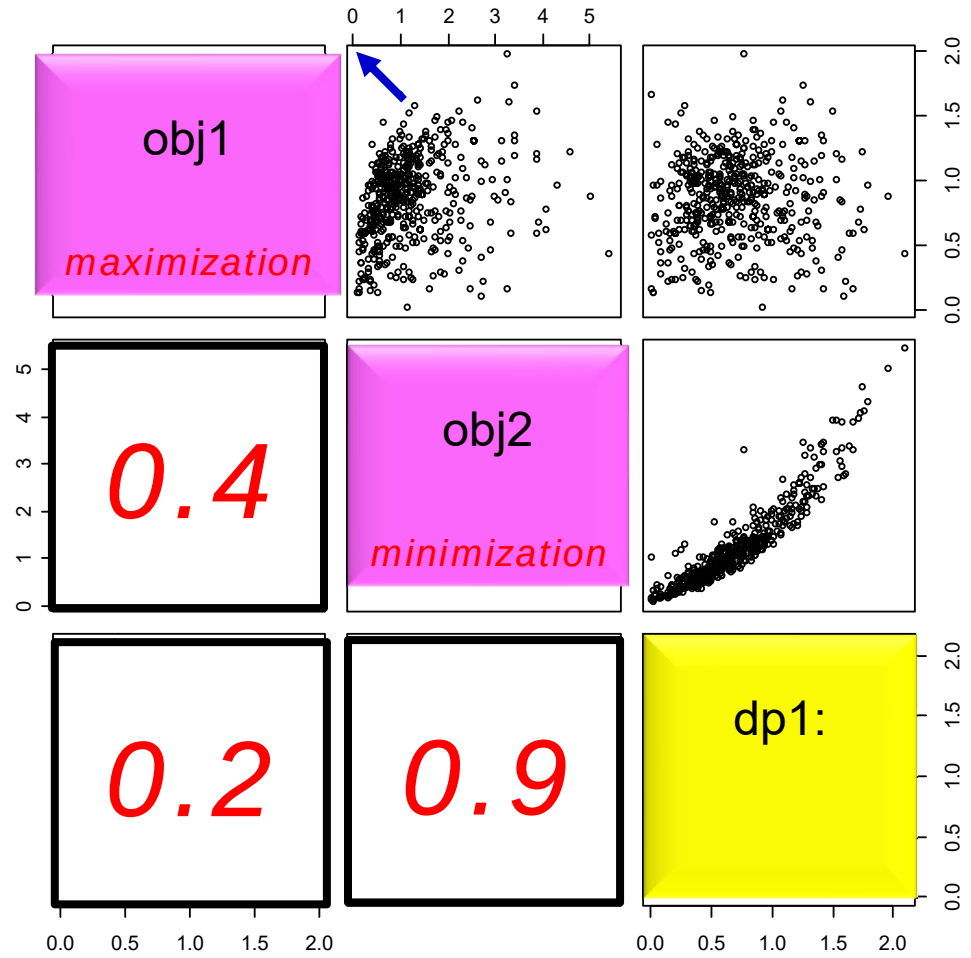
Minimum distance max



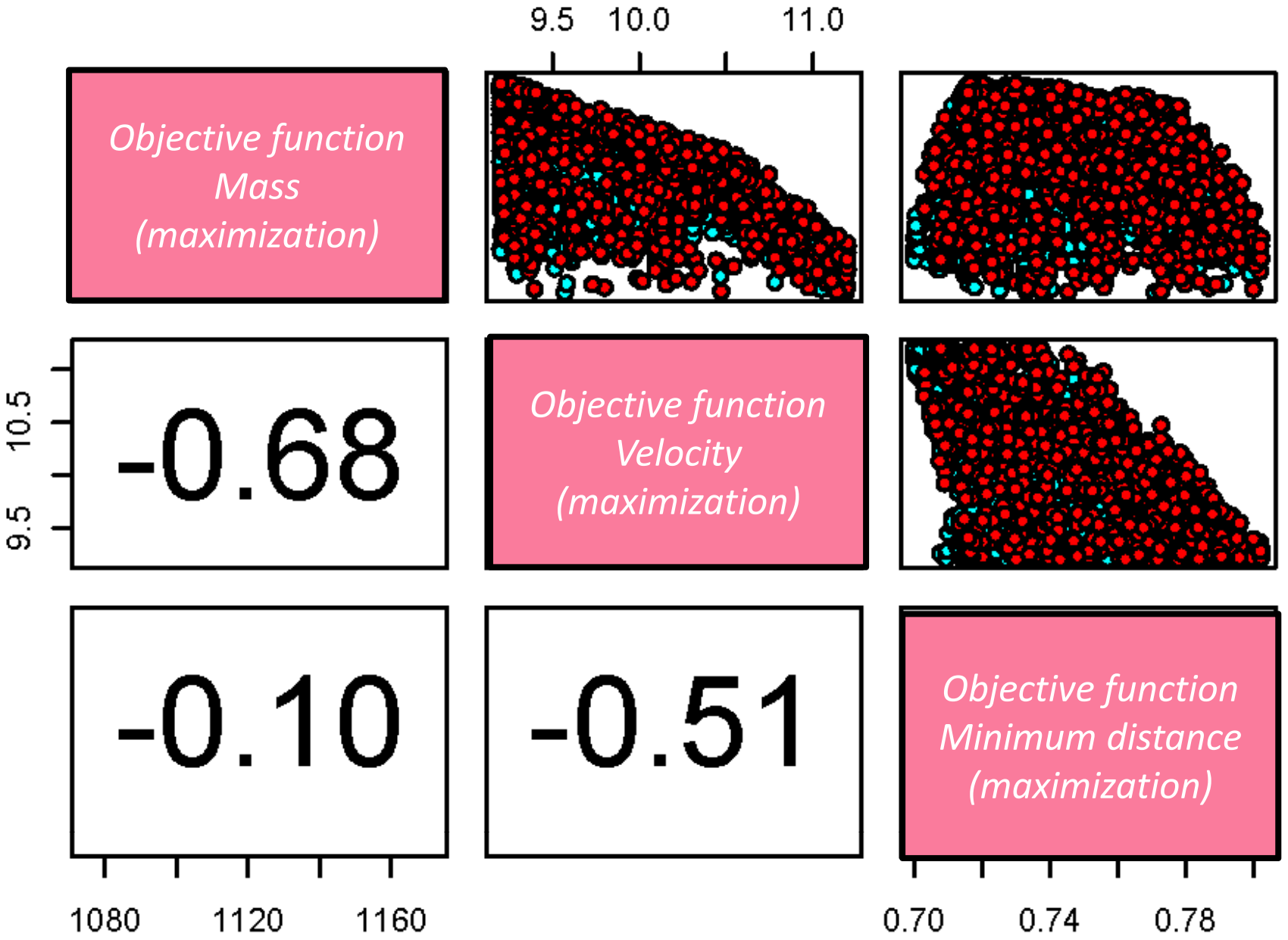
- **0.80AU**×1.19AU
- *Final mass 1100kg*
- ***V 9.2 km/s (lower limit)***
- ΔV 2.28 km/s
- **Az -103.9**
- **El 41.8 (upper limit)**
- ***Inclination 17.7-19.2***
- Isp 2678.3 s
- *Absolute thrust 134mN*
- *Thrusting Time 6565h*

Visualization using SPM

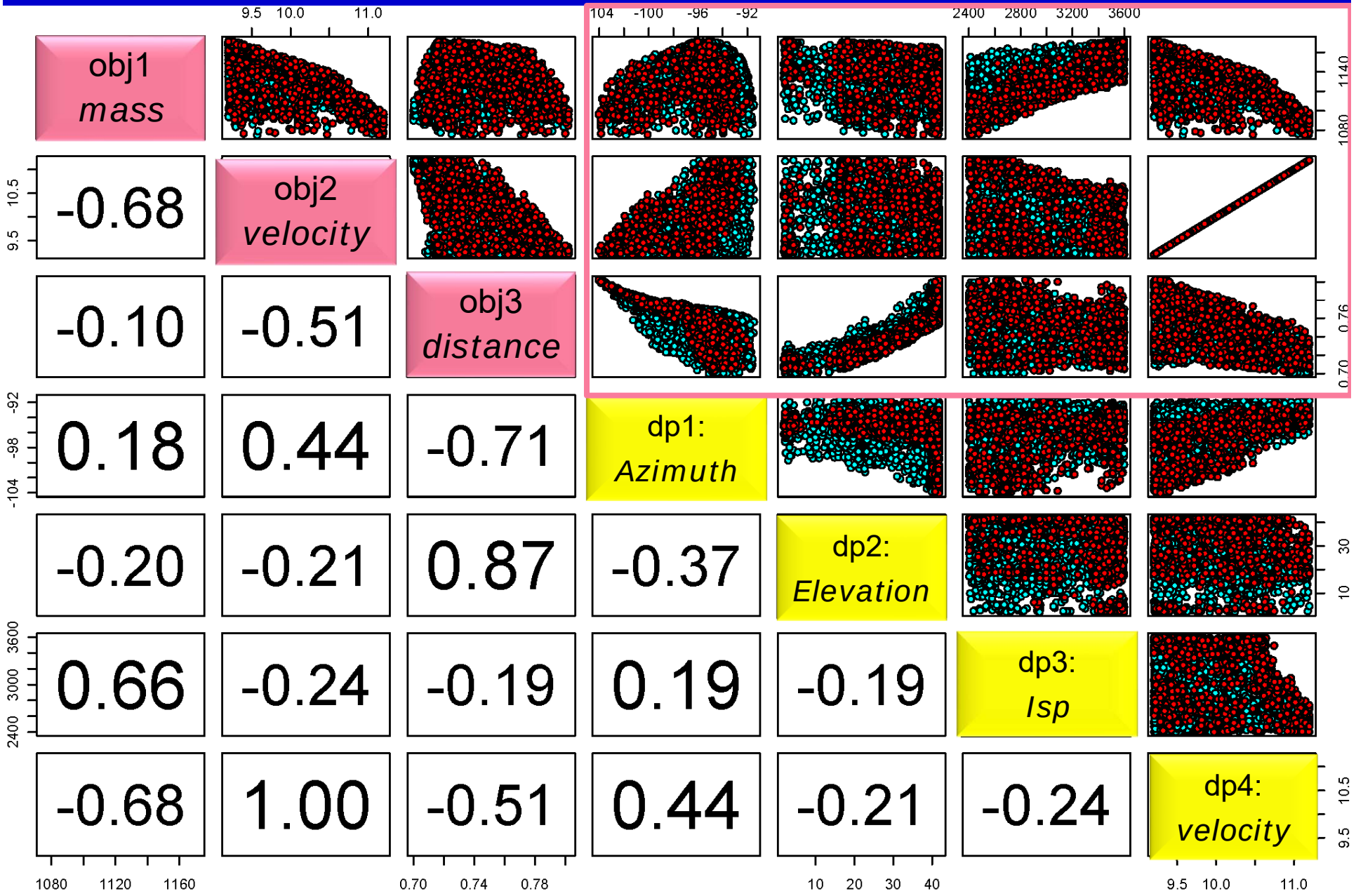
- Scatter plot matrix
  - Upper right triangular matrix: scatter plot matrix
  - Lower left triangular matrix: correlation coefficients



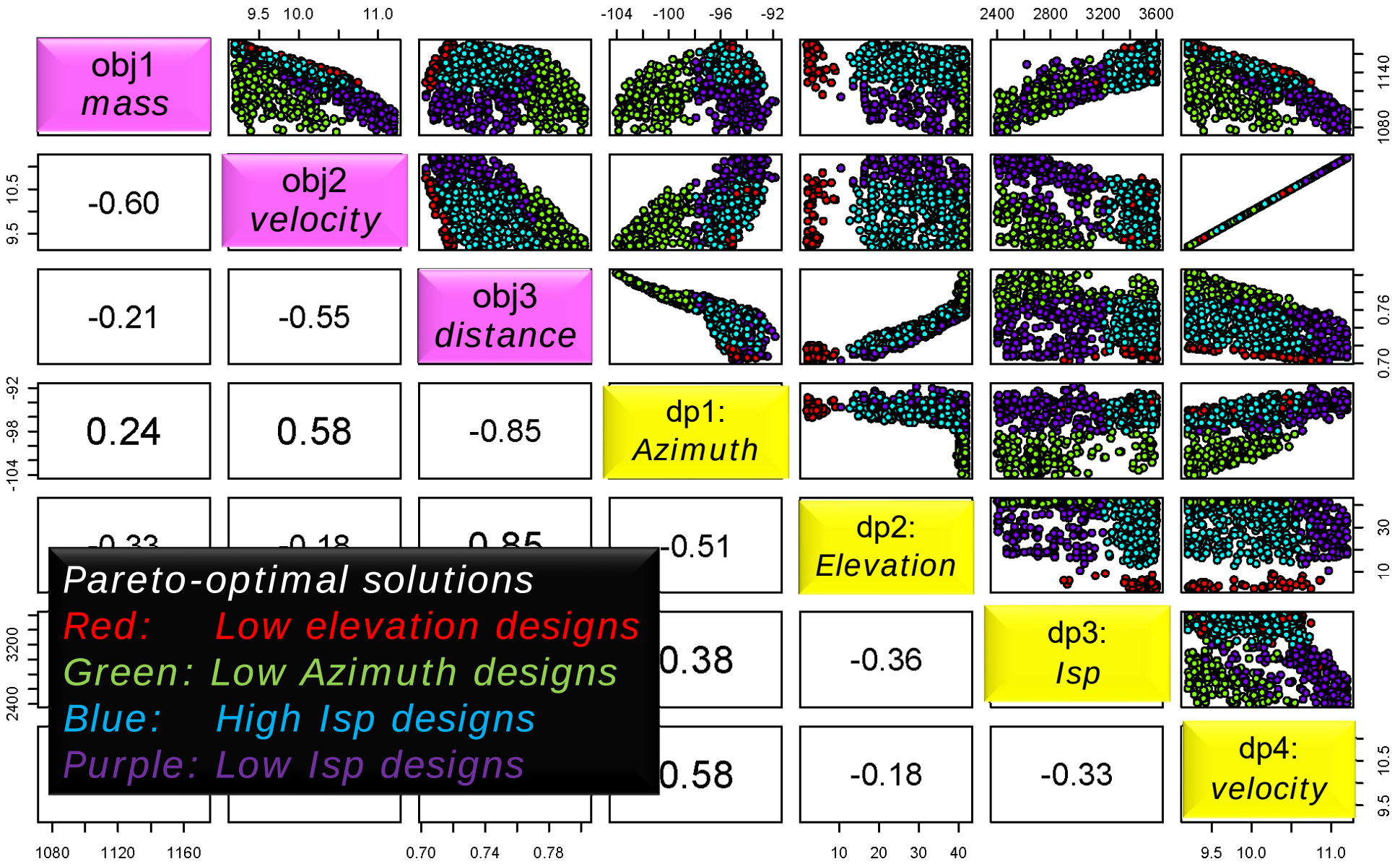
Visualization using scatter plot matrix



Visualization using scatter plot matrix



Data mining with scatter plot matrix



1周回目の軌道最適化 MODE of the first cycle

- 1周期目の軌道の多目的設計探査をこない，軌道設計への多目的設計探査の有効性をしめした

  Feasibility of MODE was shown

  - 3つの目的関数の間には強いトレードオフがあることが確認された

    Tradeoff among three objective was visualized

  - 従来の設計法で得られた設計よりも優れたパレート最適解を発見した

    Pareto-optimal designs that outperform the nominal design were found

  - 新しい設計知見を得ることが出来た.

    New knowledge was obtained.

第5周回までの軌道の多目的設計探査

MODE of five cycles

設計問題 Design problem

従来手法で設計されたノミナル軌道をベースに設計問題を定式化する

Design problem bases on nominal design

- Launch condition
  - Launcher: H2A with a solid motor upper stage
  - Initial mass: 1200kg
  - Launch date: 2017.6.7
  - The first earth swing-by date: 2018.6.7
  - Initial relative velocity magnitude: 7.7 km/s
  - Azimuth angle at the first swing-by: -94.285 deg.
  - Elevation angle at the first swing-by: 19.793 deg.
- Spacecraft
  - Operation rate is 0.875

設計目的　design objectives

- 終端質量の最大化 （搭載可能ペイロード最大化に相当）

  Maximization of mass of the spacecraft at the end (this corresponds to maximization of science payload mass)

- 終端における対地球相対速度の最大化 （軌道傾斜角最大化に相当）

  Maximization of the relative speed to the Earth at the end (this corresponds to maximization of trajectory inclination angle)

- 太陽との最小距離の最大化 （熱防御システム重量最小化に相当）

  Maximization of minimum distance to the sun (this corresponds to minimization of thermal protection system weight)

制約条件　constraint conditions

- 地球と衝突しないこと

  Do not collide with the earth

- 離心率が0．3を超えないこと

  Eccentricity should not exceed 0.3

- 最大推力 x 比推力 が一定値（ノミナル値と同値）であること（必要電力一定に相当）

  Maximum thrust x specific impulse = constant

  (this corresponds to required power is constant)

設計変数　design parameters

- イオンエンジンの推力制御プロファイル　x　5周回

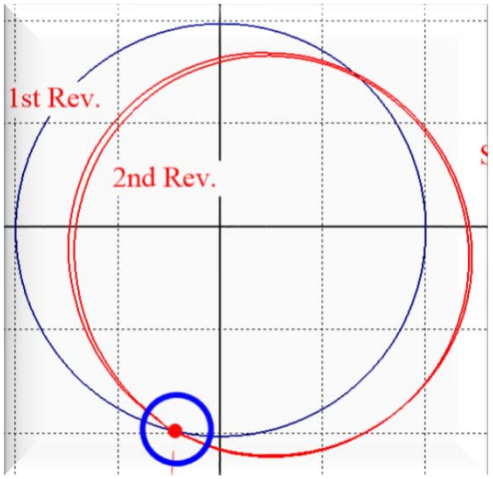
  Ion engine thrust control profiles x 5 cycles

- 各周回始点の方位角

  Azimuth angle at the start of each cycle

- 各周回始点のElevation角（仰角）

  Elevation angle at the start of each cycle



- 各周回終点の対地球相対速度

  Relative speed to the earth at the end of each cycle

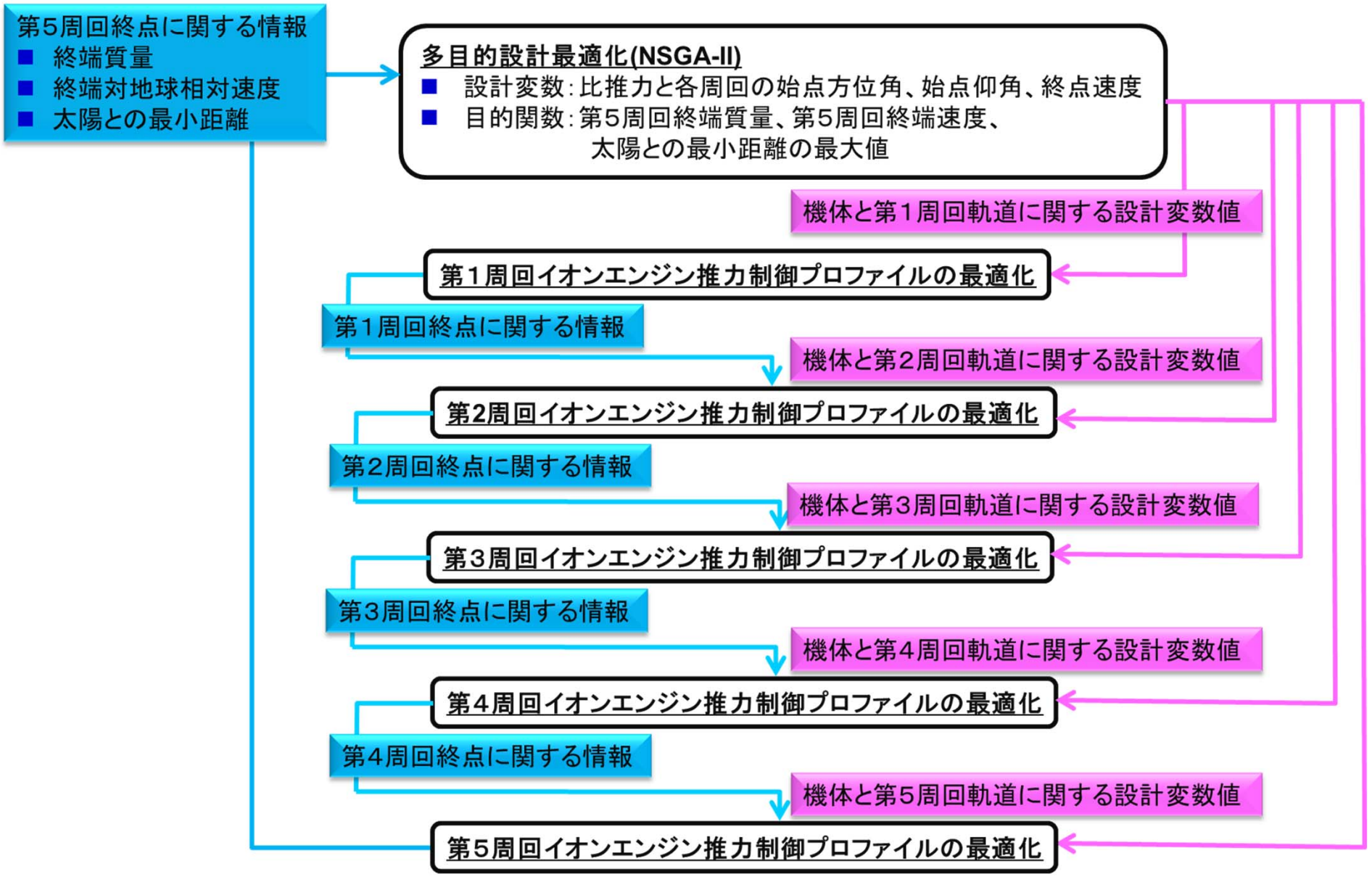
- イオンエンジン比推力　(機体の設計パラメータ)

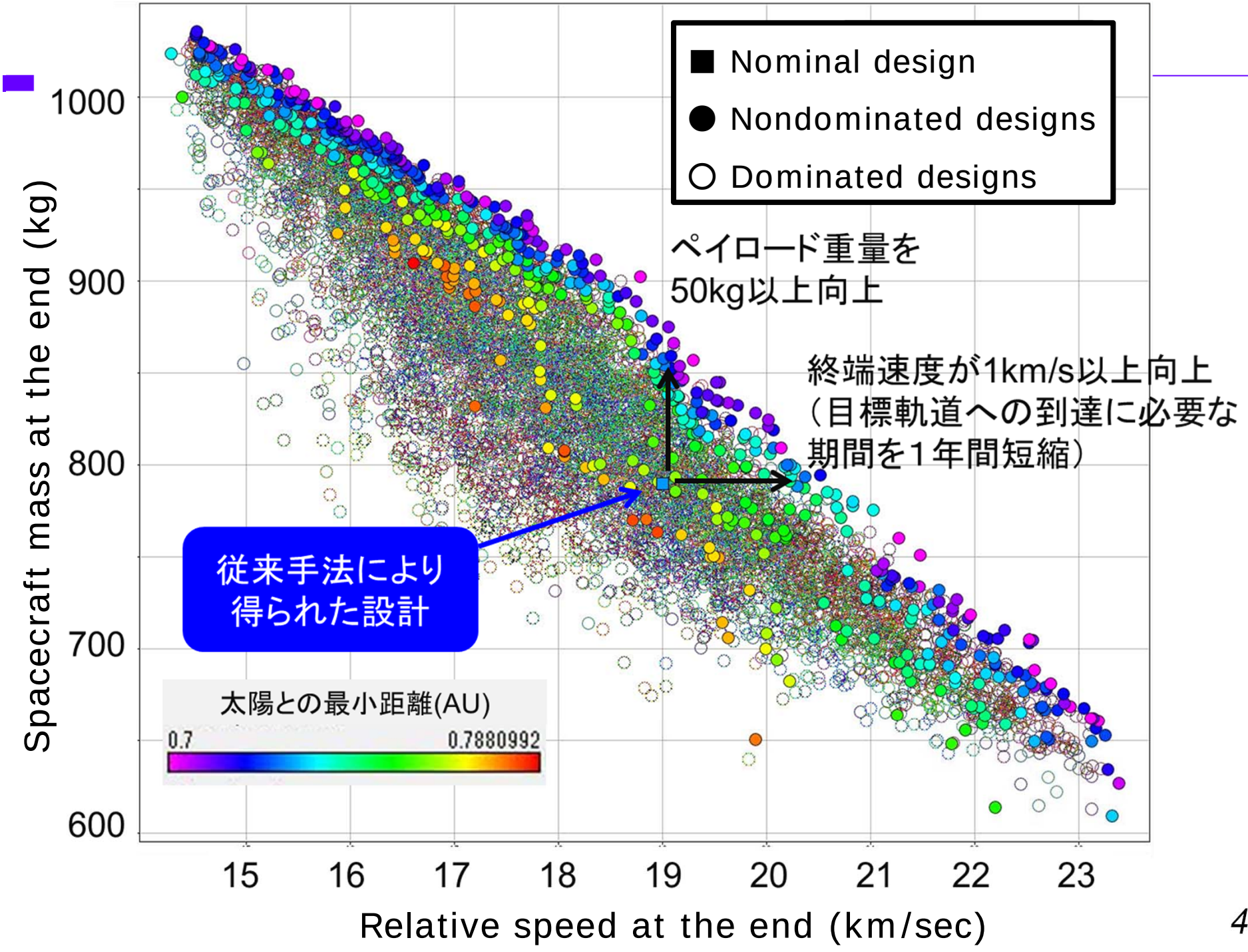
  Ion engine specific impulse (design parameter of spacecraft)

設計変数の数　number of design parameters

42x5＋3x5＋1＝226

具体的な設計手法


第5周回終点に関する情報
■ 終端質量
■ 終端対地球相対速度
■ 太陽との最小距離
多目的設計最適化(NSGA-II)
■ 設計変数：比推力と各周回の始点方位角、始点仰角、終点速度
■ 目的関数：第5周回終端質量、第5周回終端速度、
太陽との最小距離の最大値
機体と第1周回軌道に関する設計変数値
第1周回イオンエンジン推力制御プロファイルの最適化
第1周回終点に関する情報
機体と第2周回軌道に関する設計変数値
第2周回イオンエンジン推力制御プロファイルの最適化
第2周回終点に関する情報
機体と第3周回軌道に関する設計変数値
第3周回イオンエンジン推力制御プロファイルの最適化
第3周回終点に関する情報
機体と第4周回軌道に関する設計変数値
第4周回イオンエンジン推力制御プロファイルの最適化
第4周回終点に関する情報
機体と第5周回軌道に関する設計変数値
第5周回イオンエンジン推力制御プロファイルの最適化

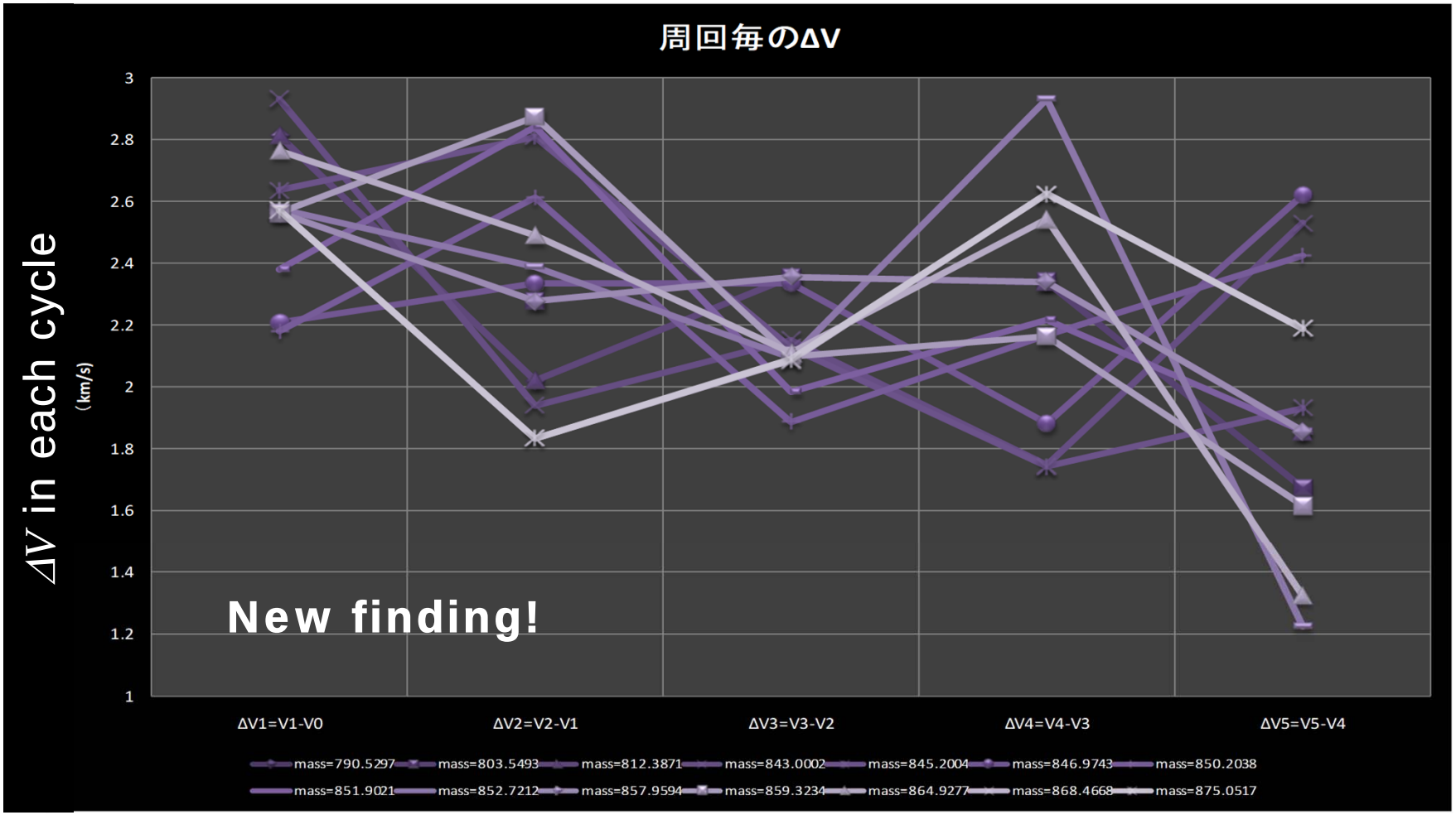

■ Nominal design
● Nondominated designs
○ Dominated designs
ペイロード重量を
50kg以上向上
終端速度が1km/s以上向上
（目標軌道への到達に必要な
期間を1年間短縮）
従来手法により
得られた設計
太陽との最小距離(AU)
0.7
0.7880992
Spacecraft mass at the end (kg)
1000
900
800
700
600
15
16
17
18
19
20
21
22
23
Relative speed at the end (km/sec)

4周回で19km/sに到達する軌道の分析

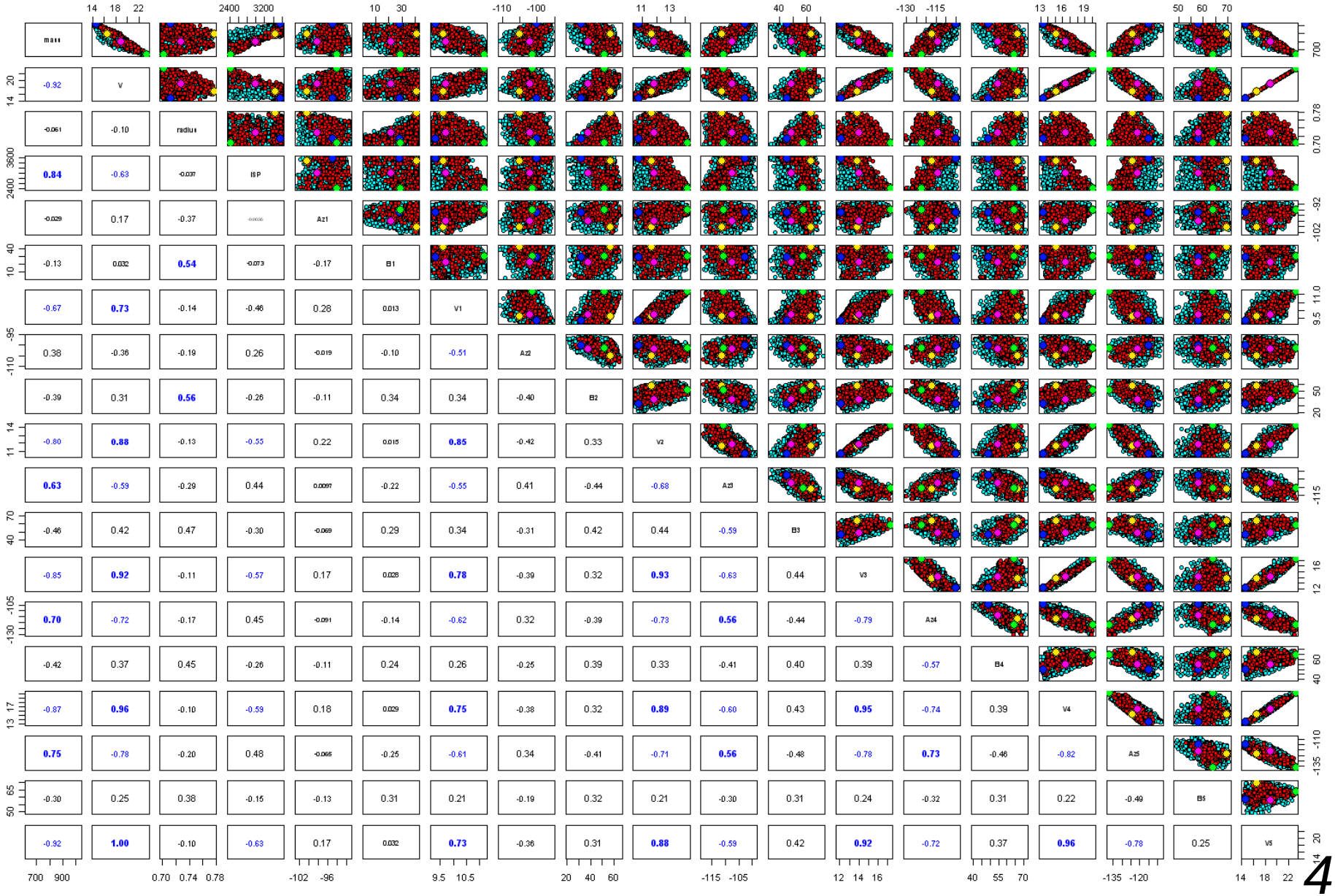
Analysis of trajectories that reach 19km/s by 4th cycle

増速量を単調に減らしていくことが必ずしも最適な解とは限らない

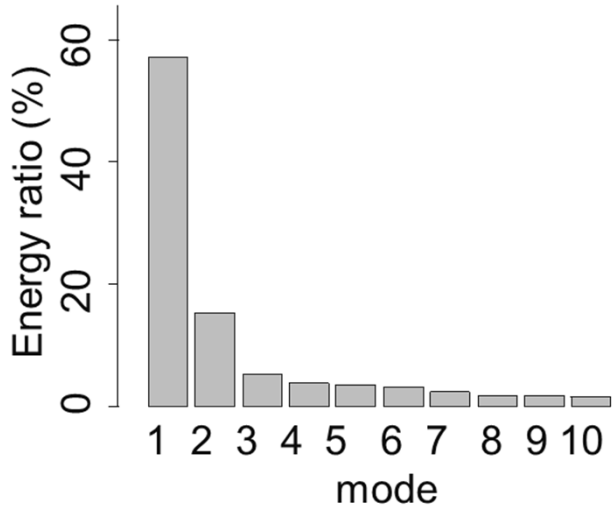
It is not always the best to decrease the incremental speed monotonically



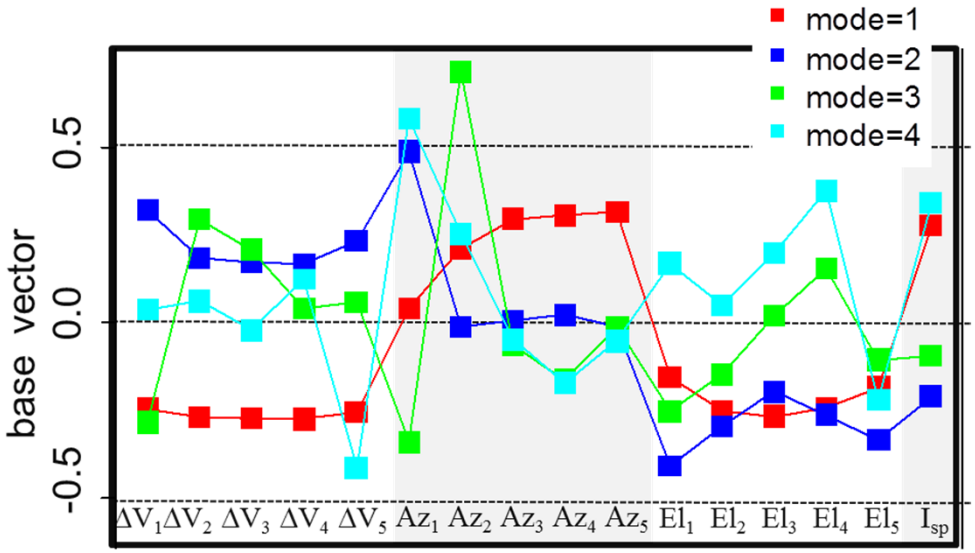
Visualization using scatter plot matrix



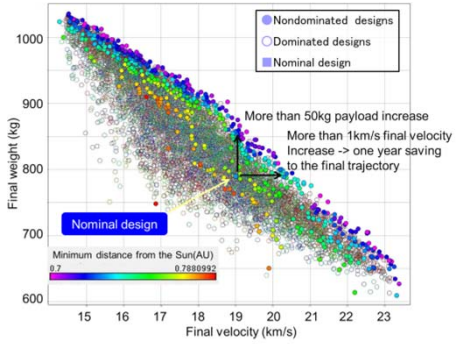
Mode analysis of design parameters using POD

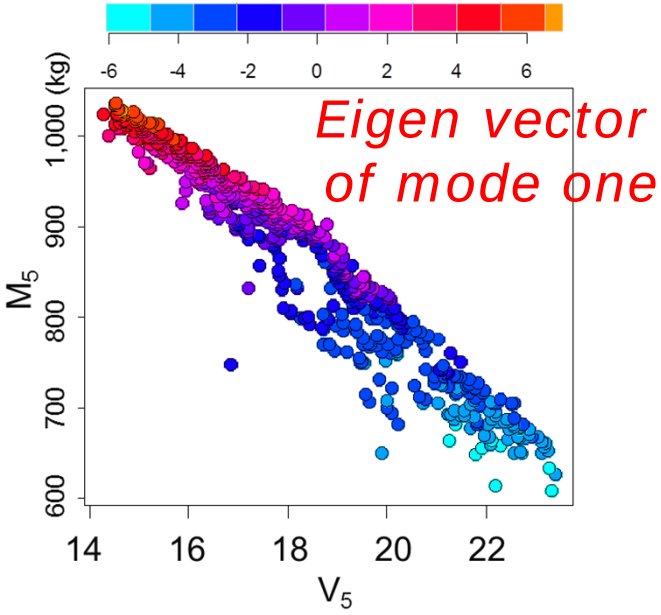


Energy ratio of top ten modes

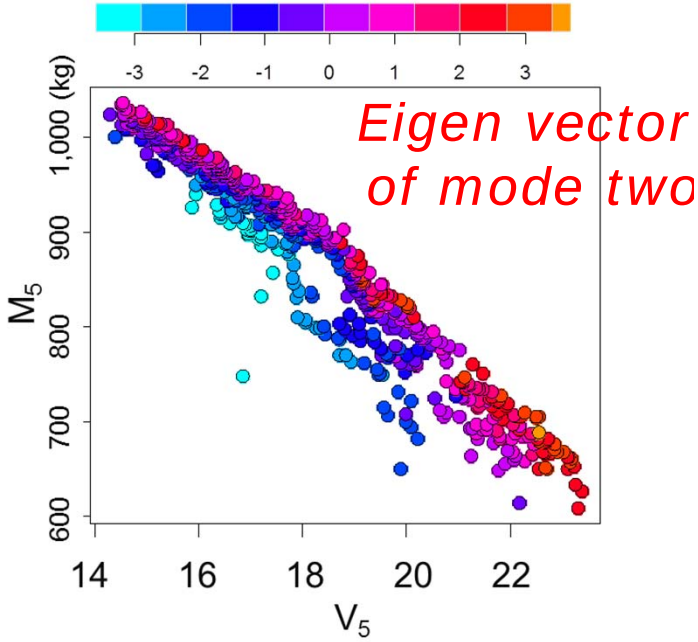


Base vector of mode one to four





Eigen vector of mode one



Eigen vector of mode two

まとめ Summary

- 従来手法により得られた設計よりも優れた設計候補を多数発見

  Many designs that outperform conventional design were found

  - 搭載可能ペイロード重量を50kg以上増加させる軌道（最小動径半径や終端速度もノミナル設計以上）

    A design that enables 50kg higher payload weight

  - 目標軌道への到達に必要な期間を１年間短縮する軌道（５年間→４年間，最小動径半径やペイロード質量もほぼノミナル設計と同じ）

    A design that reaches to the target trajectory in four years

- 軌道設計にかかる期間が大幅に短縮された（２年程度→3ヶ月程度）

  Design period is reduced significantly (from two years to three months)

まとめ Summary

- 軌道設計問題に関するさまざまな新しい知見の獲得

  New findings on trajectory design

  - 目的関数間のトレードオフ情報

    Tradeoff information among the objectives

  - 各設計変数の各目的関数への感度や各設計変数の設計可能範囲等の情報の提供

    Sensitivity of each design parameters and feasible design space were clarified

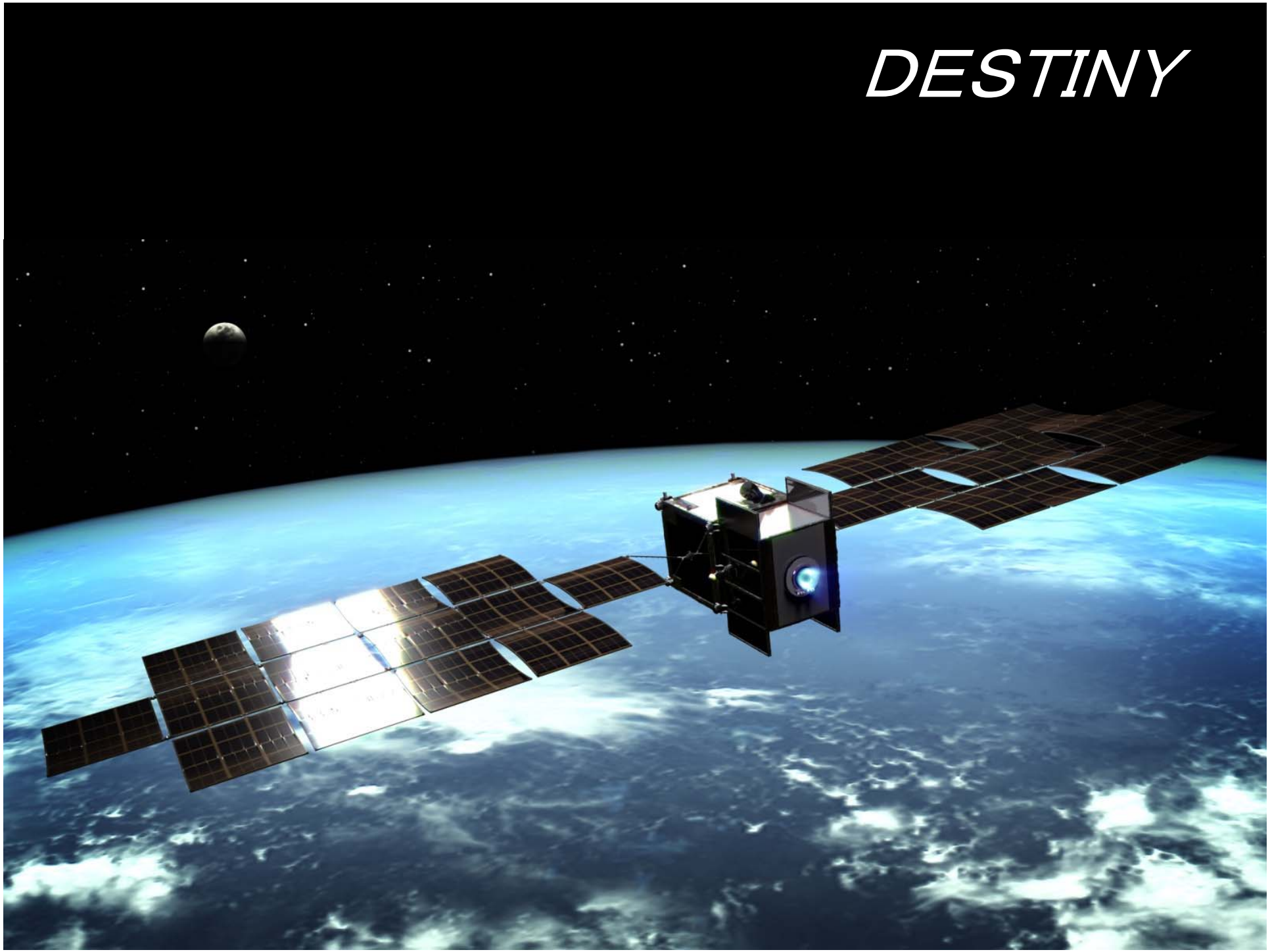
  - 最適解は周回ごとに必ずしもΔVを減らしていくわけではない

    It is not always the best to decrease the incremental speed monotonically

Trajectory designs

From sun polar region observation mission

to DESTINY mission

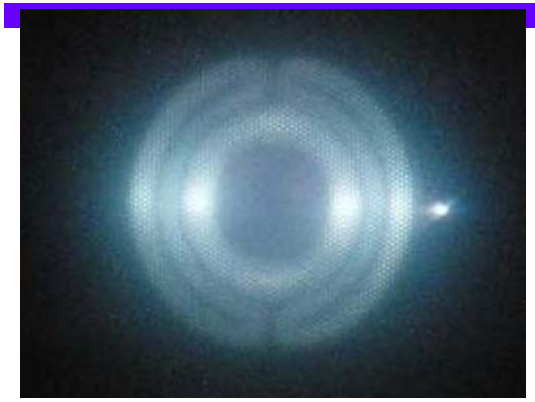

DESTINY



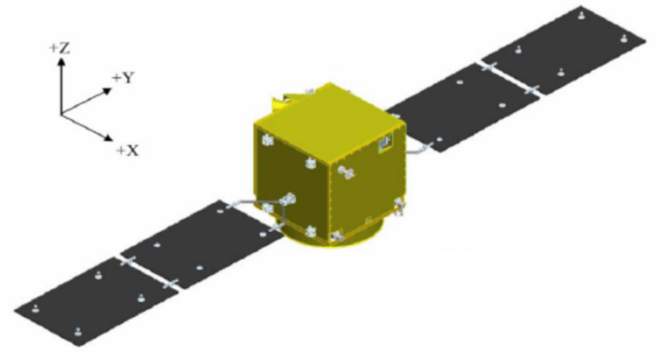
Concept of DESTINY mission:

深宇宙　敷居を下げる　DESTINY

Deep Space　　Lowers the difficulty to　　DESTINY



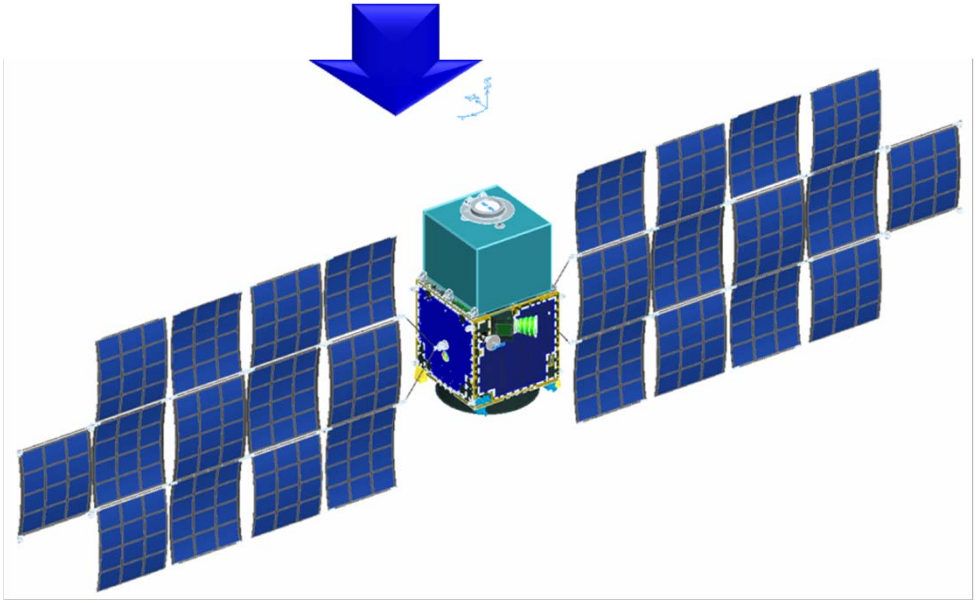
μ 20 Ion engine



standard bus system for a series of small scientific satellites

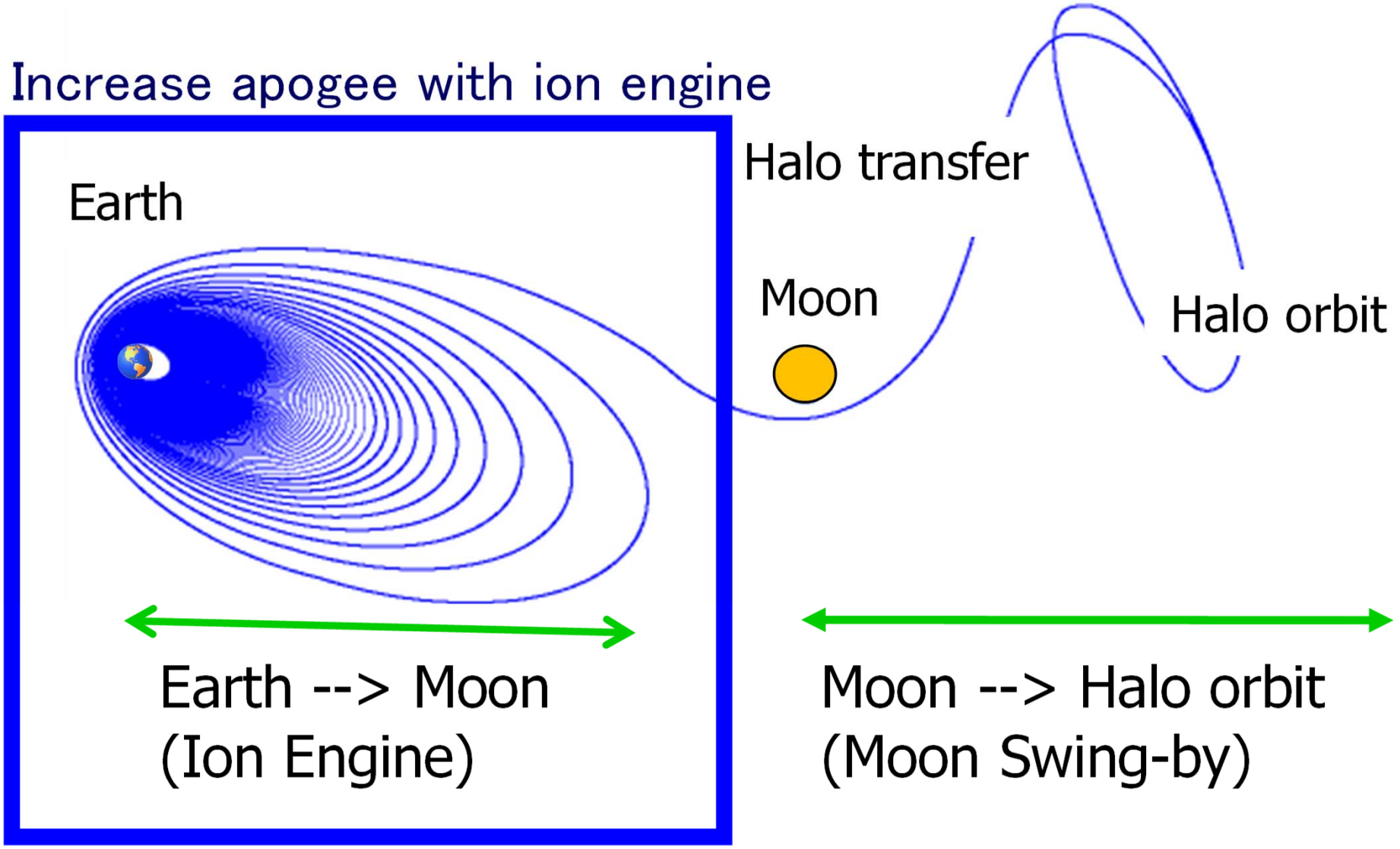


Epsilon rocket



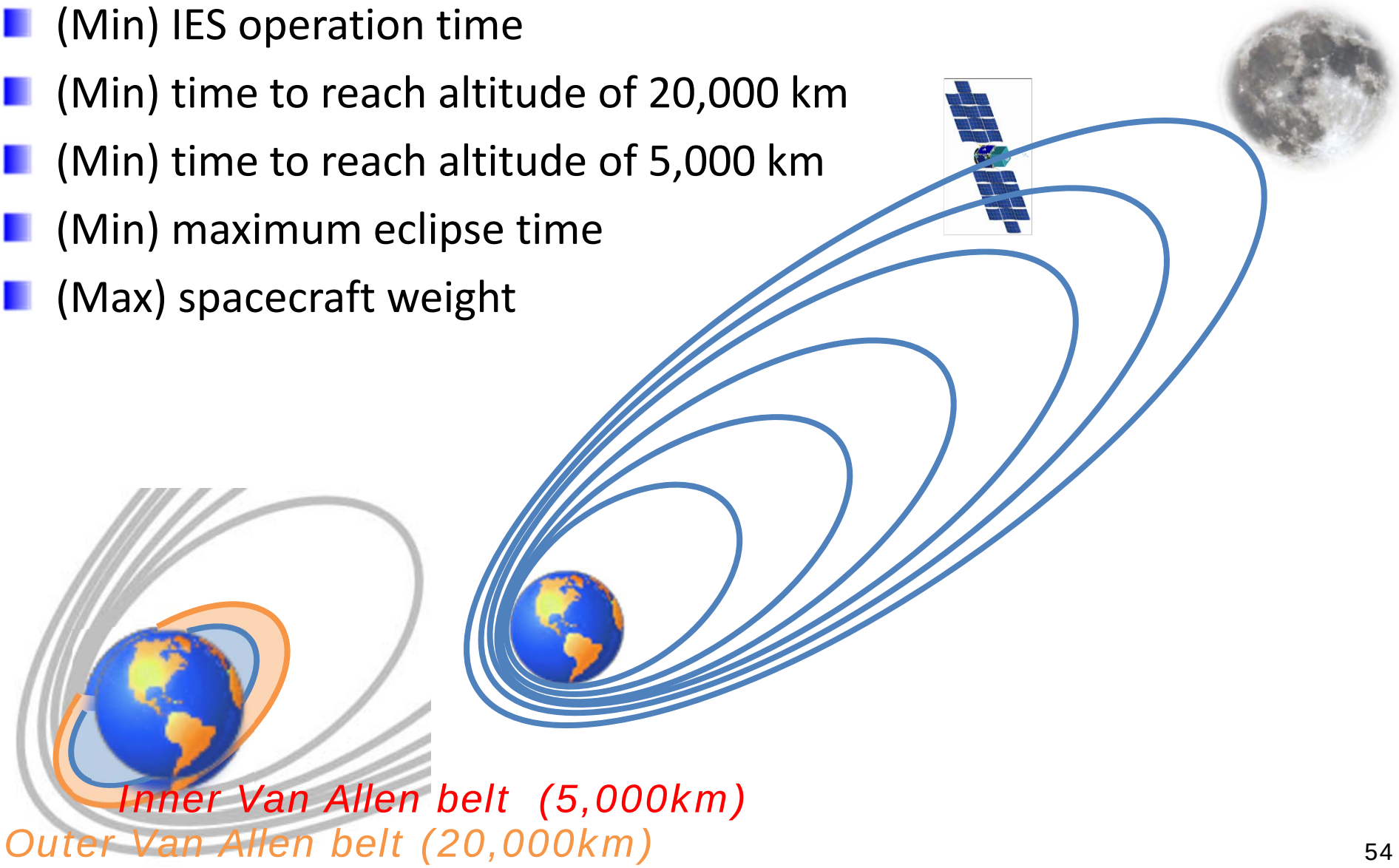
350kg of payload to L2 Halo orbit

Transfer to Sun-Earth L2 Halo



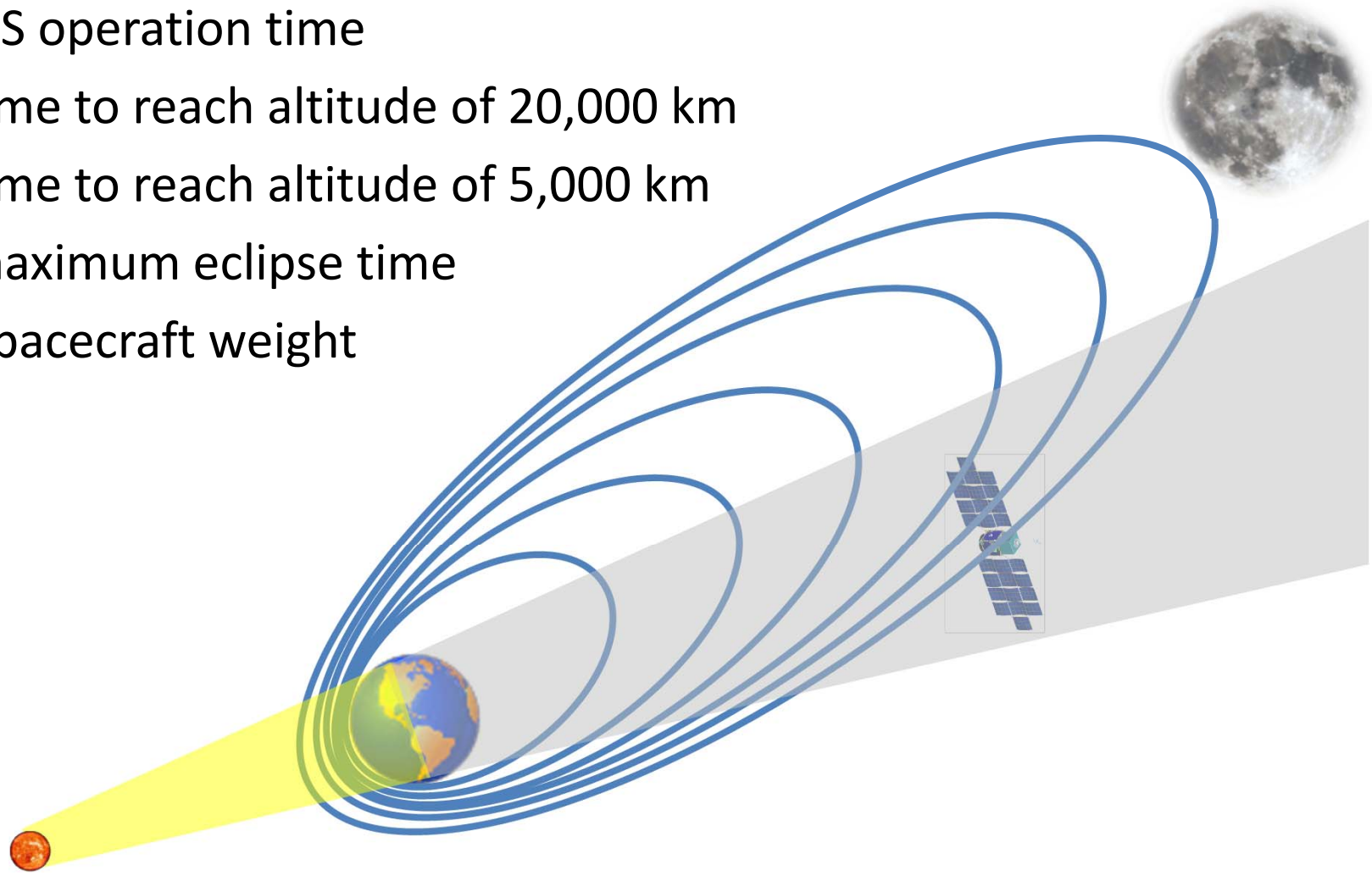
Design objectives

- (Min) time to reach Moon
- (Min) IES operation time
- (Min) time to reach altitude of 20,000 km
- (Min) time to reach altitude of 5,000 km
- (Min) maximum eclipse time
- (Max) spacecraft weight



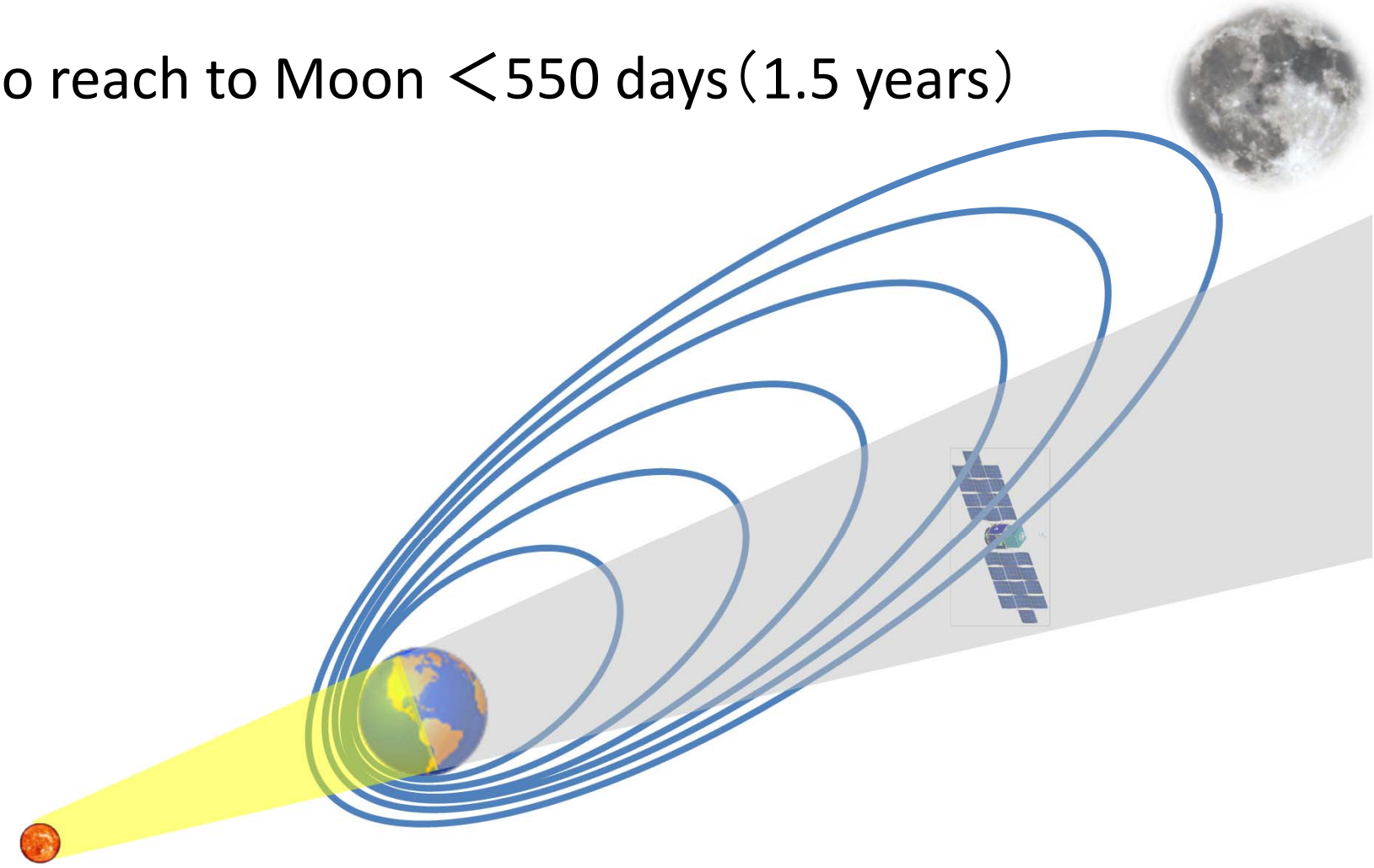
Design objectives

- (Min) time to reach Moon
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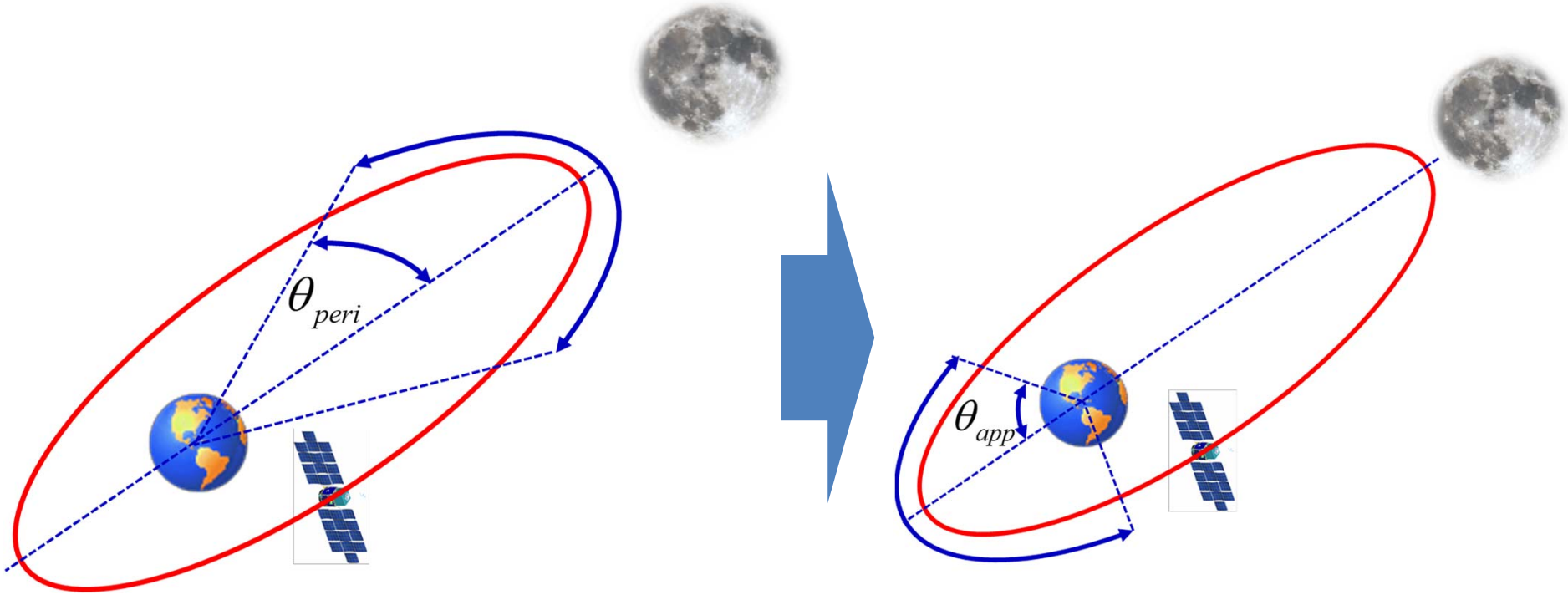


Constraint

- Time to reach to Moon ＜550 days（1.5 years）



Design parameters



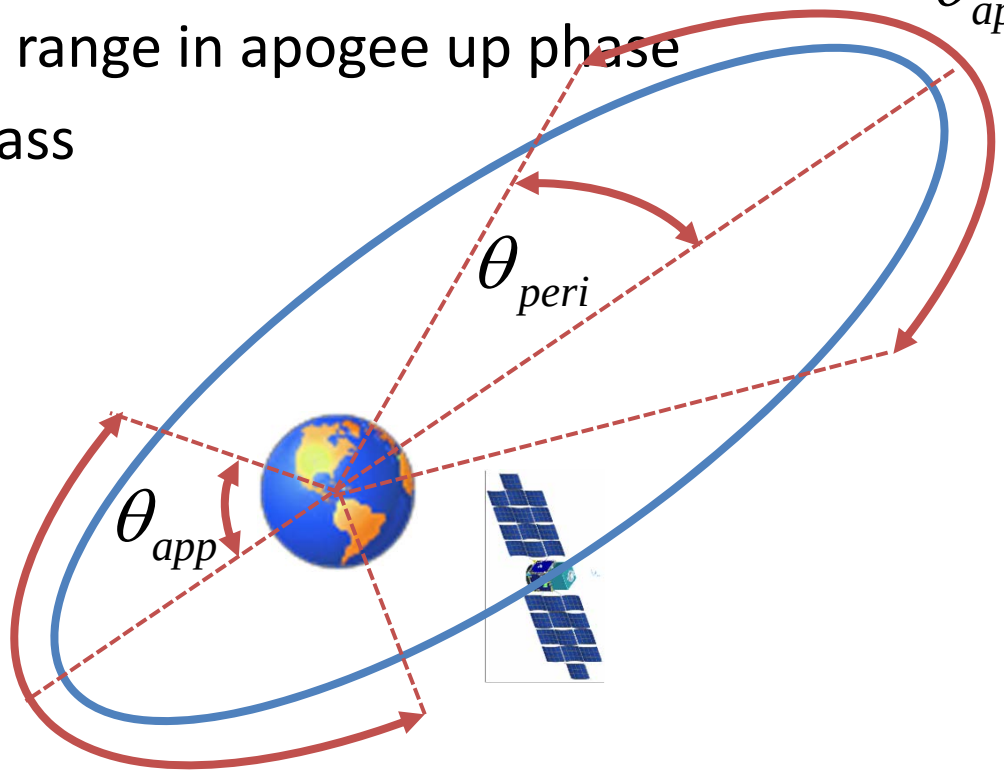
近地点上げフェーズ
Perigee raising phase

遠地点上げフェーズ
Apogee raising phase

Design parameters

- Start date
- Start time
- Day when switching the perigee raising phase to apogee raising phase
- IES operation range in perigee up phase
- IES operation range in apogee up phase
- Spacecraft mass

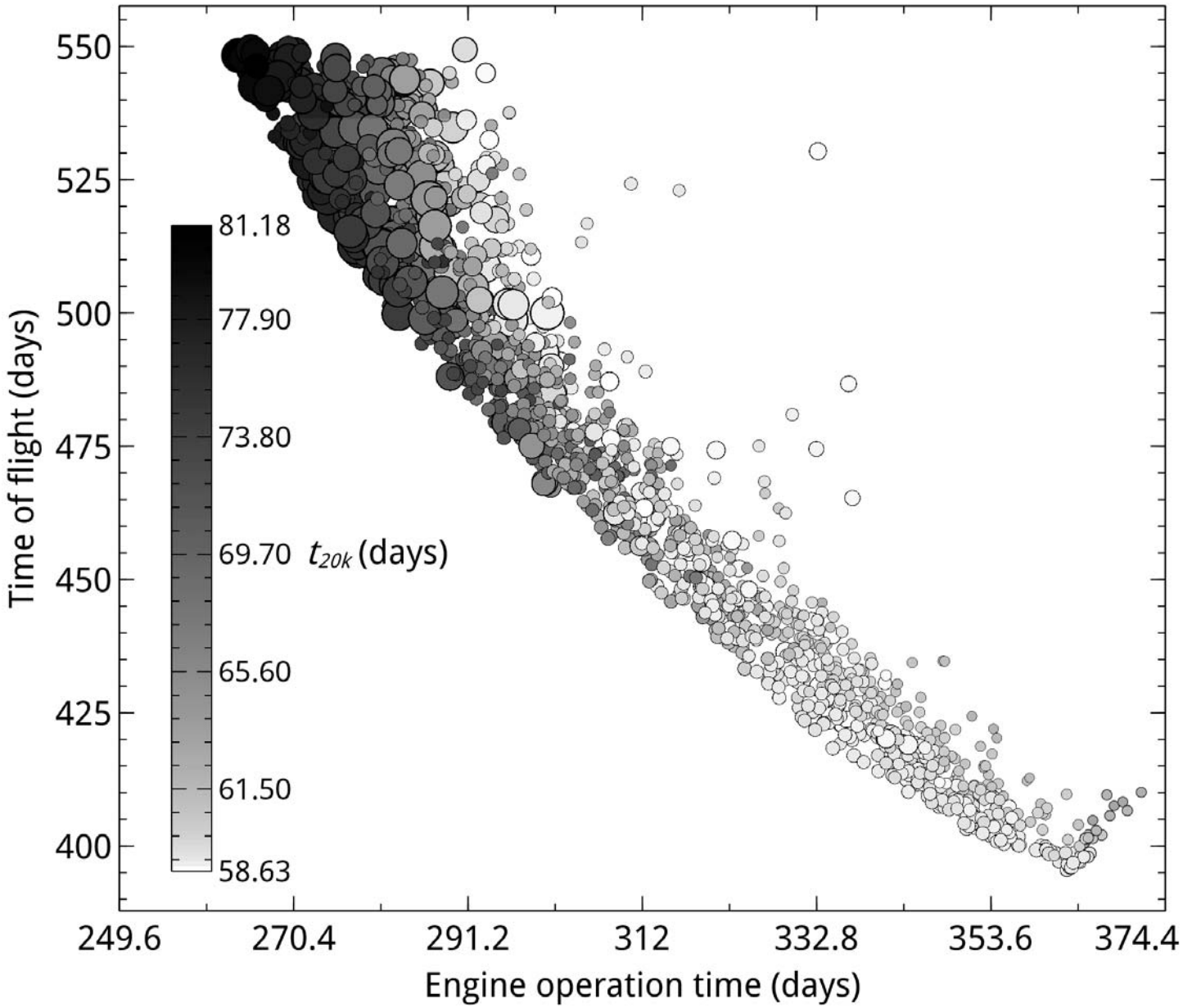




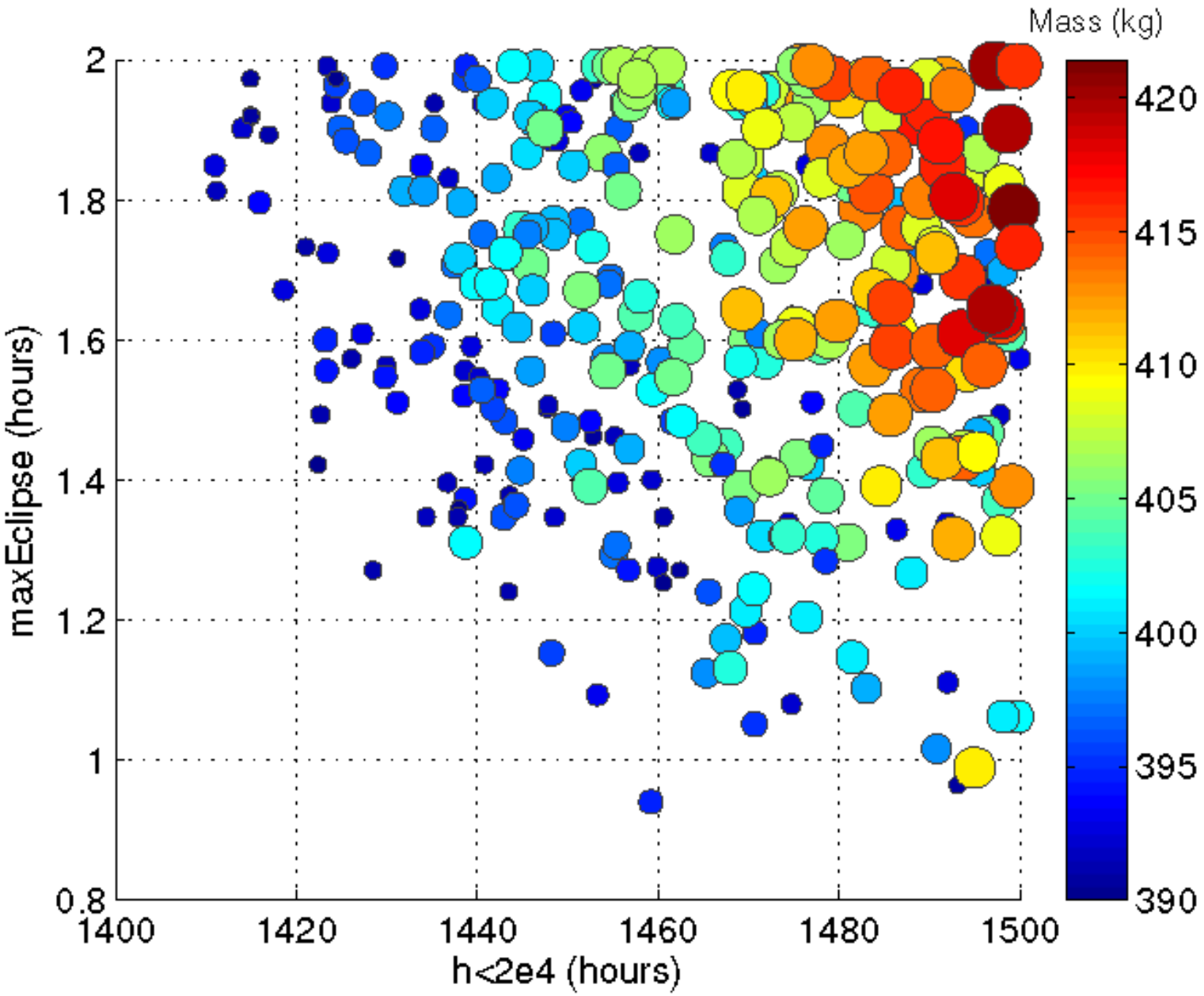
Design parameters

Parameter	Range	Unit
Start date	[2017/01/01 , 2018/01/01]	day
Start time	[00:00:00 , 24:00:00]	second
Switching apogee-perigee date	[90, 365]	day
Range apogee rise	[0 , 180]	degree
Range perigee rise	[0 , 180]	degree
Mass	[350 , 450]	kg

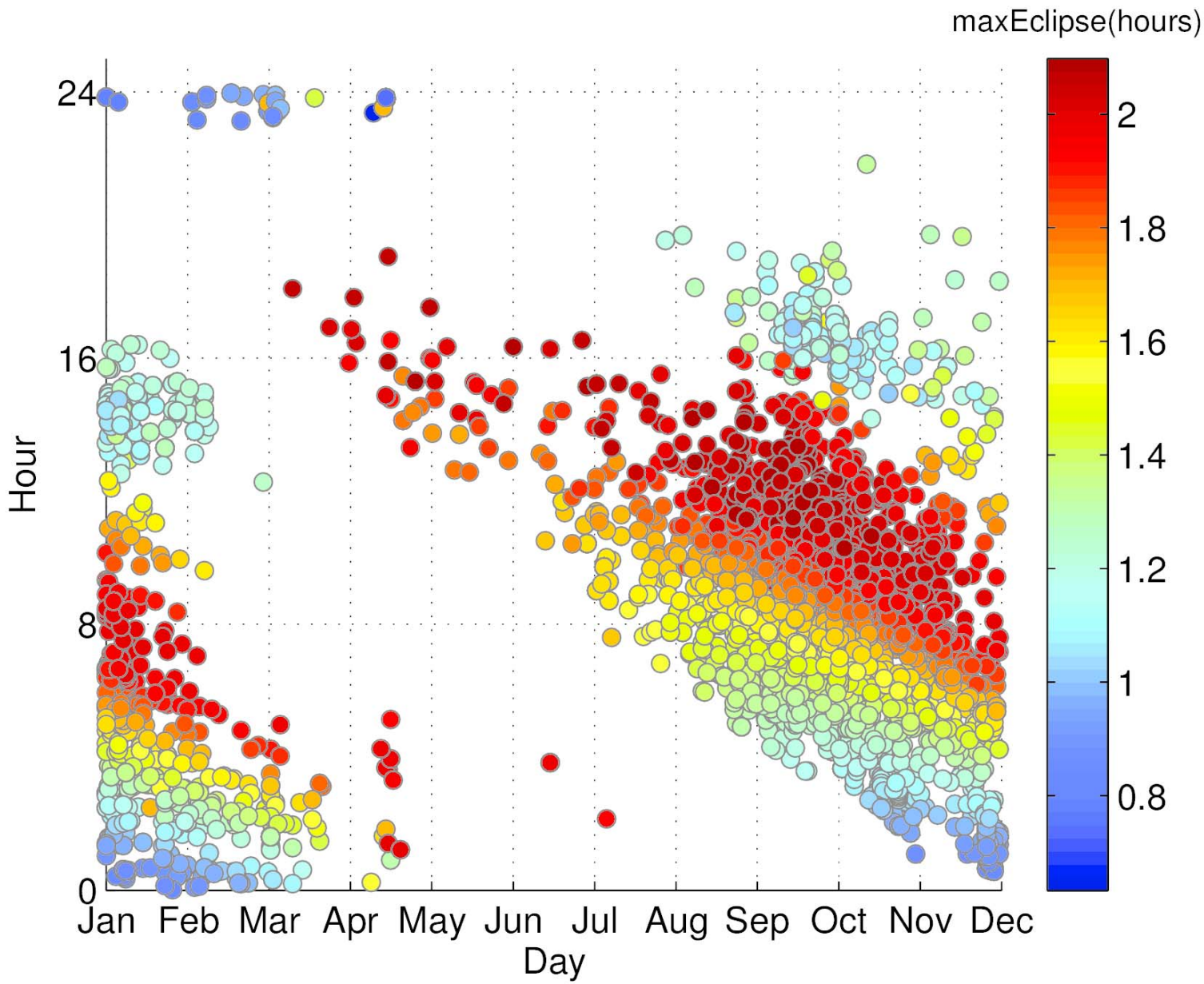
Tradeoff


550
525
500
475
450
425
400
Time of flight (days)
81.18
77.90
73.80
69.70
t_{20k} (days)
65.60
61.50
58.63
249.6
270.4
291.2
312
332.8
353.6
374.4
Engine operation time (days)

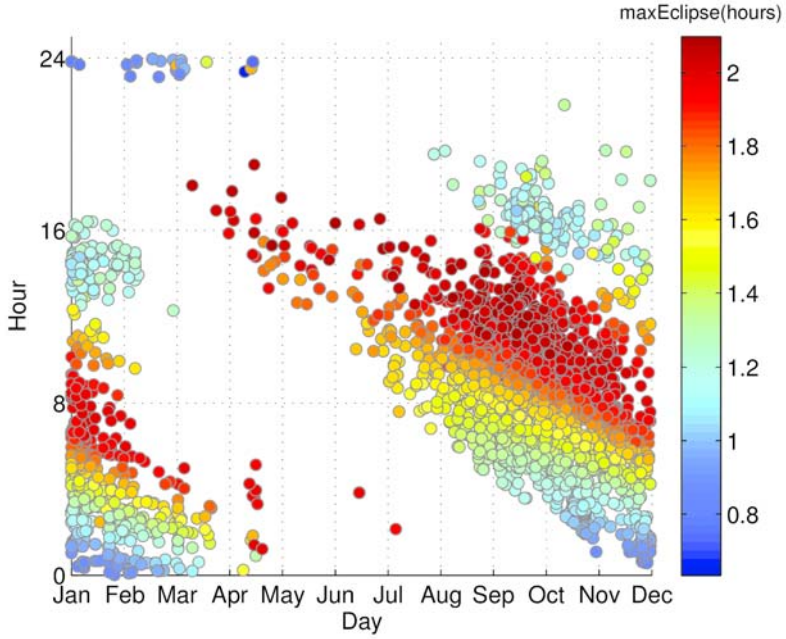
Designs for small maximum eclipse time



Launch date and time



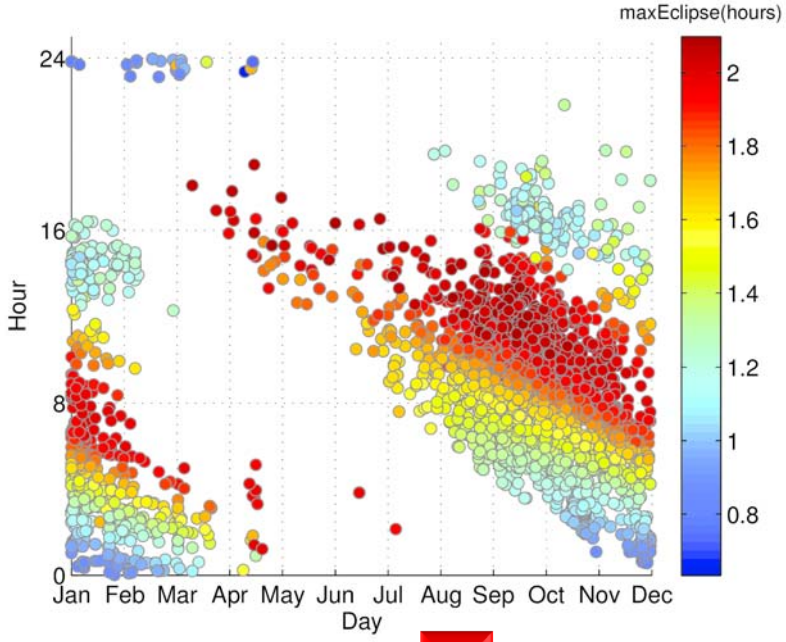
Launch date and time



始めは，１月１日0:00からの経過時間で打ち上げ日時を設計変数化していたが，よい解が得られなかった．なぜか？

When launch date and time are represented by one parameter, i.e. time past Jan. 1 0:00, good solutions were not obtained. Why?

Launch date and time

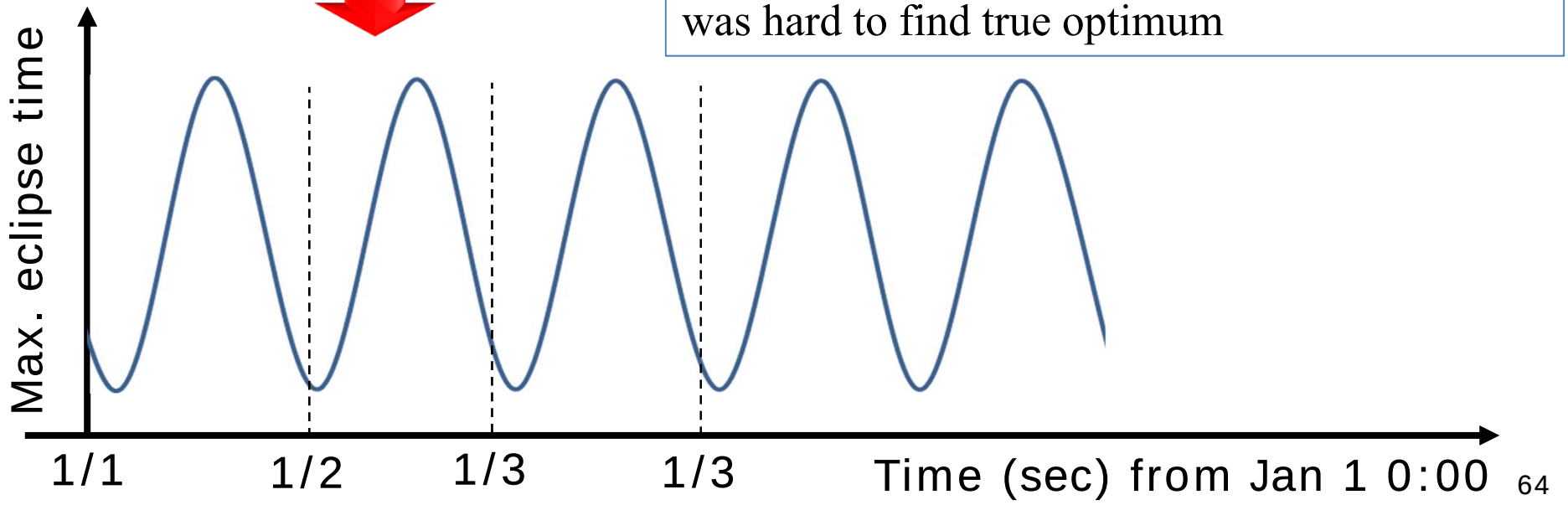


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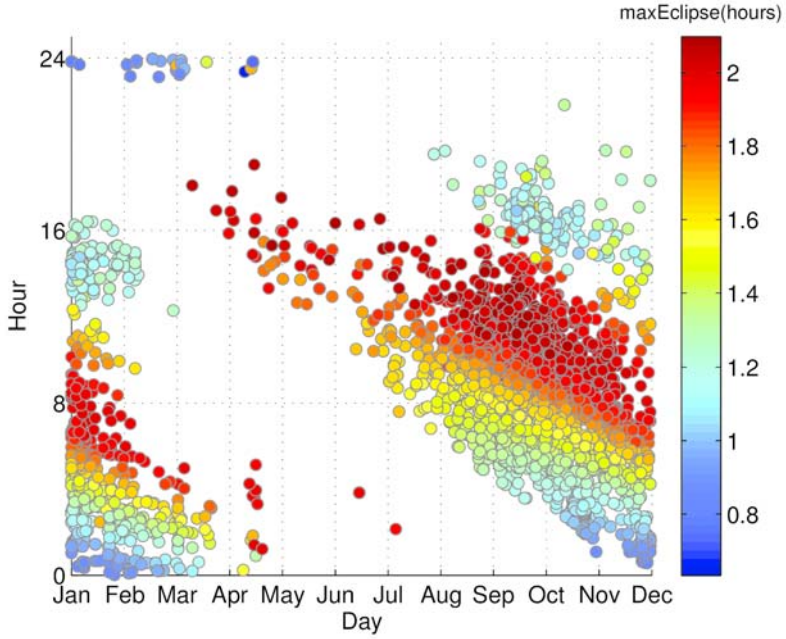
When launch date and time are represented by one parameter, i.e. time past Jan. 1 0:00, good solutions were not obtained. Why?

局所解が多数(365個）ある分布になっており，最適解を見つけることが難しかった．

Because many (365) local optimums exist, it was hard to find true optimum

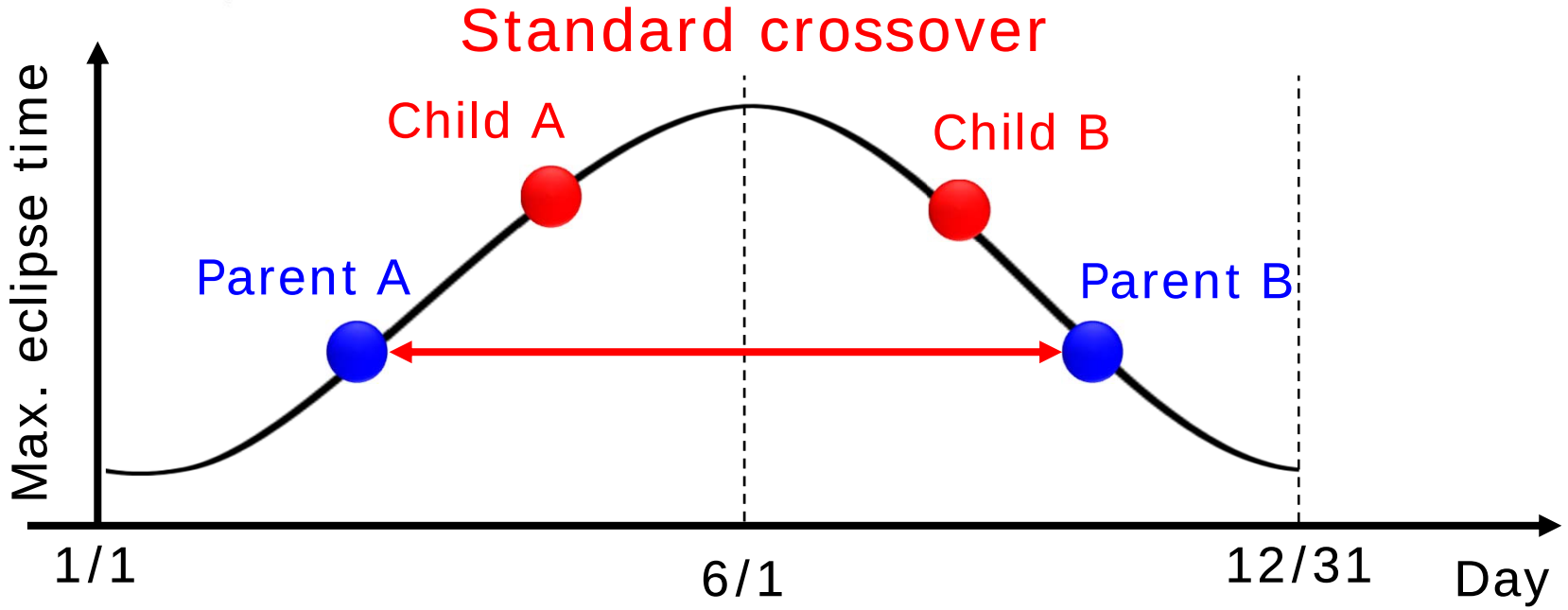


Launch date and time

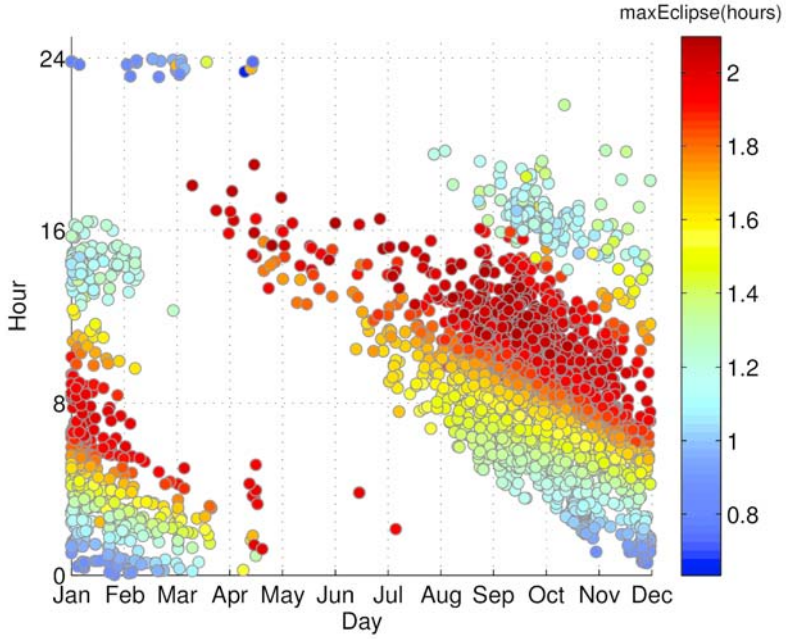


それでも，なぜか多目的進化アルゴリズムは夏ばかり探査してしまった．なぜか？

However, MOEA search for solutions in summer. Why?

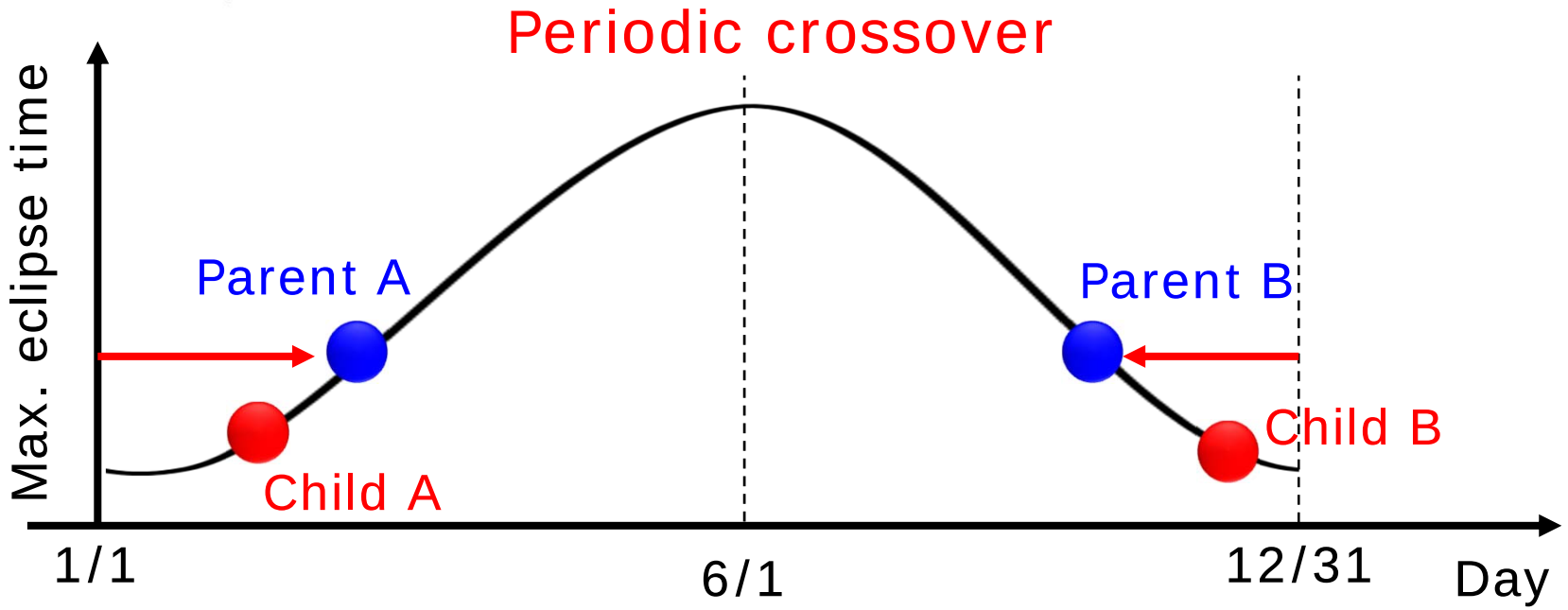


Launch date and time



それでも，なぜか多目的進化アルゴリズムは夏ばかり探査してしまった．なぜか？

However, MOEA search for solutions in summer. Why?



Lessons learned

最適化問題の定式化の見直しが必要になる場合がある.
Review of problem definition may be required.

最適化アルゴリズムのカスタマイズが必要になる場合がある
Customization of optimization algorithm may be required.

まとめ Summary

- 多数の優れたパレート最適解を得ることが出来た.

  Many Pareto-optimal designs were obtained.

- パレート最適解に関する情報を得ることが出来た

  Useful information about Pareto-optimal designs was obtained.

Trajectory design optimization for DESTINY mission

No. 2

Design Exploration of a Low-Thurst Space Trajectory Problem for DESTINY Mission

Takeshi Watanabe, Tomoaki Tatsukawa, Takayuki Yamamoto,
Akira Oyama, and Yasuhiro Kawakatsu

THURSDAY, JANUARY 07, 2016
2:00 PM - 2:30 PM

IS-11. Intelligent Mission Design and Vehicle Control
2:00 PM – 5:30 PM; Regatta B

SCITECH2016 4–8 January 2016 San Diego, California

Optimization problem

Objective functions

Four objective functions.

- Minimize ion engine system (IES) operation time (t_IES).
- Minimize time of flight (ToF).
- Minimize maximum eclipse (Ecl) time (t_Ecl).
- Maximize initial mass of the satellite (Mass).

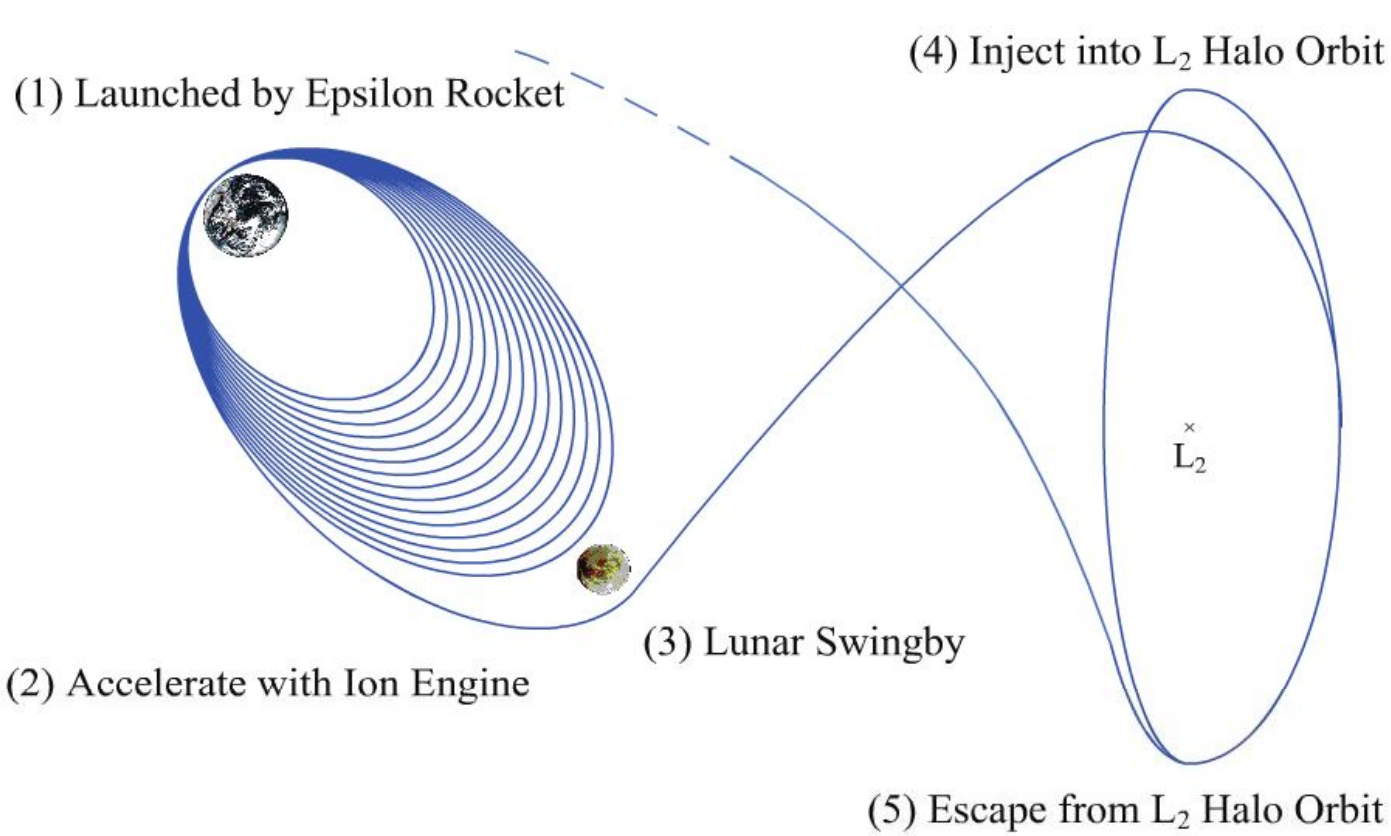
One constraint.
$C_{\mathrm{eq}} + 0.1 \geq 0, \quad C_{\mathrm{eq}} = l_{\mathrm{Moon}} - l_{\mathrm{target}}$.
$l_{\mathrm{target}} = 300,000$ [km]: The target radius on the Moon's orbital plane.
l_{Moon} [km]: The distance from the intersection of the final orbit and the Moon's orbital plane to the Earth.

Optimization problem

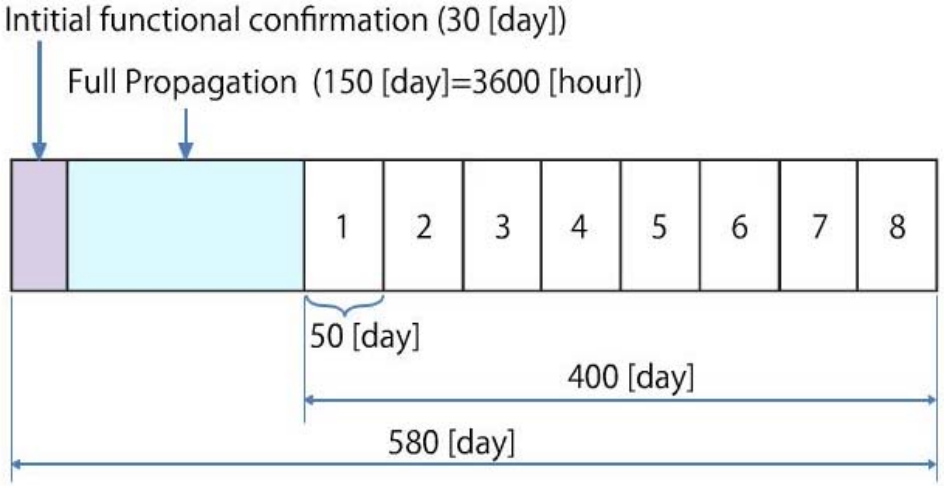
Design parameters

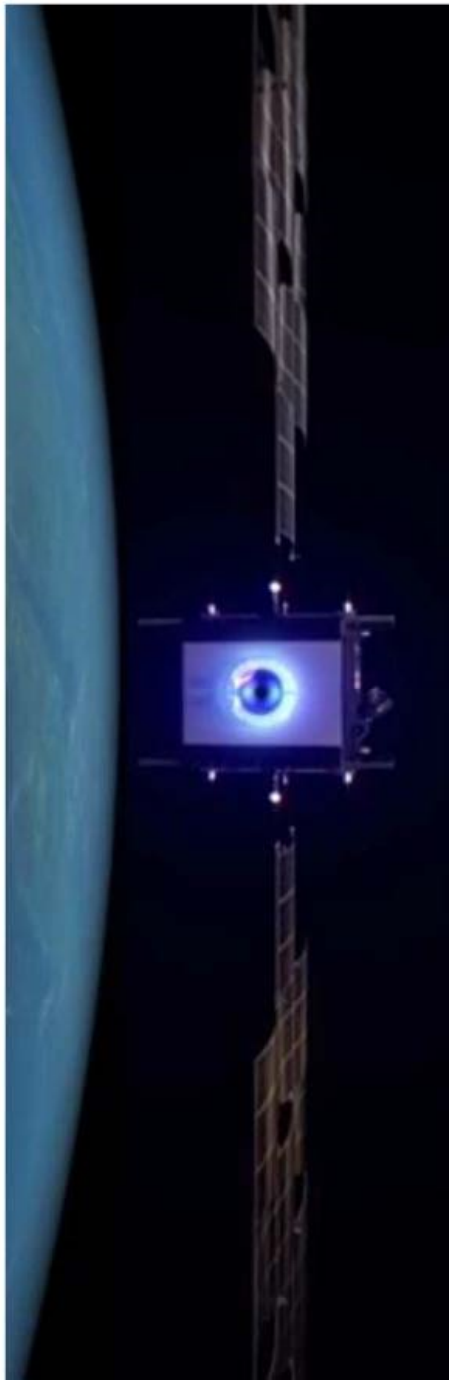
Twenty seven design parameters.
$(D, t, x_1, y_1, z_1, \cdots, x_8, y_8, z_8, M)$
$D \in [0, 366] \cap \mathbb{Z}$ [day]: Launch date.
$t \in [0, 86400] \cap \mathbb{Z}$ [s]: Launch time.
Launch at 2019–07–01 00:00:00 $+D$ [day] $+(t - 180)$ [s] UTC.
$x_i, y_i, z_i \in [0, 1]$: Parameters to determine operation profile of the ion engine system.
$M \in [400, 450] \cap \mathbb{Z}$ [kg]: Initial mass of the spacecraft.

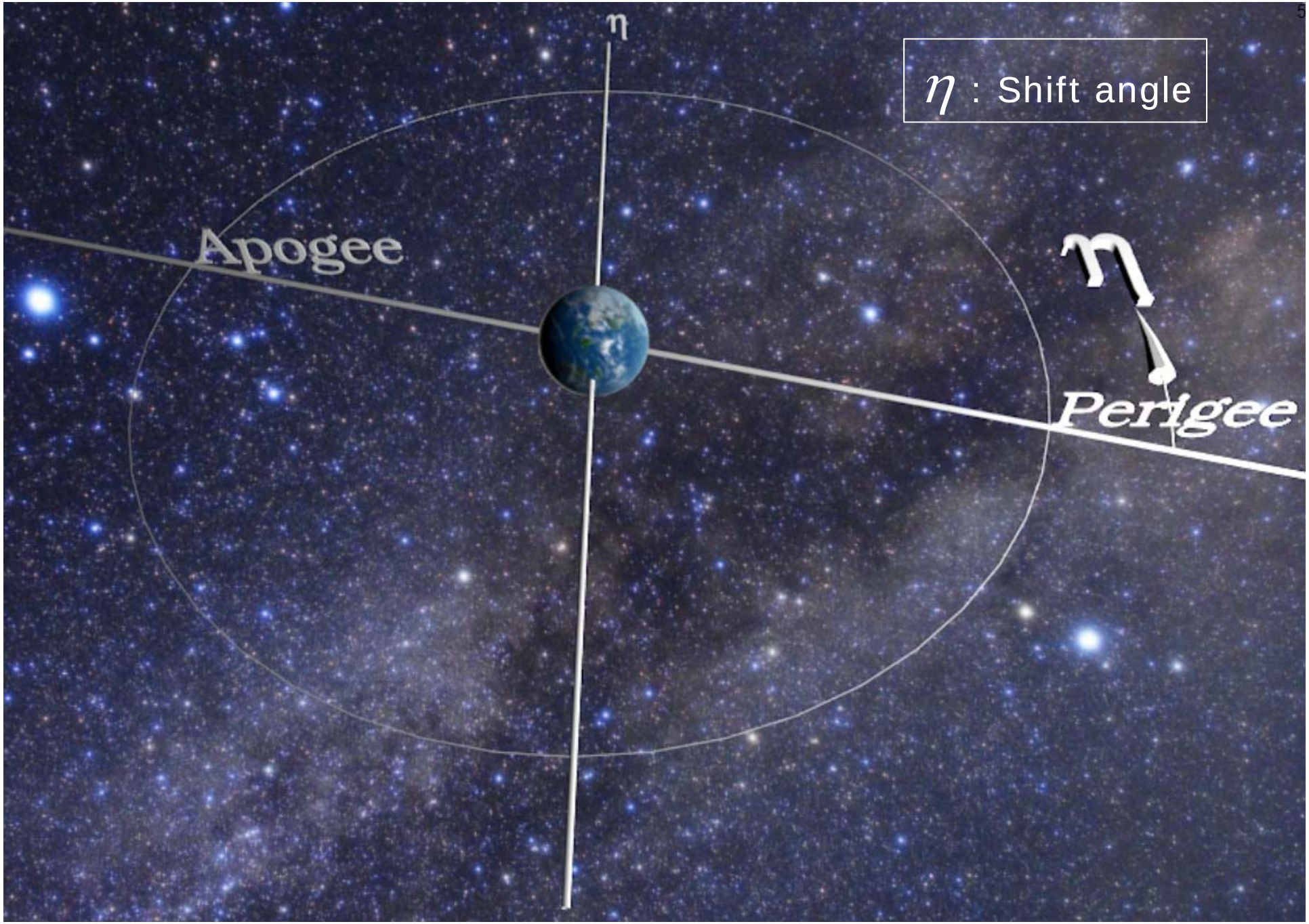
$x_i, y_i, z_i \in [0, 1]$: Parameters to determine operation profile of the ion engine system.

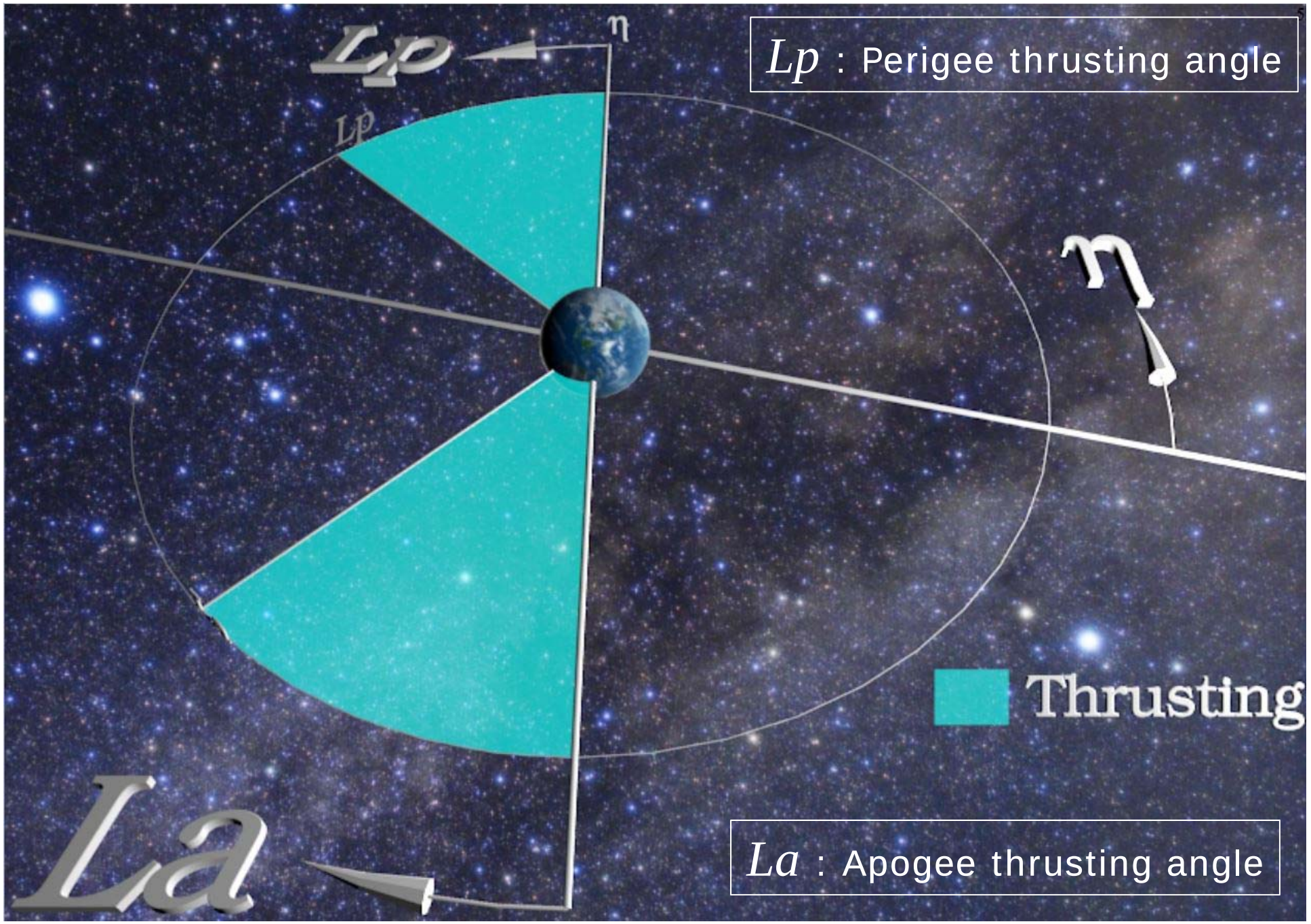


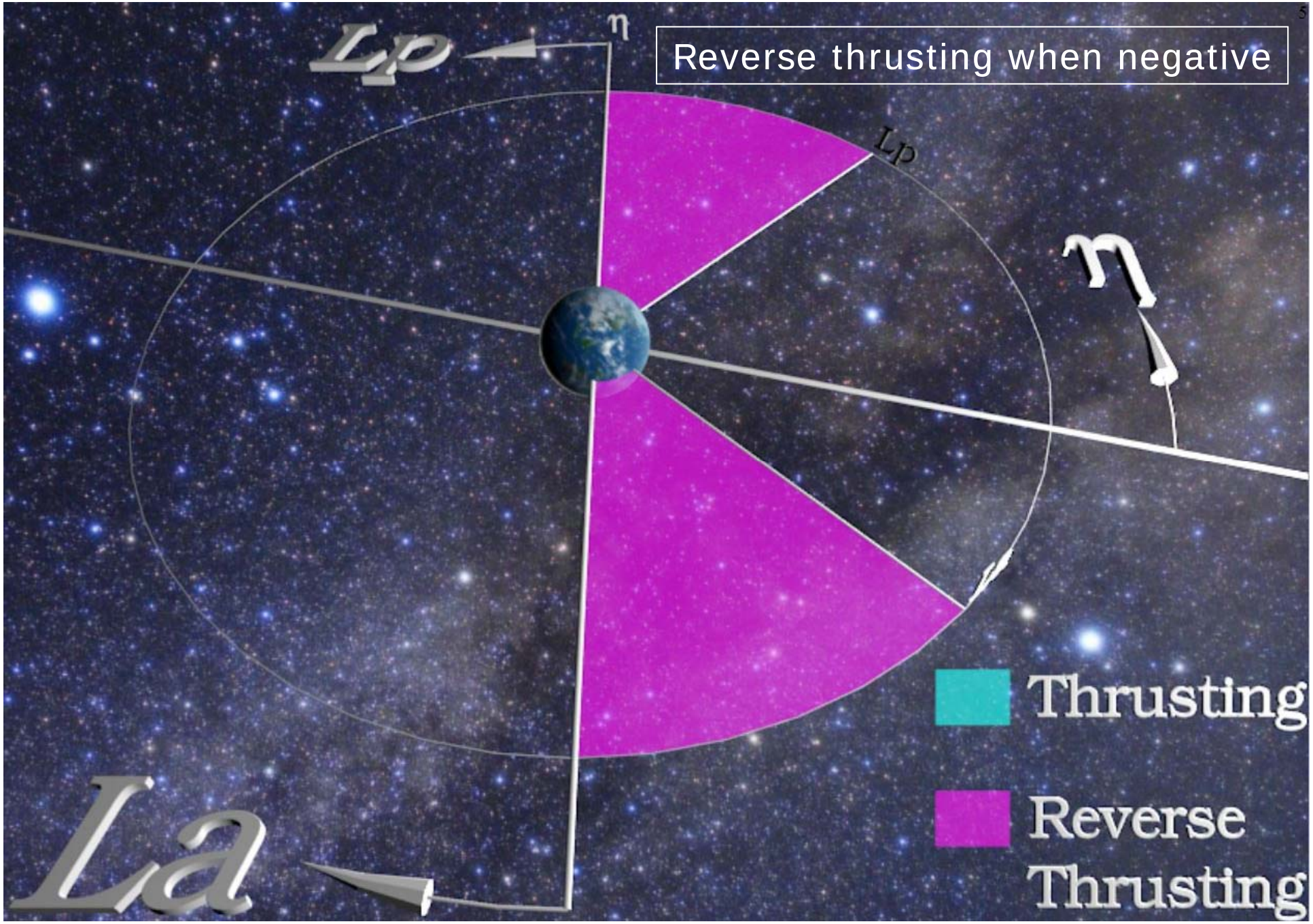
Spiral raising phase

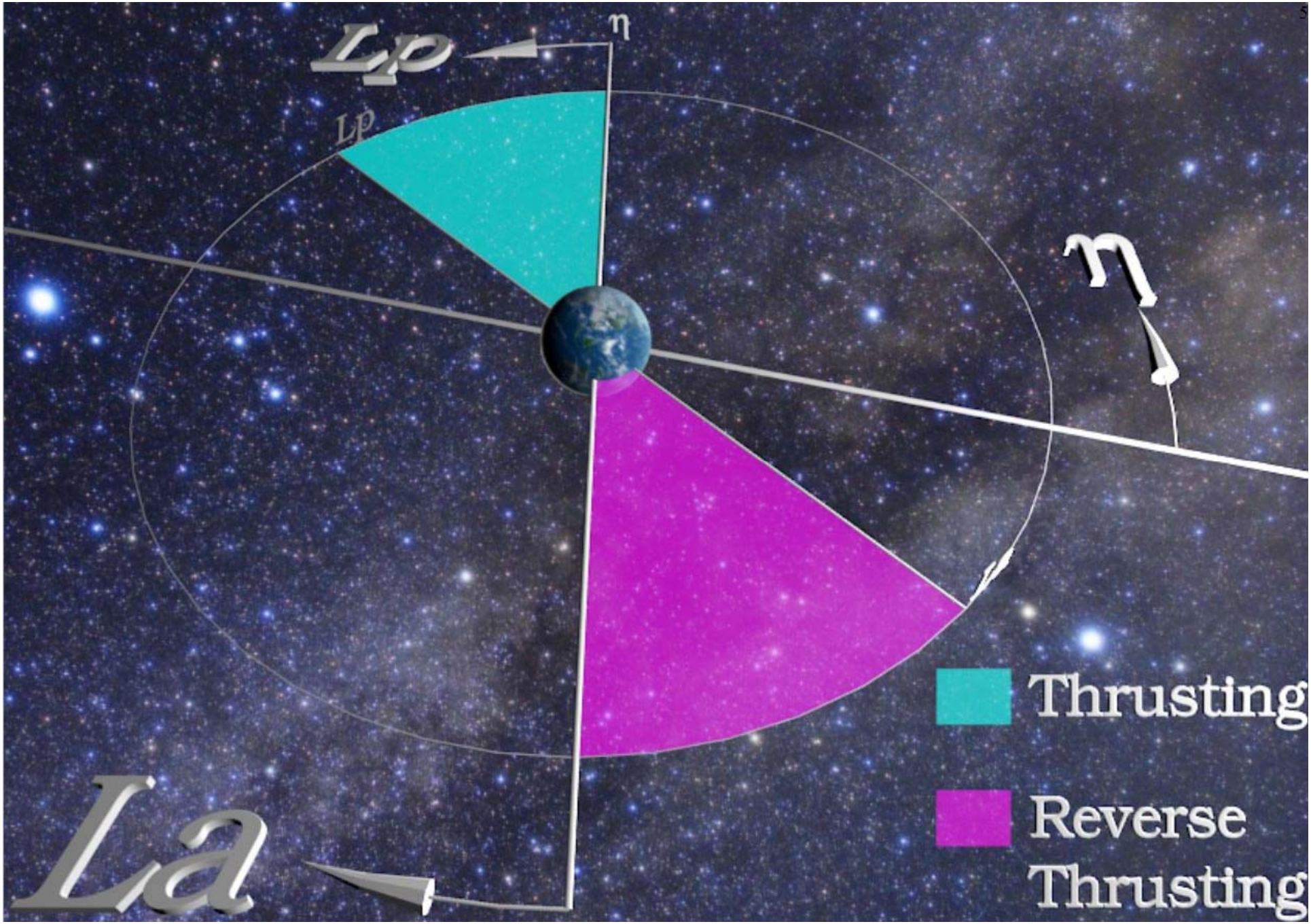


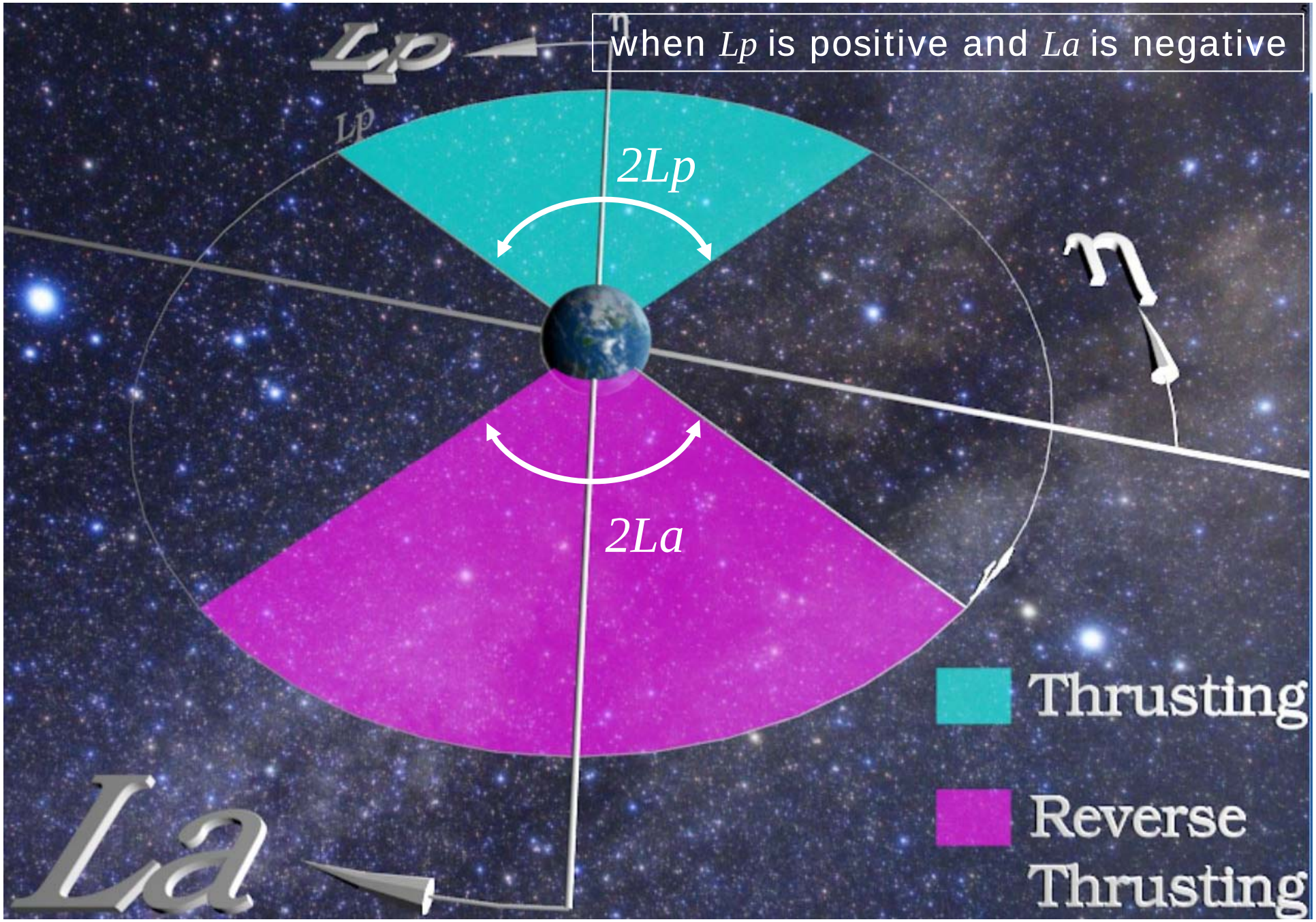


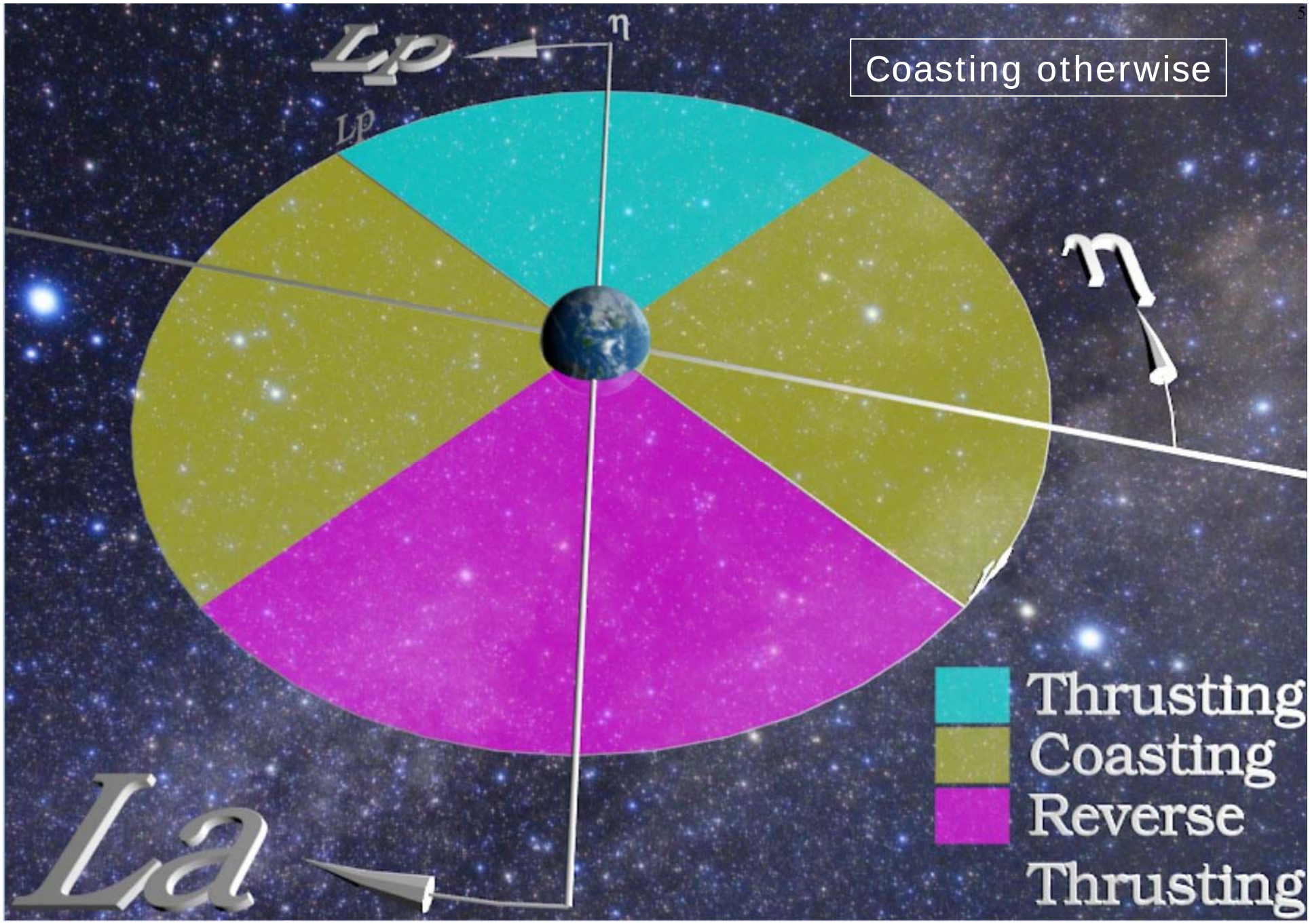

η
η : Shift angle
Apogee
Perigee

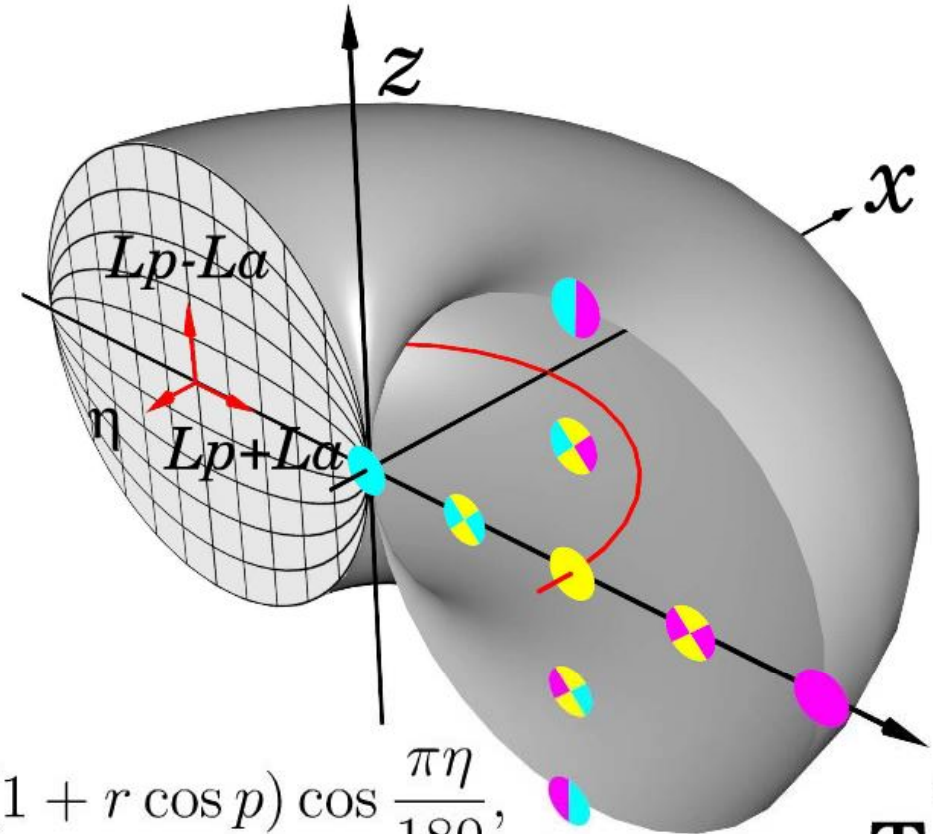

Lp
η
Lp
Lp : Perigee thrusting angle
η
Thrusting
La
La : Apogee thrusting angle


Reverse thrusting when negative
Lp
η
Lp
η
La
Thrusting
Reverse Thrusting


Lp
η
Lp
η
Thrusting
Reverse
Thrusting
La


when Lp is positive and La is negative
Lp
Lp
2Lp
2La
Thrusting
Reverse Thrusting
La


Lp
η
Lp
Coasting otherwise
η
Thrusting
Coasting
Reverse
Thrusting
La



: Thrusting

: Coasting

: Reverse thrusting

Transformation of coordinate system

$$
\begin{aligned}
x &= (1 + r\cos p)\cos\frac{\pi\eta}{180}, \\
y &= (1 + r\cos p)\sin\frac{\pi\eta}{180}, \\
z &= r\sin p, \\
\Delta L_a &= 90 - \frac{180}{\pi}\cos^{-1}\left(-\sin\frac{\pi r}{2}\cos p\right) - \frac{180}{\pi}\tan^{-1}\left(\tan\frac{\pi r}{2}\sin p\right), \\
\Delta L_p &= 90 - \frac{180}{\pi}\cos^{-1}\left(-\sin\frac{\pi r}{2}\cos p\right) + \frac{180}{\pi}\tan^{-1}\left(\tan\frac{\pi r}{2}\sin p\right).
\end{aligned}
$$

Optimizer

CHEETAH (CHEbyshev-Epsilon opTimization AlgoritHm).
A many-objective optimizer with "Region of interest (RoI)" set in the central part of the non-dominated solution set and solutions in RoI are given priority in being chosen as parents, and ϵ-indicator for ranking individuals.

Population size: 10,000.

Number of generations: 100.

Crossover rate: 1.0, crossover index: 15,
mutation rate: 0.1, and mutation index: 25.

Objective functions

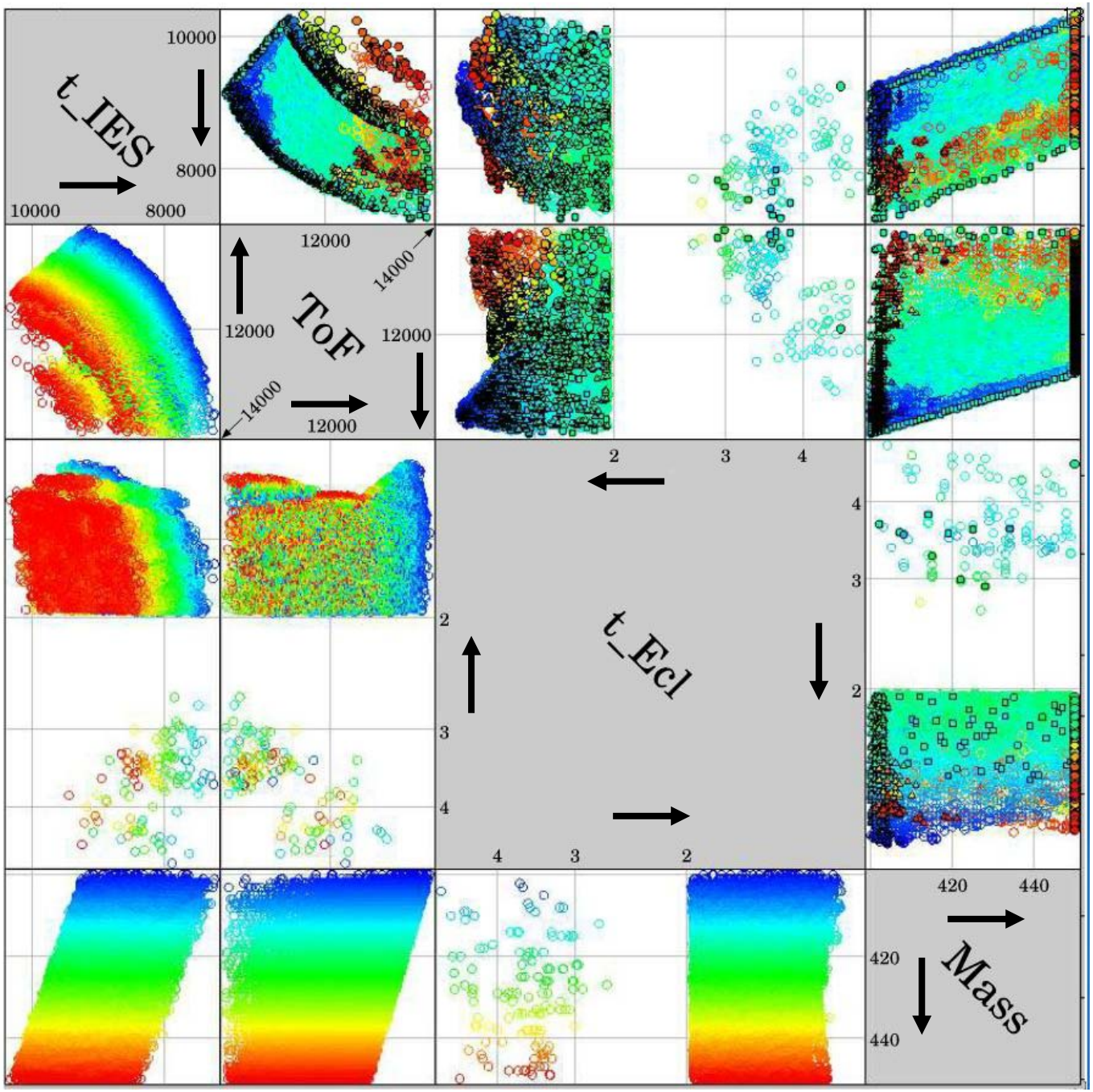
Lower triangle: Colored by Mass.

t_IES & ToF: Proportional to Mass.

t_Ecl: Almost uniform with respect to Mass.

Upper triangle: Edge extraction, colored by orbit injection time.

Distribution of non-dominated solutions are represented by two extreme cases: Heavy & light cases.

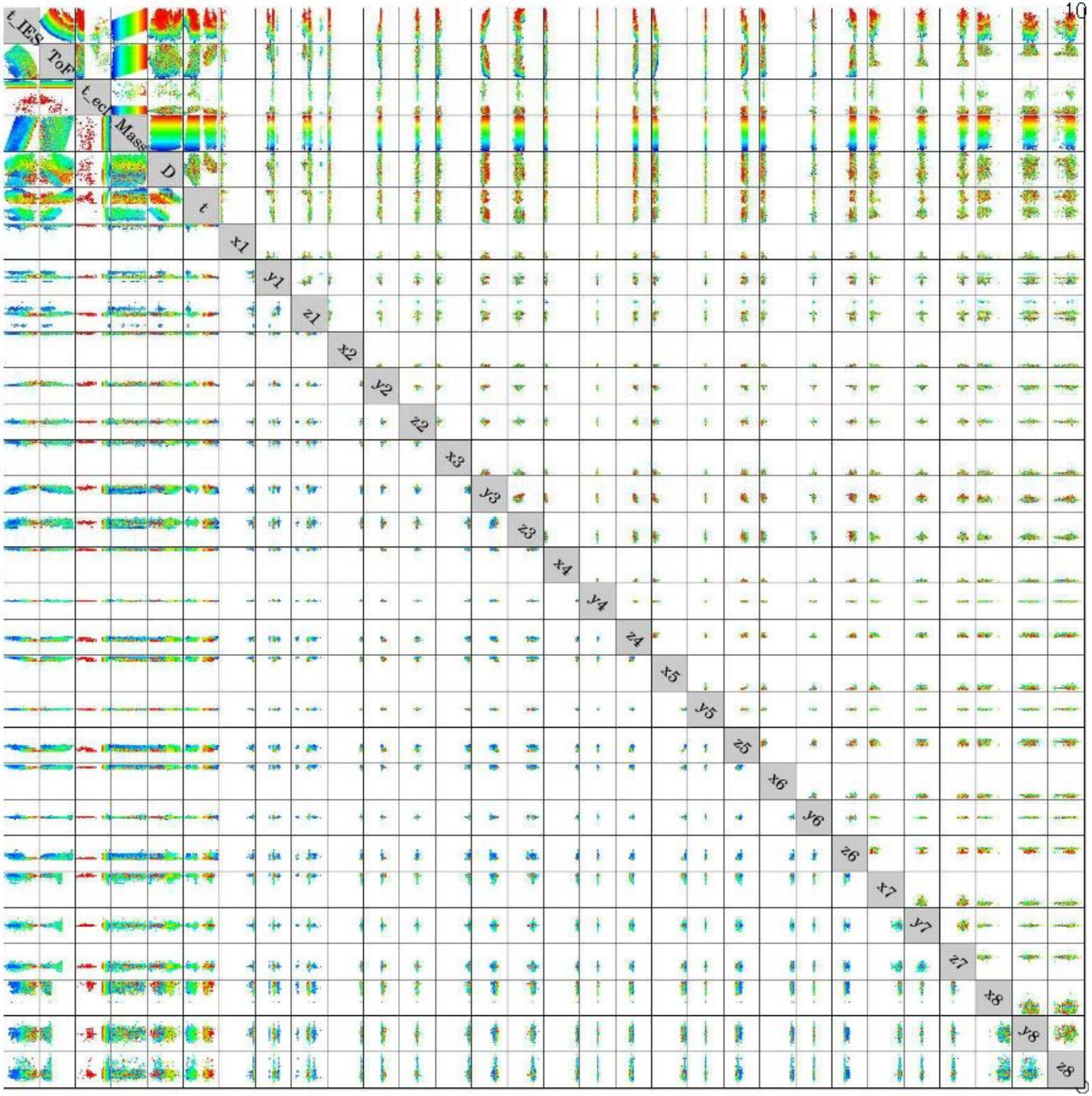


Result

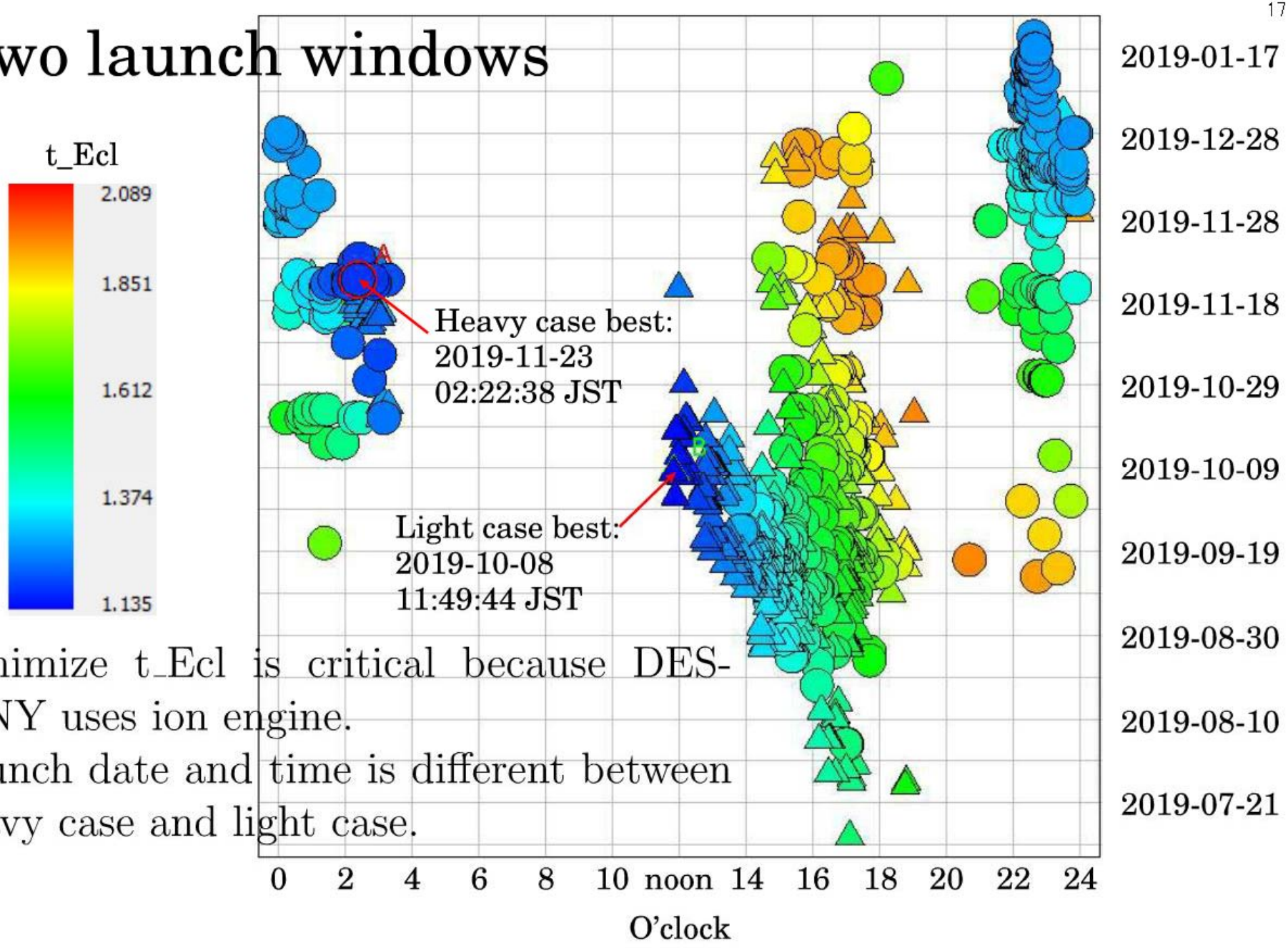
35,272 non-dominated solutions.

Scatter plot matrix (SPM)

Matrix of two-dimensional scatter plots of all binary combinations among design parameters and objective functions. Upper triangle is colored by Mass, and lower, t_Ecl.



Two launch windows

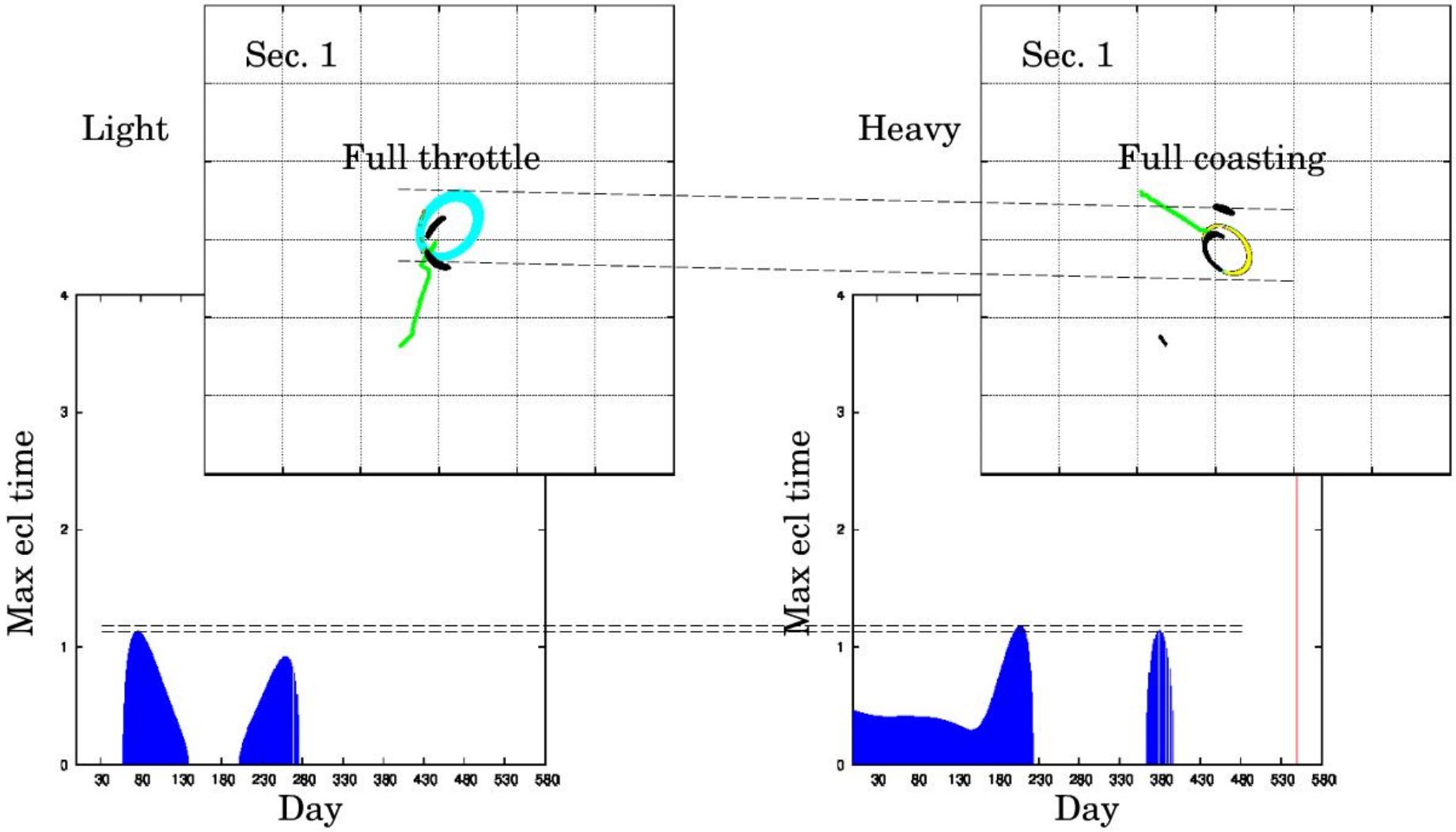


Minimize t_Ecl is critical because DESTINY uses ion engine.
Launch date and time is different between heavy case and light case.

Strategy to avoid long eclipse period

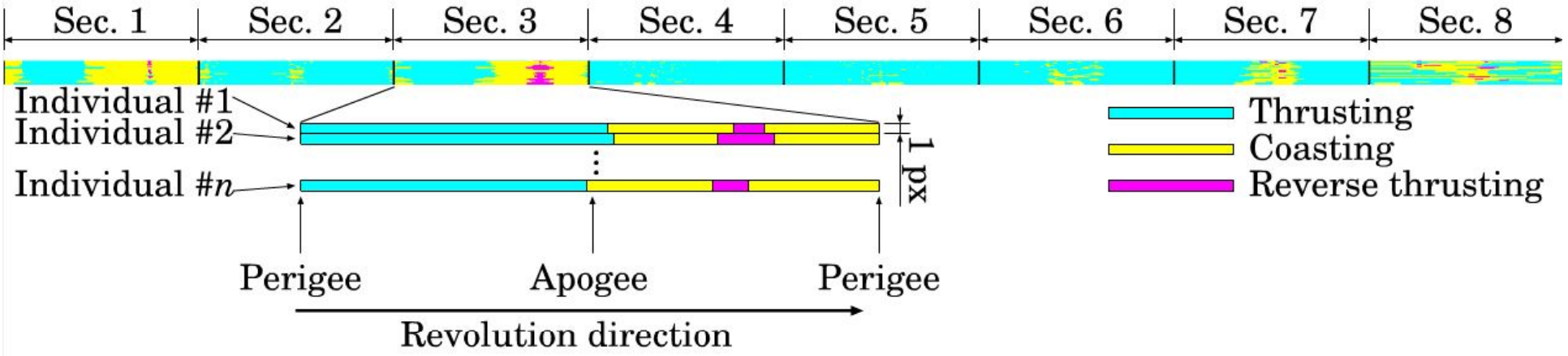
Light: Fast escape.

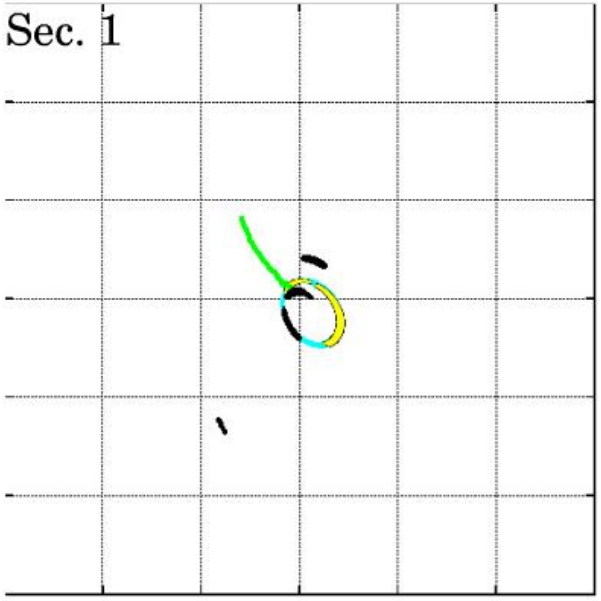
Heavy: All things come to those who wait.

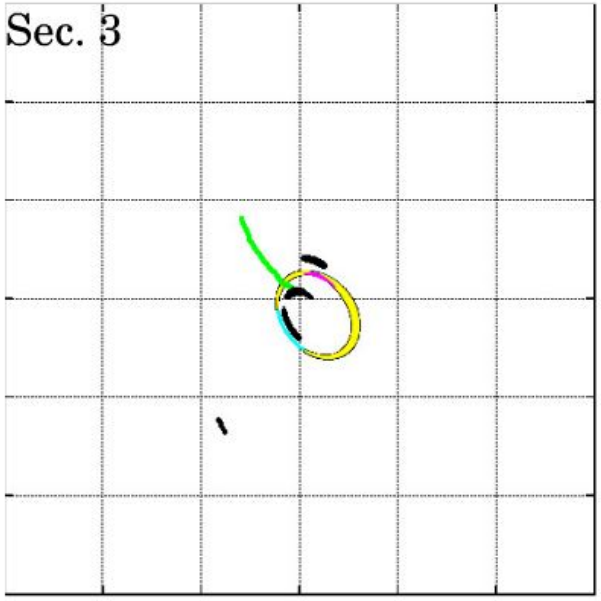


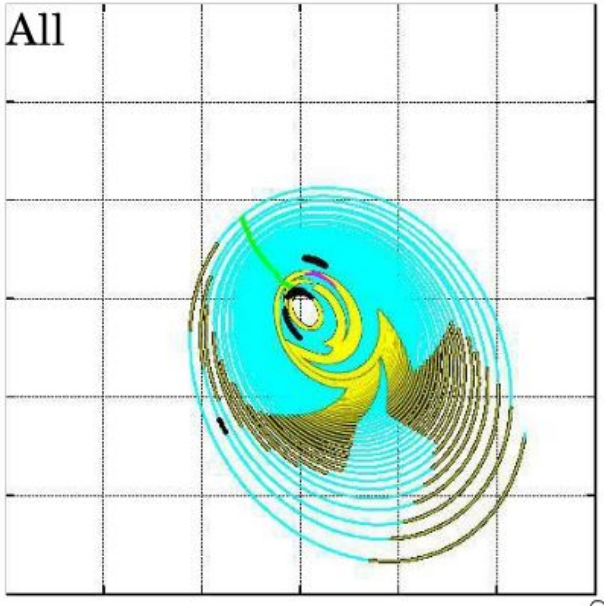
Heatmap

One-pixel horizontal line represents an individual.







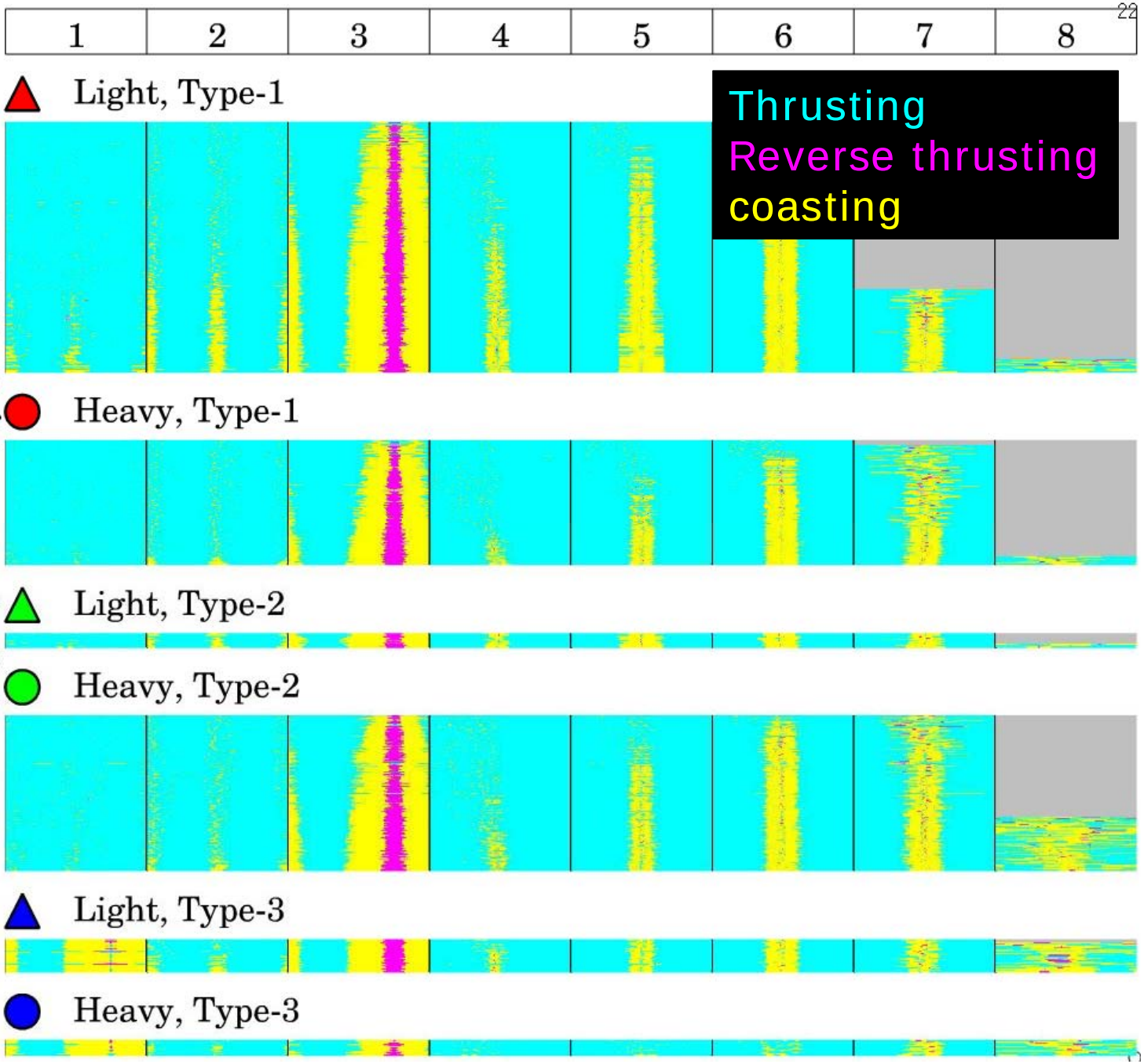


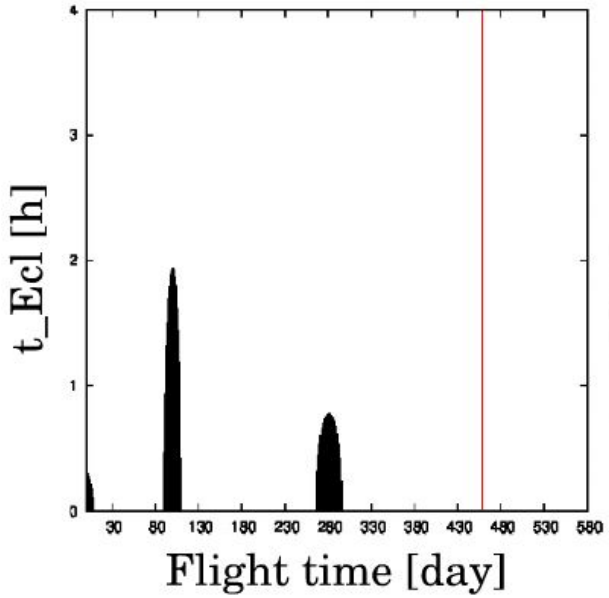
Heatmap

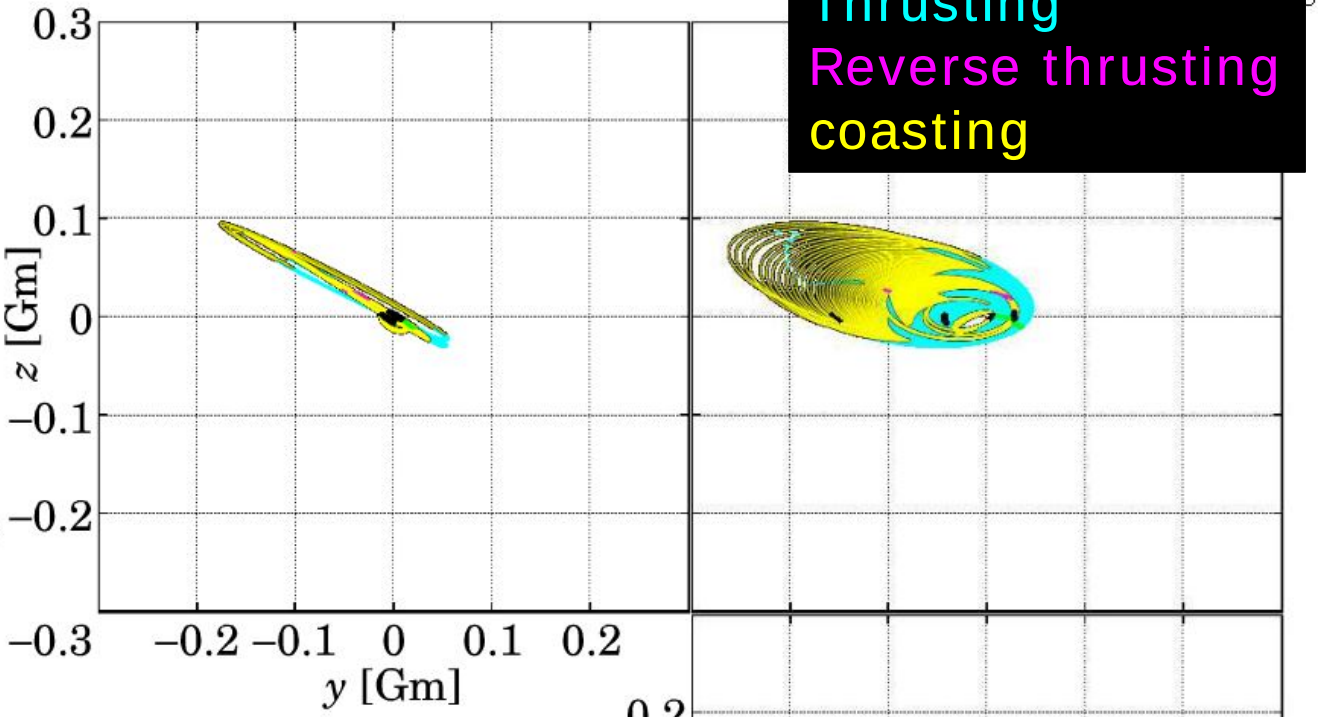
Sorted by ToF.

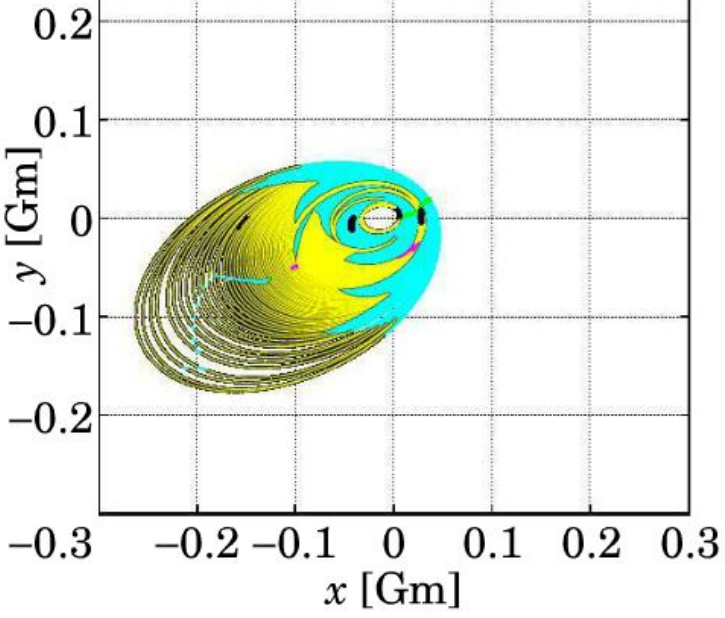
IES operation profile is almost the same except for Sec. 1 of Type-3.

Coasting region becomes larger when ToF becomes longer.



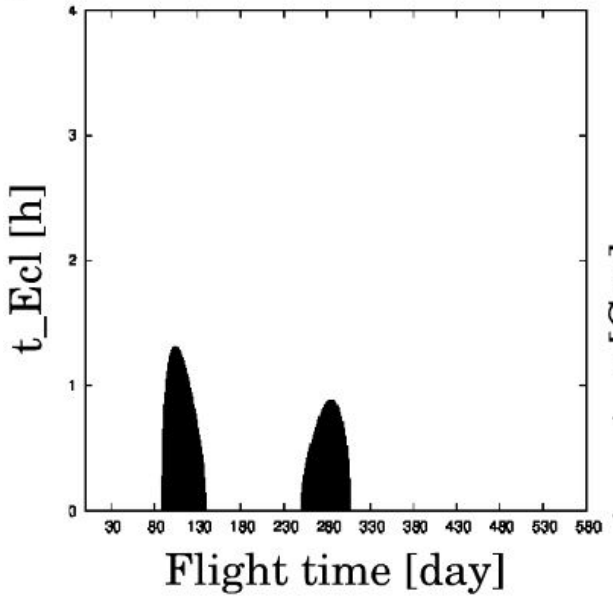


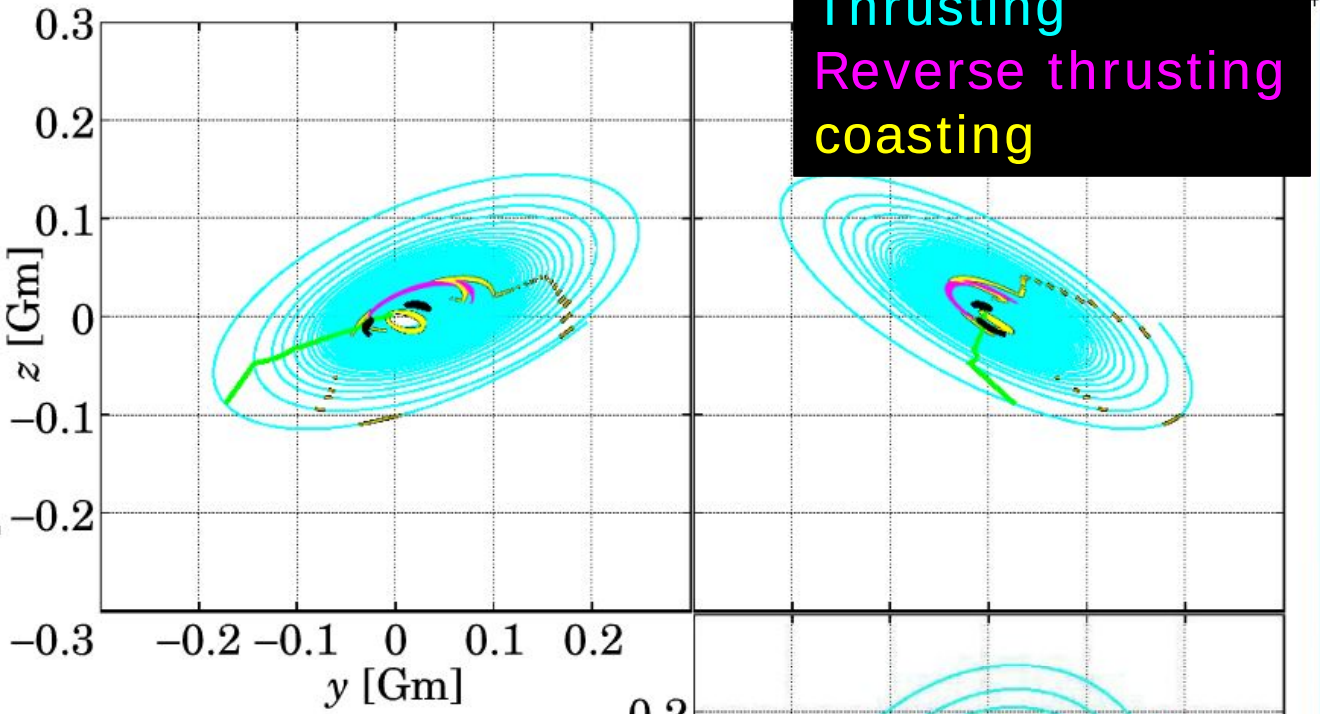


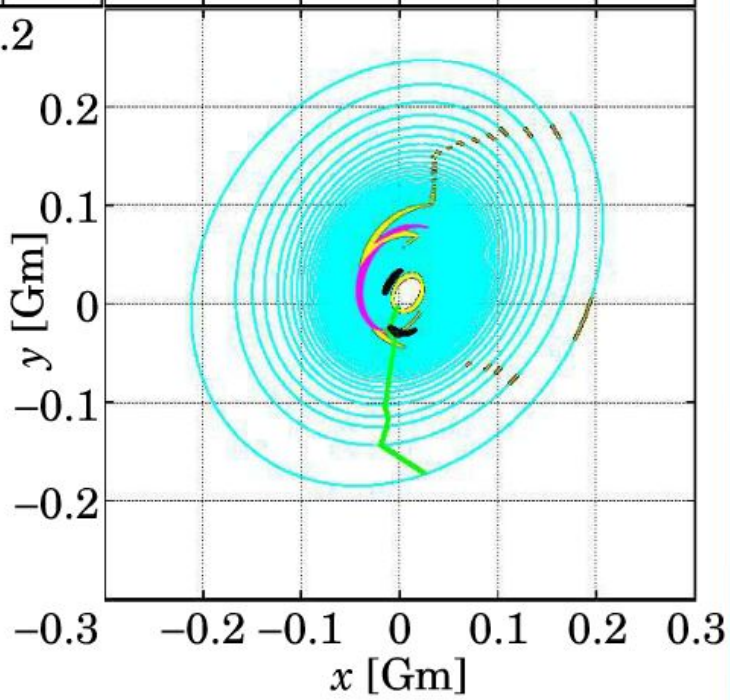


Min. t_IES

t_IES	7325.063114 [h]
ToF	13126.14272 [h]
t_Ecl	1.93842833 [h]
Mass	402 [kg]
Launch date	2019-12-20
Launch time	16:26:52
Gen	95
Num	8205

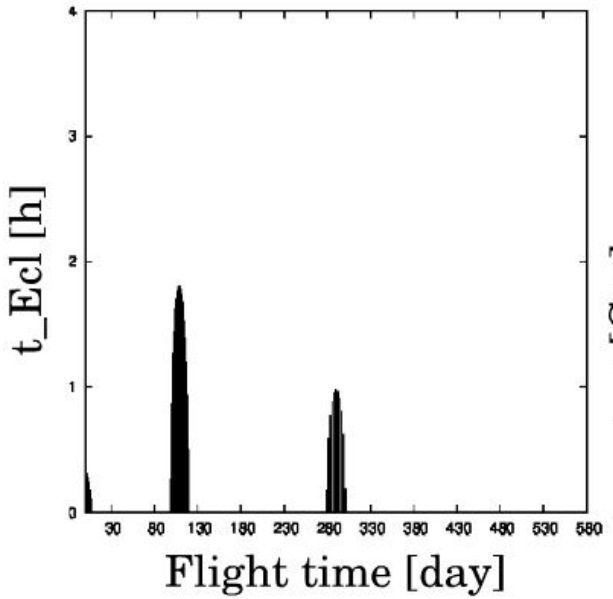




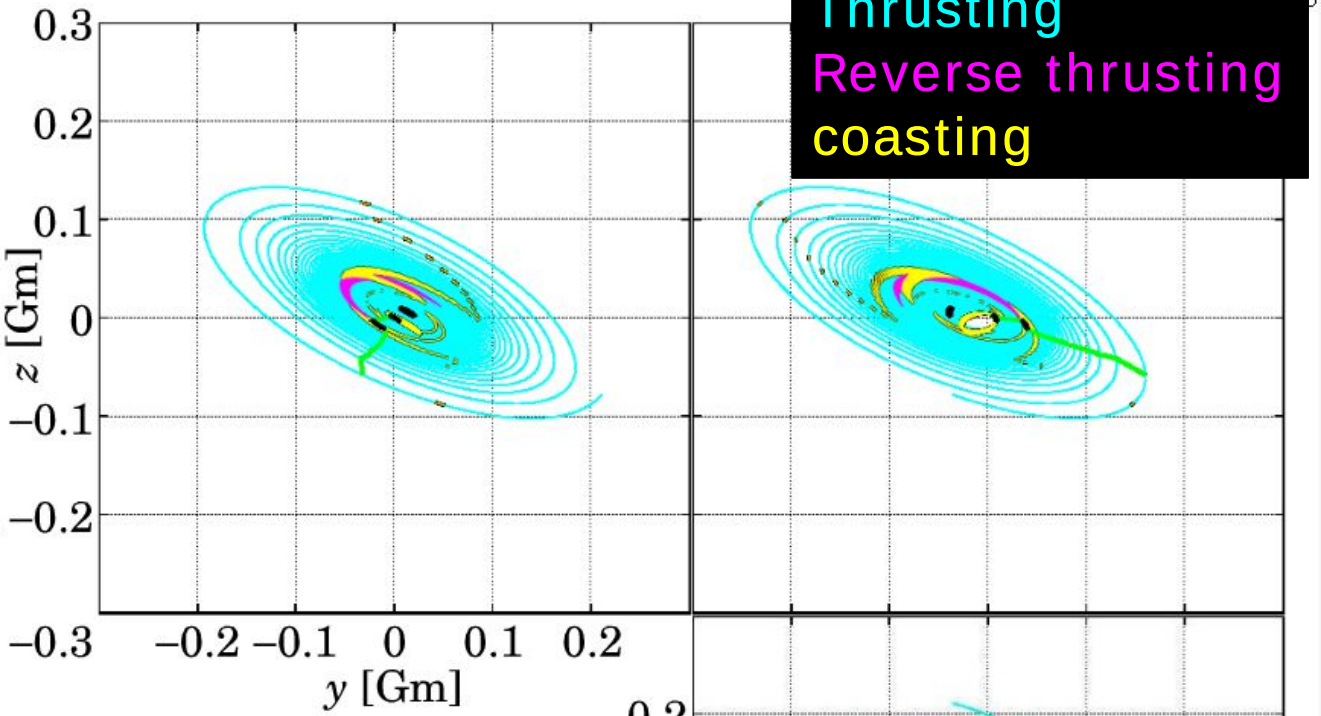


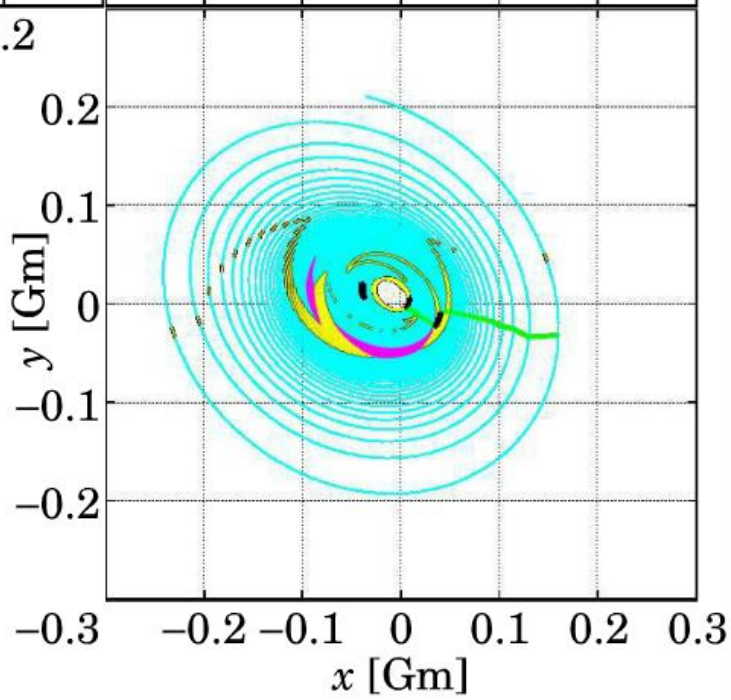
Max. t_IES

t_IES	10436.64689 [h]
ToF	11526.0888 [h]
t_Ecl	1.31445847 [h]
Mass	450 [kg]
Launch date	2019-09-18
Launch time	14:01:02
Gen	99
Num	1983



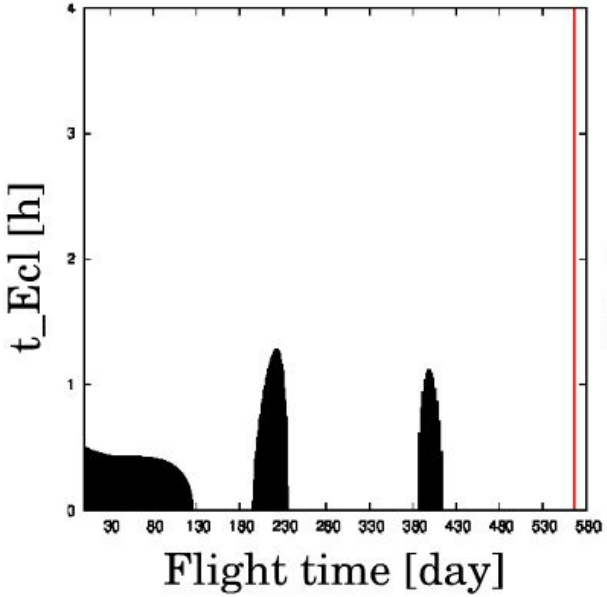
Thrusting
Reverse thrusting
coasting



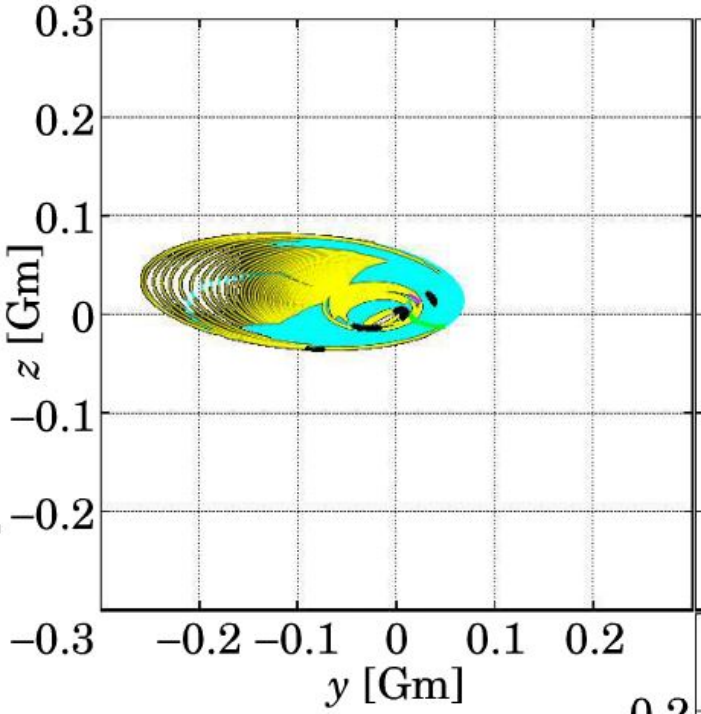


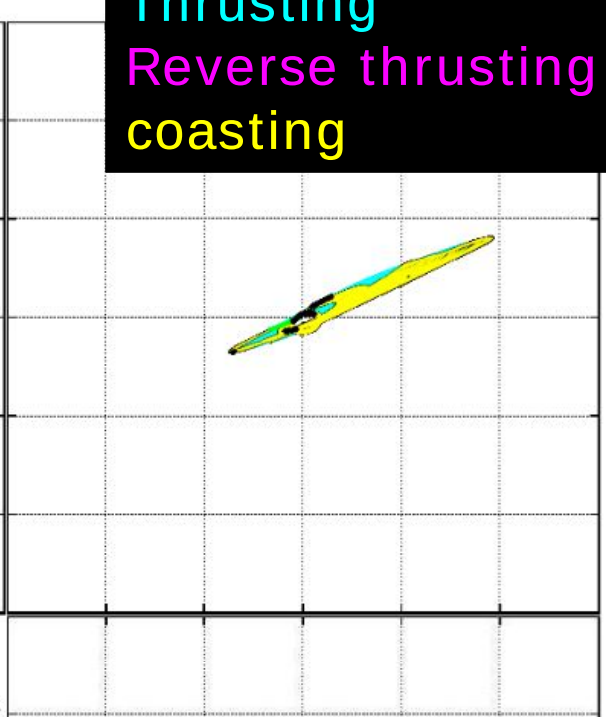
Min. ToF

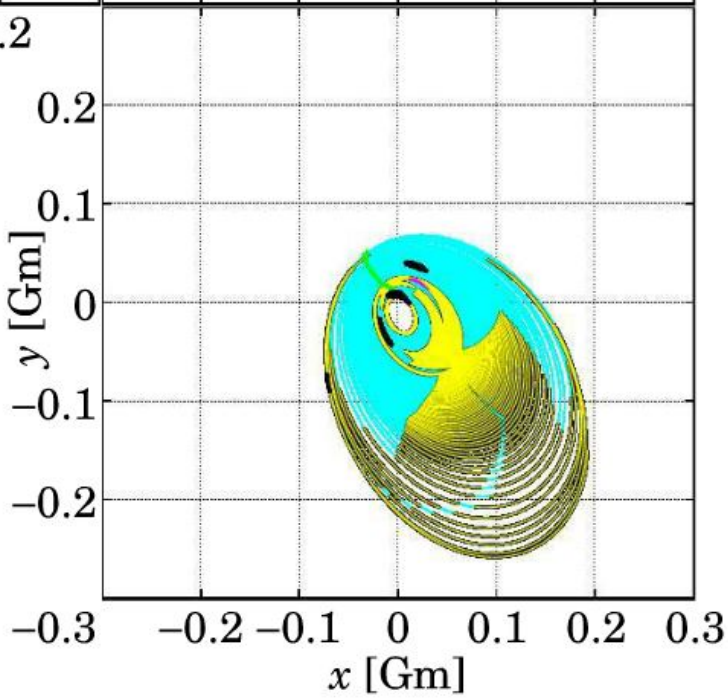
t_IES	9116.760022 [h]
ToF	10311.43781 [h]
t_Ecl	1.806989799 [h]
Mass	400 [kg]
Launch date	2019-11-11
Launch time	15:59:24
Gen	82
Num	1273



Thrusting
Reverse thrusting
coasting

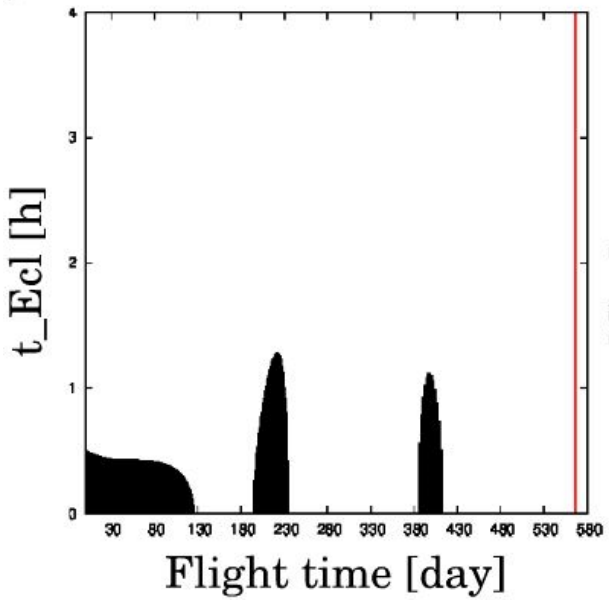


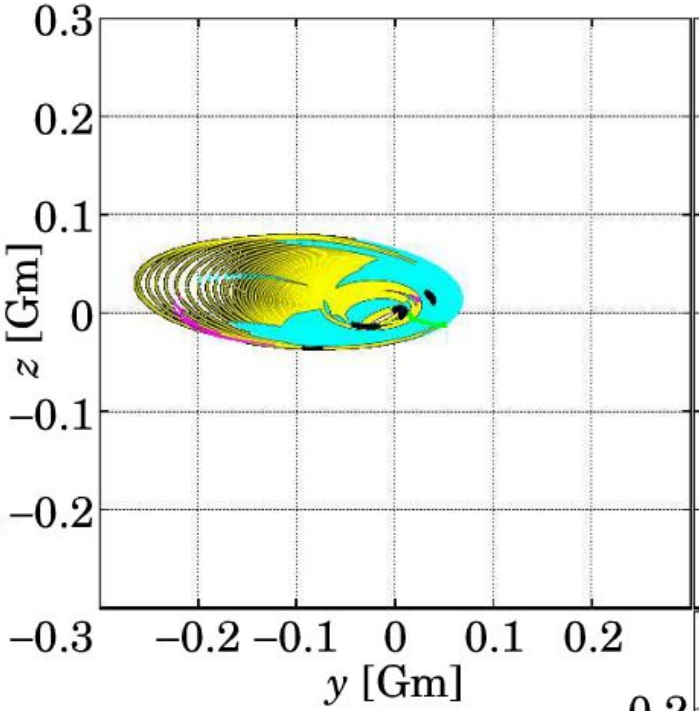




Max. ToF (1)

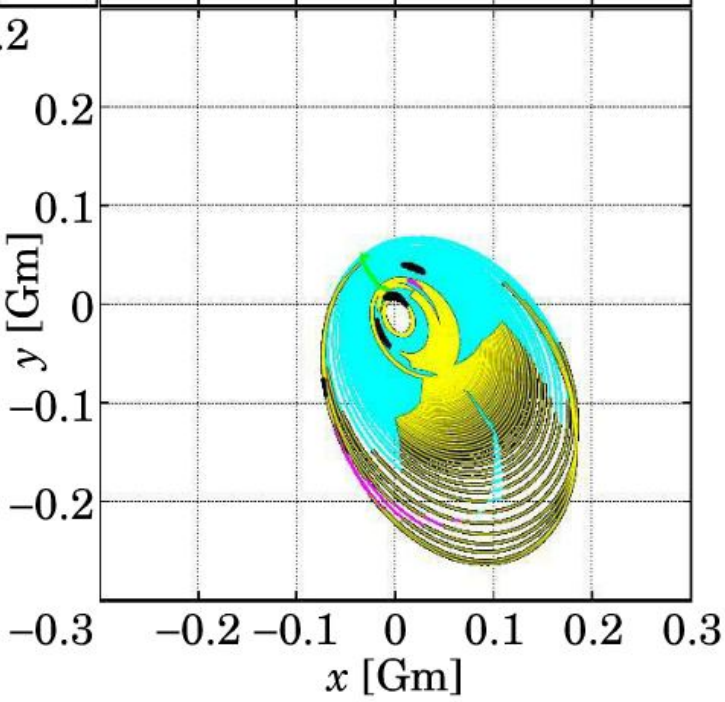
t_IES	7657.769718 [h]
ToF	13920 [h]
t_Ecl	1.287534065 [h]
Mass	406 [kg]
Launch date	2019-10-24
Launch time	03:10:42
Gen	95
Num	9704



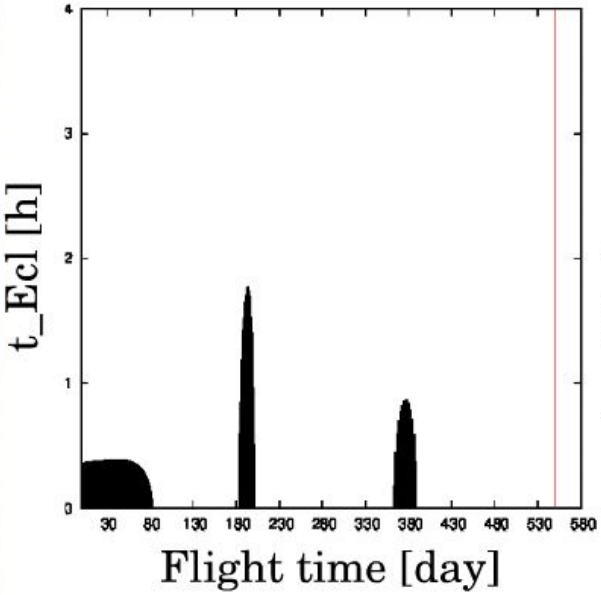


Thrusting
Reverse thrusting
coasting

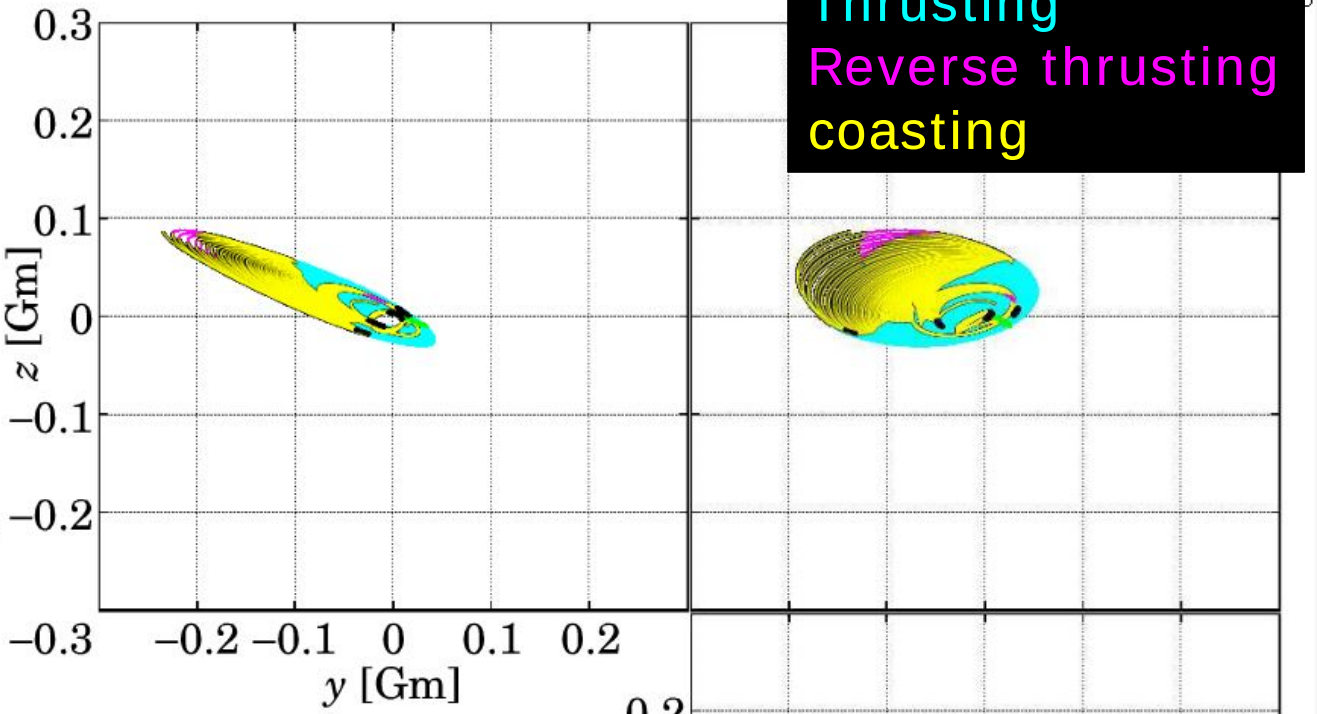
Max. ToF (2)

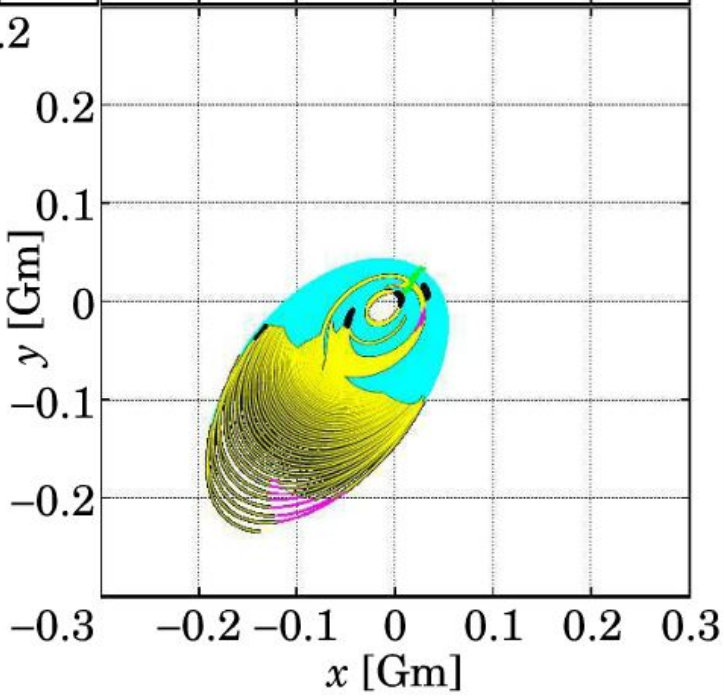


t_IES	7756.867608 [h]
ToF	13920 [h]
t_Ecl	1.284921206 [h]
Mass	406 [kg]
Launch date	2019-10-25
Launch time	03:07:25
Gen	99
Num	9462



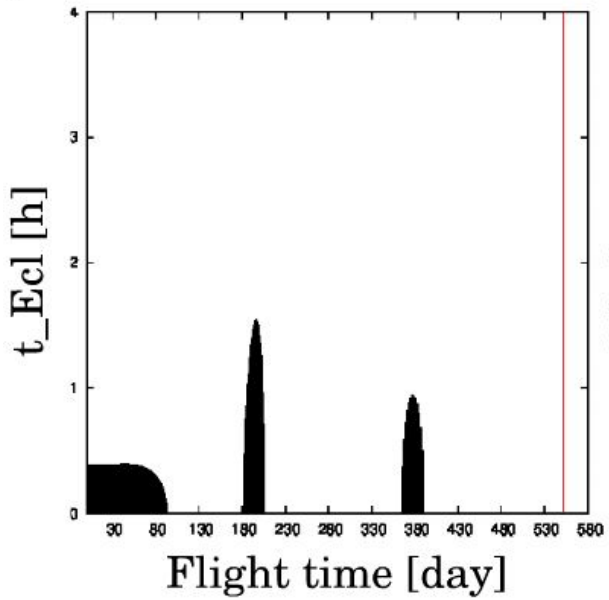
Thrusting
Reverse thrusting
coasting



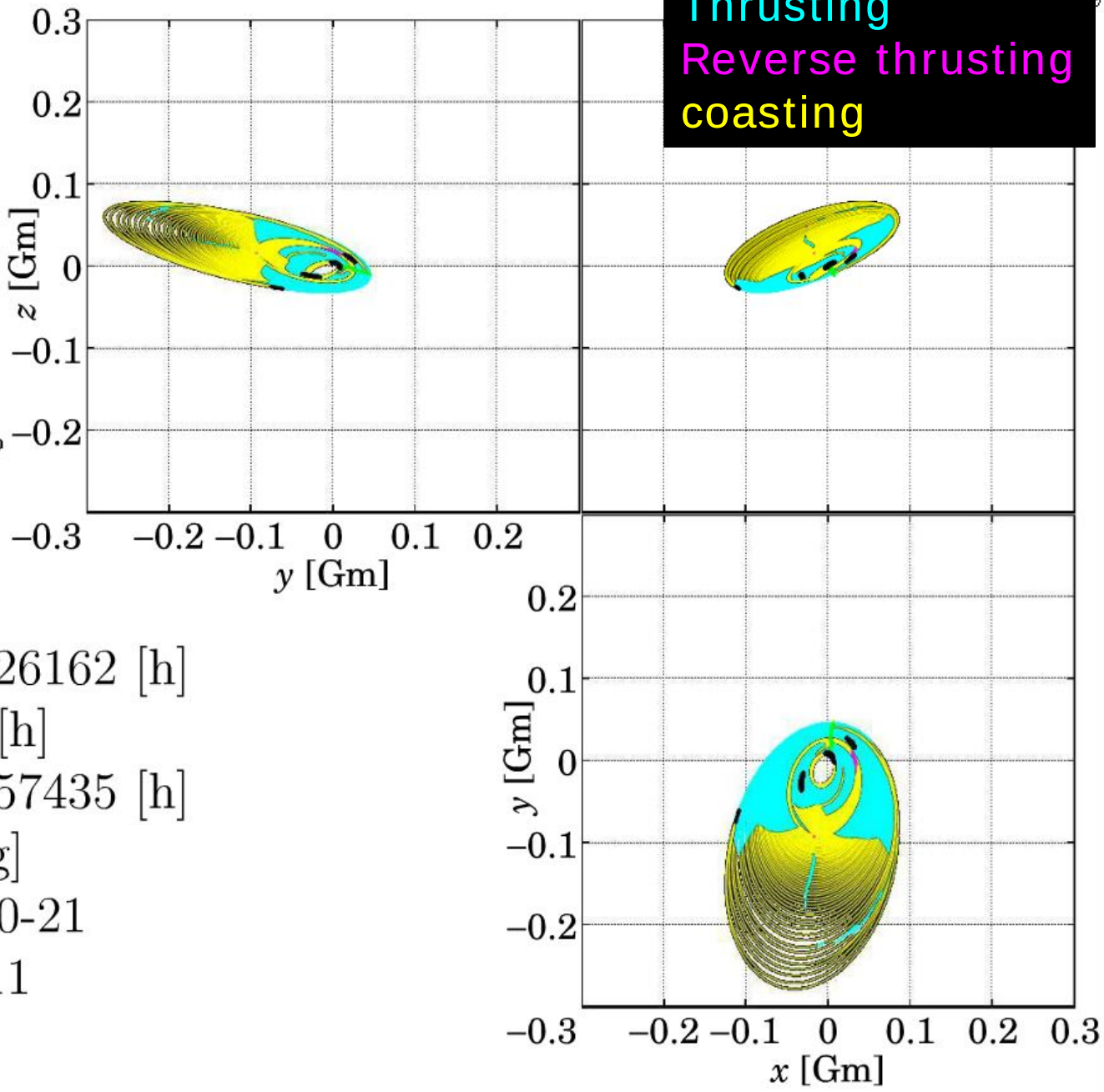


Max. ToF (3)

t_IES	8390.648521 [h]
ToF	13920 [h]
t_Ecl	1.77163302 [h]
Mass	450 [kg]
Launch date	2019-10-01
Launch time	23:42:54
Gen	96
Num	8971

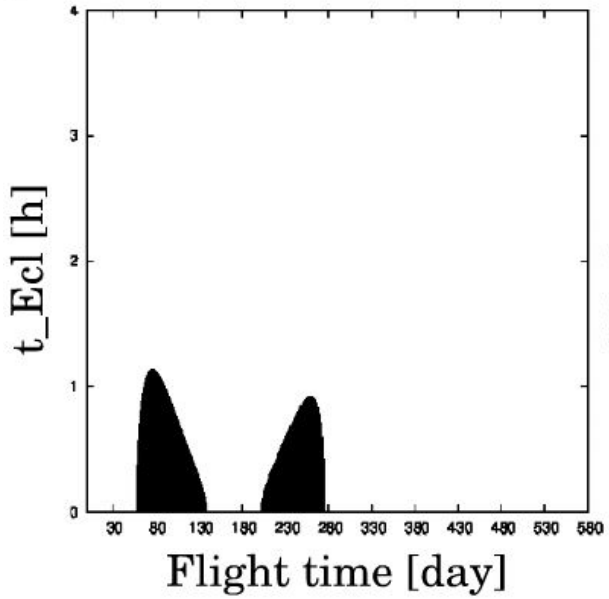


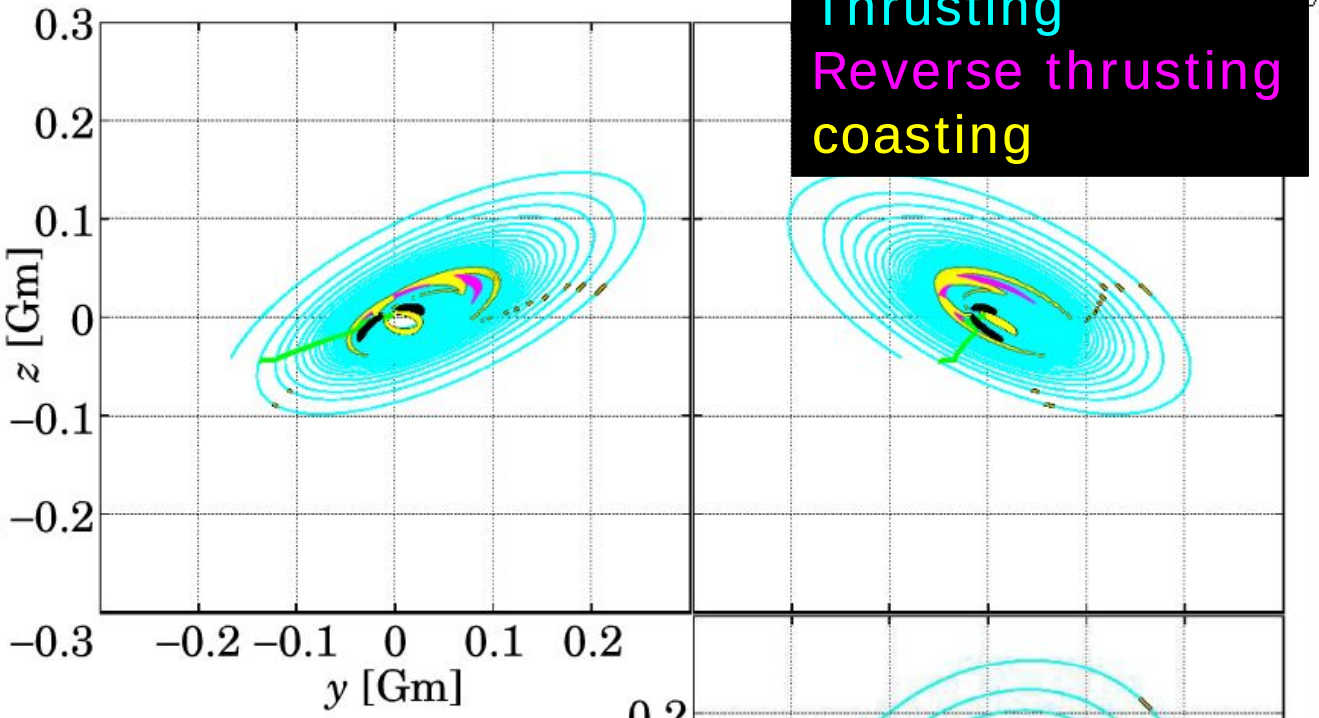
Thrusting
Reverse thrusting
coasting

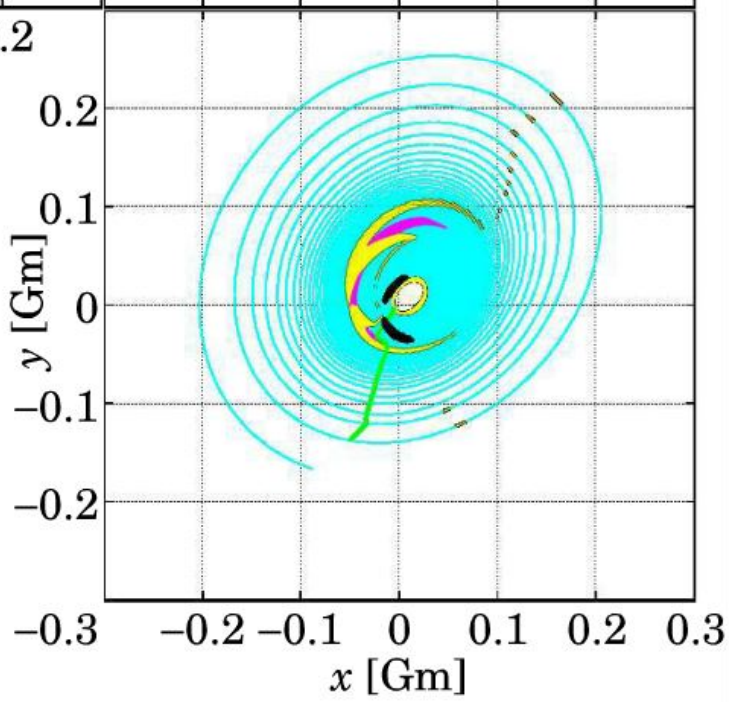


Max. ToF (4)

t_IES	8405.226162 [h]
ToF	13920 [h]
t_Ecl	1.542657435 [h]
Mass	450 [kg]
Launch date	2019-10-21
Launch time	00:41:11
Gen	94
Num	7307

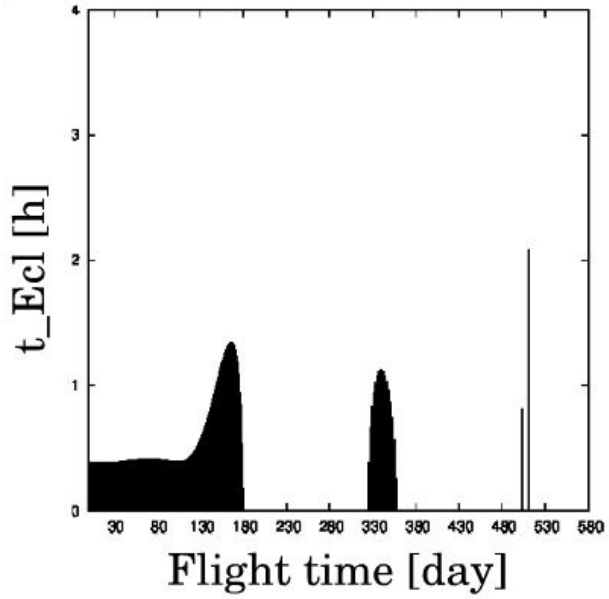


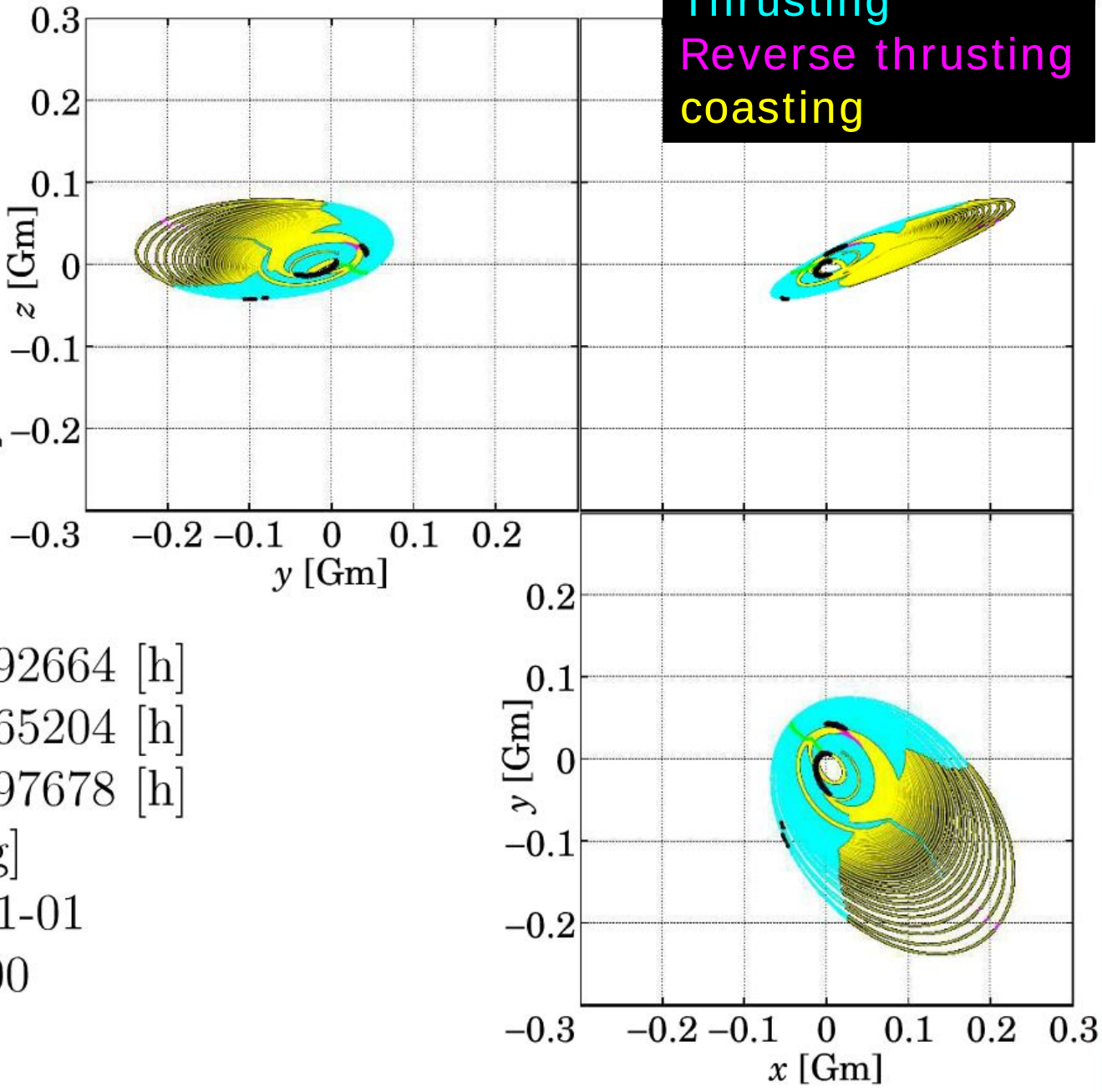




Min. t_Ecl

t_IES	9312.434023 [h]
ToF	10636.69752 [h]
t_Ecl	1.135058149 [h]
Mass	404 [kg]
Launch date	2019-10-08
Launch time	11:49:44
Gen	100
Num	6081





Max. t_Ecl

t_IES	7897.892664 [h]
ToF	12621.65204 [h]
t_Ecl	2.088997678 [h]
Mass	407 [kg]
Launch date	2020-01-01
Launch time	23:21:00
Gen	100
Num	7009

Concluding remarks

- Optimization of spiral raising orbit.
- Fast approximation calculations for 1,000,000 evaluations.
- Non-dominated solutions: 35,272.
- Edge extraction: Light case (t_IES, ToF, t_Ecl) and heavy case (Mass). 1,750 edge solutions.
- Slow but exact calculation: 844 non-dominated solutions.
- Two launch windows: Maximum eclipse time is mostly determined by launch date and time.
- Coasting and reverse propulsion are effective if it is well-timed.
- Two strategies to avoid long eclipse period.
  - Light: Fast escape — Full throttle.
  - Heavy: All things come to those who wait — Full coasting.

まとめ
Summary

まとめ Summary

軌道設計最適化問題に対する多目的設計探査の有用性が示された

Usefulness of MODE for trajectory design optimization problems was shown

- 従来手法により得られた設計よりも優れた設計候補を多数発見

  Many designs that outperform conventional design were found
- 軌道設計問題に関するさまざまな新しい知見の獲得

  New findings on trajectory design
- 軌道設計にかかる期間を大幅に短縮

  Design period is reduced significantly